《基础物理实验》课程完整报告

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实验 E4. 基于光纤光谱仪的原子发射光谱实验

1　实验目的

(1) 掌握光谱仪的工作原理。

(2) 观测钠原子光谱，了解碱金属原子光谱的一般规律。

(3) 观测汞原子光谱，加深对碱金属原子中外层电子与原子核相互作用的了解。

(4) 观测 LED、氢氘灯、溴钨灯等多种光源的发射光谱，学习光谱的异同。

2　仪器用具

仪器	数量	样式
光纤光谱仪	1	低压汞灯、低压钠灯、氢氘灯、溴钨灯 红、黄、蓝、绿、白色的发光二极管
光源	9	
实验测控用计算机	1	

3　实验原理

3.1　碱金属原子光谱

碱金属原子和氢原子一样，核外只有一个价电子，但在碱金属原子中除了一个价电子外，还有内封闭壳层的电子，这些内封闭壳层电子与原子核统称为原子实。价电子是在原子核和内部电子共同组成的力场中运动。原子实作用于价电子的电场与电荷的电场有显著不同，特别是当价电子轨道贯穿原子实时（称贯穿轨道），这种差别就更为突出。碱金属原子光谱公式为

$$\tilde{\nu} = R\left(\frac{1}{n_2^{*2}} - \frac{1}{n_1^{*2}}\right) = \frac{R}{(n' - \mu_{l'}')^2} - \frac{R}{(n - \mu_l)^2} \tag{E4.1}$$

其中，$\tilde{\nu}$ 为光谱线的波数；R 为里德伯常量；n' 与 n 分别为始态和终态的主量子数；n_2^* 与 n_1^* 分别为始态和终态的有效量子数；l' 与 l 分别为该量子数决定的能级的轨道量子数；$\mu_{l'}'$ 与 μ_l 分别为始态和终态的量子数（也称量子改正数、量子亏损、屏蔽常数）。

根据玻尔理论，电子轨道越接近原子中心，μ 的数值越大；轨道互相贯穿时，μ 的数值更大，此时原子核有效电荷 Z_{eff} 会显著改变。在原子核附近，Z_{eff} 接近核电荷数；距离很远时，$Z_{\text{eff}} \to 1$。因此 s 项的 μ 值最大，p 项次之，d 项更小，以此类推。屏蔽常数 μ 的大小直接反映原子实电场与点电荷电场的偏离程度。

钠原子光谱分以下四个线系：

- 主线系：$\tilde{\nu} = np \to 3s$（$n = 3, 4, 5, \cdots$）
- 锐线系：$\tilde{\nu} = ns \to 3p$（$n = 4, 5, 6, \cdots$）
- 漫线系：$\tilde{\nu} = nd \to 3p$（$n = 3, 4, 5, \cdots$）
- 基线系：$\tilde{\nu} = nf \to 3d$（$n = 4, 5, 6, \cdots$）

对于任一线系，谱线的波数为：

$$\tilde{\nu} = A_{n'l'} - \frac{R}{(n - \mu_l)^2} \tag{E4.2}$$

其中，$A_{n'l'}$ 对一个线系来说为常数，称为**固定项**。

从钠原子光谱中可观察到各线系的明显特征，这些特征也适用于其他碱金属原子光谱：

3.1.1 各线系的共同特点

(1) 同一线系内，越靠近短波方向，相邻谱线的波数差越小，最后趋于连续谱（因能量越高，能级越密，最终趋于连续）。

(2) 同一线系内，越靠近短波方向，谱线光强越小（因能级越高，原子从基态激发至该能级的难度越大）。

3.1.2 各线系的区别

(1) 光谱区域不同：主线系仅 $\tilde{\nu} = 3p \to 3s$ 的两条谱线（钠双黄线）在可见光区，其余在紫外区；锐线系和漫线系除第一条线（$\tilde{\nu} = 4s \to 3p$ 和 $\tilde{\nu} = 3d \to 3p$）在红外区外，其余在可见区；基线系在红外区。主线系的下能级是基态（$3s_{1/2}$ 能级），因此连续光谱通过钠原子蒸气后会出现主线系的吸收光谱，其第一组双线（钠双黄线，波长 589.0 nm 和 589.6 nm）称为**共振线**。

(2) 能级兼并与分裂不同：s 能级无兼并，p、d、f 能级因电子自旋-轨道相互作用发生谱项分裂，为双重兼并，且分裂随能级增高而变小。根据选择定则，主线系和锐线系是双线，漫线系和基线系是复双重线。主线系双线的波数差随短波方向减小（由 p 能级分裂大小决定），而锐线系各双线波数差相等（由内 3p 能级与各 s 能级跃迁决定）。

(3) 主线系谱线的光强较大/锐线系谱线轮廓清晰，漫线系谱线显得模糊，一般复双重线连成一片

3.2 单色仪的色散系统

记录光谱的仪器称为**光谱仪**，其核心部件是**单色仪 (monochromator)**。如图 1所示，若分光元件为棱镜，称为**棱镜单色仪**；若为光栅，称为**光栅单色仪**。

根据**瑞利定理**，两谱线可分辨的判据是：一条谱线波长的中央极大与另一条谱线波长的第一极小刚好重叠。此时分辨能力 R 定义为：

$$R = \frac{\lambda}{\Delta\lambda} \tag{E4.3}$$

其中，λ 为两波长的平均值，$\Delta\lambda$ 为波长差。

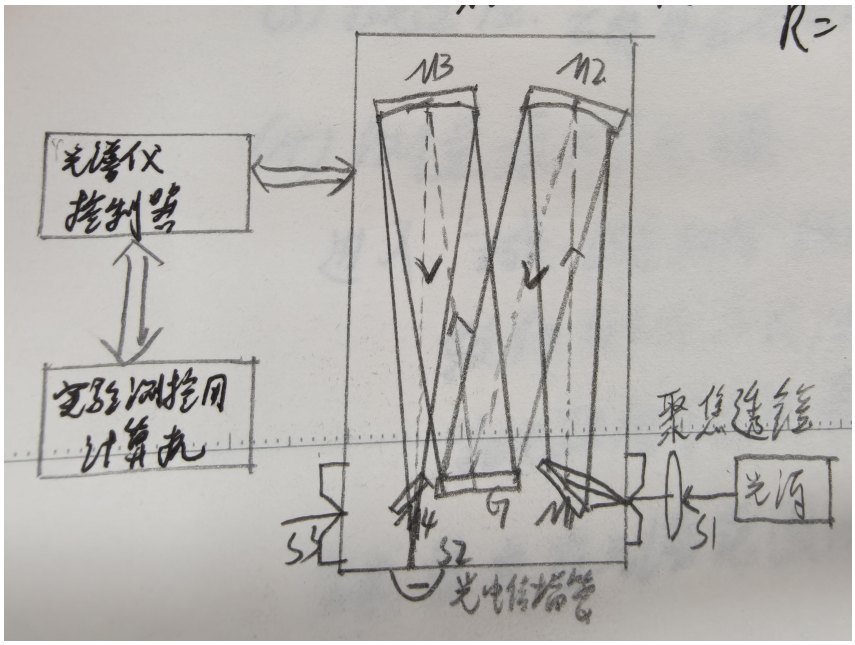


图 1: E4.1

3.2.1 棱镜单色仪

棱镜单色仪的色散元件通常为一块如图 2所示的等边三棱镜（紫外区用石英棱镜，可见光区用玻璃棱镜）。用**角色散** $\frac{\mathrm{d}\theta}{\mathrm{d}\lambda}$ 描述其分光能力，由几何关系近似为：

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \approx \frac{\Delta\theta}{\Delta\lambda} = \frac{\theta_2 - \theta_1}{\lambda_2 - \lambda_1} \tag{E4.4}$$

进一步可表示为：

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \frac{\mathrm{d}\theta}{\mathrm{d}n} \cdot \frac{\mathrm{d}n}{\mathrm{d}\lambda} \tag{E4.5}$$

其中，$\frac{\mathrm{d}\theta}{\mathrm{d}n}$ 由棱镜几何形状决定，$\frac{\mathrm{d}n}{\mathrm{d}\lambda}$ 为**色散率**（随波长变短而增大）。

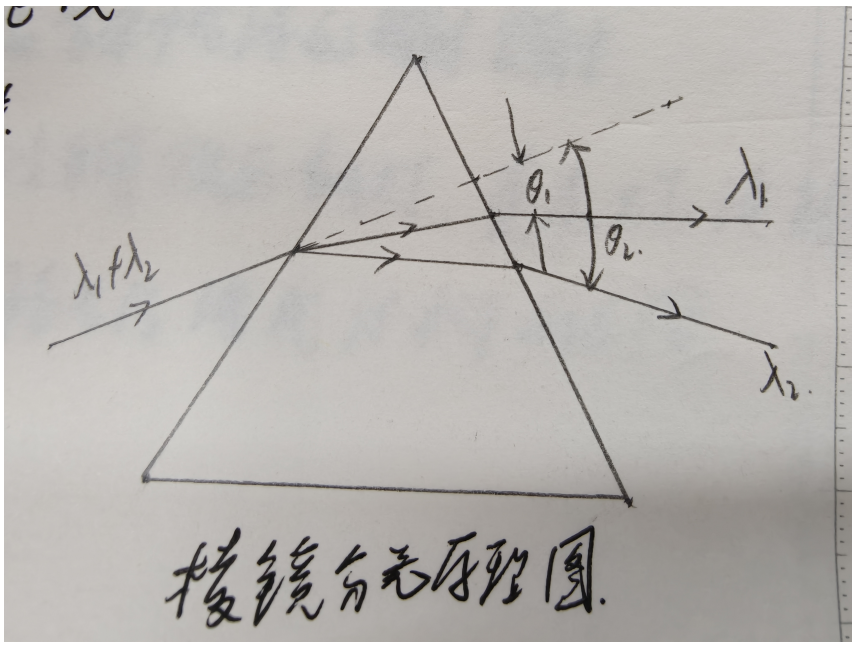


图 2: E4.2

3.2.2 光栅单色仪

如图 3所示，光栅的分光原理基于衍射，平面透射光栅的光栅方程为：

$$k\lambda = d\sin\theta, \quad k = 0, 1, 2, 3, \cdots \tag{E4.6}$$

若入射光与光栅法线成 i 角，光栅方程为：

$$k\lambda = d(\sin\theta + \sin i) \tag{E4.7}$$

对 λ 微分得角色散：

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \frac{k}{d\cos\theta} \tag{E4.8}$$

式 (E4.8) 表明:①$d\lambda$ 较小时,$d\theta$ 与 k 成正比, 即衍射的级数越高, 色散越大.②$d\theta$ 与 d 成正比,d 越小 (即总刻线数越多), 色散越大. ③当 $\cos\theta$ 变化不大时,$d\theta/d\lambda$ 近似为一常数, 即光栅的色散近似线性, 不随波长而变. 这给设计带来很大的方便. 不过通常均使用反射光栅, 其原理与透射光栅类似, 同时, 全息照相技术的出现使高密度光栅的制作更为简便.

为区分窄线宽谱线（如钠双黄线），需单色仪具有高色散率（如光栅单色仪一级光谱分辨率可达 0.02 nm）。由于单色仪色散的非线性，测量未知谱线波长时需拍摄**标准光谱**，通过插值法计算未知波长。

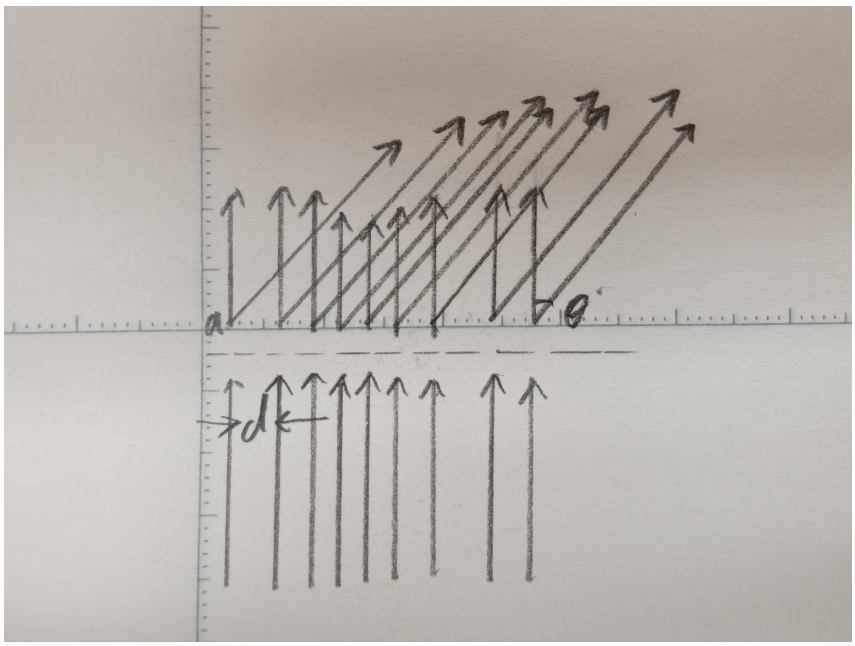


图 3: E4.3

3.3 光谱仪和光学多道分析器

光谱仪由单色仪和记录系统组成：采用棱镜单色仪的称为**棱镜光谱仪**，采用光栅单色仪的称为**光栅光谱仪**。

早期用感光胶片记录光谱（摄谱仪），现多采用光电元件：- 光电倍增管/光电二极管：每次仅记录一条谱线，灵敏度高（接近单光子测量），但速度慢；- 线阵 CCD/CMOS（**光学多道分析器 (OMA)**）：可同时记录一段波长范围内的所有谱线，速度快但灵敏度稍低。

OMA 集光学、机械、电子学和计算机技术于一体，可自动记录光谱强度。典型反射式光栅光谱仪光路（图 E4.1）：光源经入射狭缝 S1、准光镜 M2 后成为平行光，经光栅 G 衍射后，由物镜 M3 会聚到出射狭缝 S2（或经反射镜 M4 到 S3），最终由 CCD 或光电倍增管记录。

3.4 光纤光谱仪

光纤光谱仪内部结构如图 4所示，与光栅光谱仪不同，光纤光谱仪内部的所有结构都是固定的，狭缝宽度不可调节，光栅不可转动，所以出厂的时候就已经确定了光谱响应范围和光谱分辨率，不可调节。本实验拟使用的光纤光谱仪的光谱响应范围为 200nm-850nm，最高光谱分辨率 2nm，无法分辨钠双黄线。

光谱仪测量的原始数据是 CCD 探测器的计数值“S”随波长的变化曲线，再据此计算透射谱（T）、反射谱（R）吸收谱（A），标定后还可以得到光强谱（I）、色度值等。“积分时间”（即 CCD 记录光谱的曝光时间）可调节谱线峰值高度，确保全光谱范围内读数较大且不出现饱和的情况，参考值 1ms；平均次数增加可减小曲线的波动，参考值 3 次。

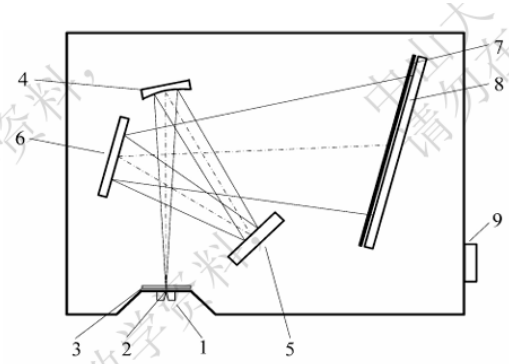


图4 光纤光谱仪结构图

1、光纤连接器，2、入射狭缝，3、滤光片，4、准直镜，5、反射光栅，6、聚焦镜，7、探测器聚光透镜，8、一维 CCD 探测器，9、USB 连接口

图 4: E4.4

4 实验装置

1. 本实验将采用如图 4 所示的光纤光谱仪来测量光源的发射光谱。光谱仪的品牌或型号可能不一样，如海洋光学或杏林睿光，测控程序界面也不一样，但操作方法基本相同，请实验前认真阅读操作说明。中山大学物理实验教学中心教学资料，

2. 采用如图 5 所示的电路来改变 LED 光源的发光强度.

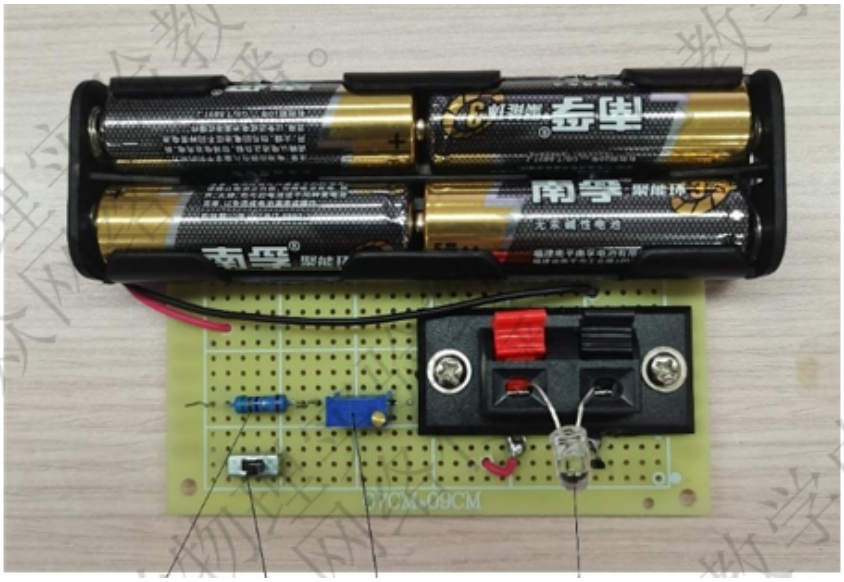


图 5: 待测量 LED 样品线路图

5 实验内容

(1) 测量各种光源的发射光谱，包括低压汞灯、低压钠灯、氢氘灯、溴钨灯、各色 LED 灯。

(2) 测量汞灯在波长 $\lambda = 546nm$、钠灯在波长 $\lambda = 589nm$ 处谱线强度从刚点亮开始随时间逐渐增加过程的变化曲线。

(3) 分别测出白光 LED（$\lambda = 545nm$）、红光 LED（$\lambda = 627nm$）的发光相对强度随工作电流强度 I 变化的曲线，电流强度通过图 5 所示电路的固定电阻两端的电压来计算，该电阻也称为采样电阻。为保护电路，电流的最大值 $I_{max} < 40mA$。

6 操作方法

6.1 使用海洋光学的光纤光谱仪进行测量

6.1.1 测发射光谱

(1) 双击“SpectraSuite”图标，得如图 6所示界面。

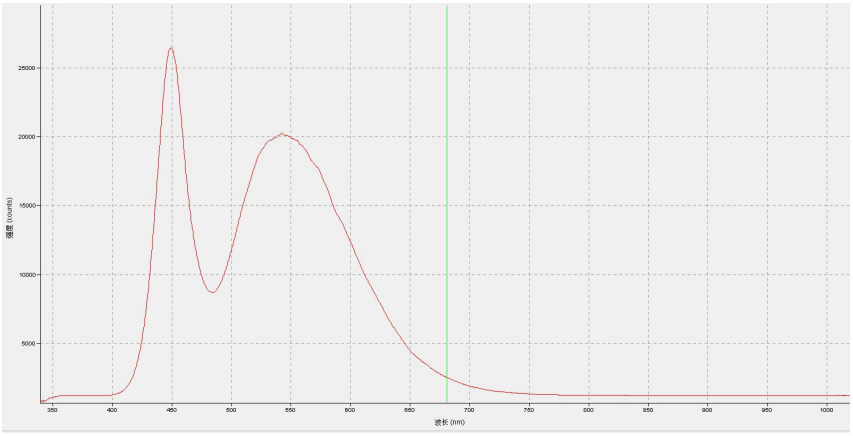


图 6: 海洋光学光谱仪软件界面

(2) 点亮光源，把固定在支撑杆上的光纤指向光源，点击运行“▷”图标采集光谱，并调整光纤入光端与光源的距离、方向，以及积分时间、平均次数等参数，确保全光谱曲线峰值不饱和。

(3) 保存光谱的数据和曲线截图。

6.1.2 测光谱强度随时间变化曲线

(1) 点击工具栏中的“时序图”按钮，出现如图 7所示的界面。



图 7: 时序图界面

(2) 选择测量参数，如“每次扫描后更新”、“One 波长”（单波长）546.18nm 等。最后点击“接受”。

(3) 把光纤头固定在支座上并指向待测的光源，点击“扫描”开始测量，并立刻点亮光源，测量谱线强度随时间变化曲线直至光强稳定（约 3 分钟）。

(4) 点击“暂停扫描”停止测量，保存光谱的数据和曲线截图。

6.1.3 测白光 LED 和红光 LED 峰值光强随电流变化关系曲线

(1) 把 LED 测量电路放在升降台上，光纤固定在支架上，调整高度对准 LED 灯。

(2) 用万用表测量采样电阻（阻值 51Ω）两端电压，计算流过 LED 的电流值。通过精密可调电阻可改变电流的大小。从 $0mA$ 至 $I_{max} = 40mA$，每隔约 $2mA$ 测量图 6 所示曲线峰值光强值并作记录，作出变化关系曲线。

6.2 使用杏林睿光的光纤光谱仪进行测量

操作方法同上，软件界面不同。

(1) 双击“BSV”图标，得如图 8 所示界面。点击菜单栏的“文件 > 导出数据”功能，可把光谱数据导出为 txt 文件。

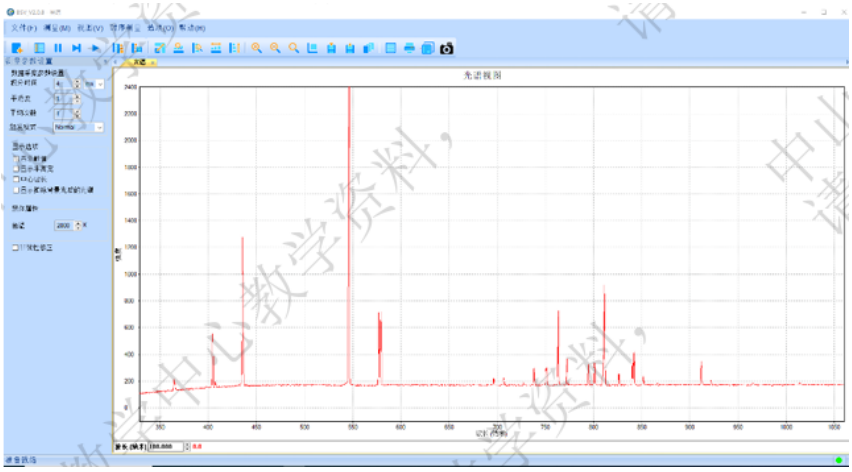


图 8: 杏林睿光光谱仪软件界面

(2) 点击菜单栏的“时序测量 > 时序测量配置”，可得图 9 所示的界面。设置单点波长数据（如 546.18nm）、初始等待时间、保存数据时间间隔、保存数据记录的时间（即测量总时间）等参数，点“确定”接受这些参数后，再点击“时序测量 > 启用”，进行时序测量，数据保存在图 9 界面中配置的文件里。

(3) 改变 LED 工作电流，测量图 8所示曲线峰值光强随电流变化关系数据并画曲线。

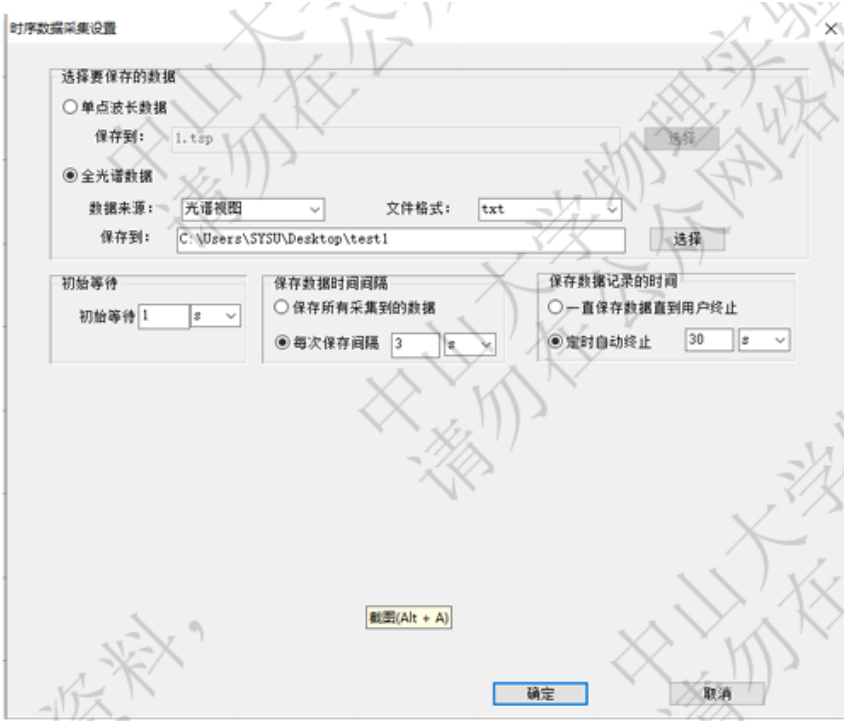


图 9: 时序测量配置界面

7 数据记录及处理

7.1 (1)

- 低压汞灯

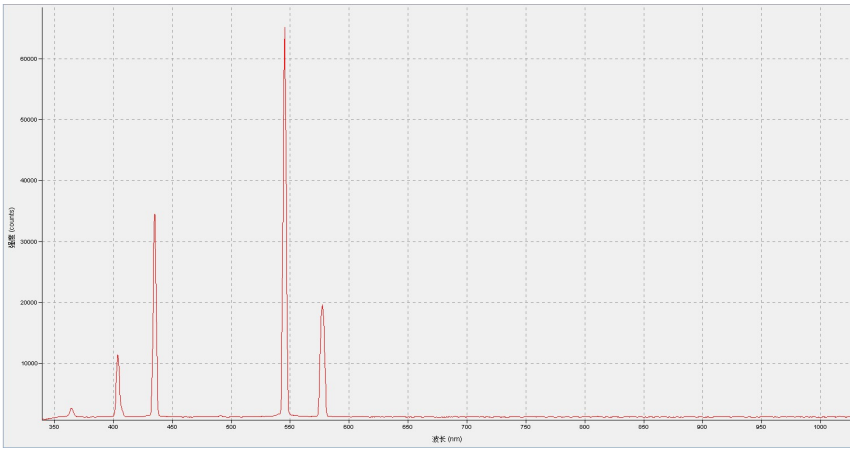


图 10: Hg

- 低压钠灯

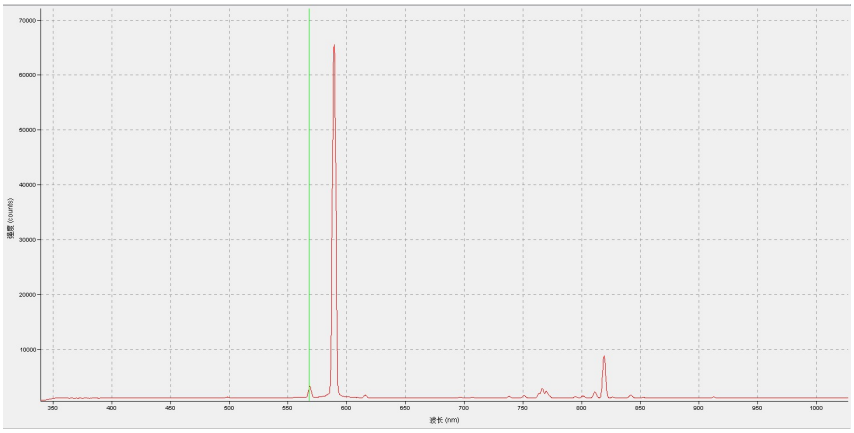


图 11: Na

- 氢氘灯

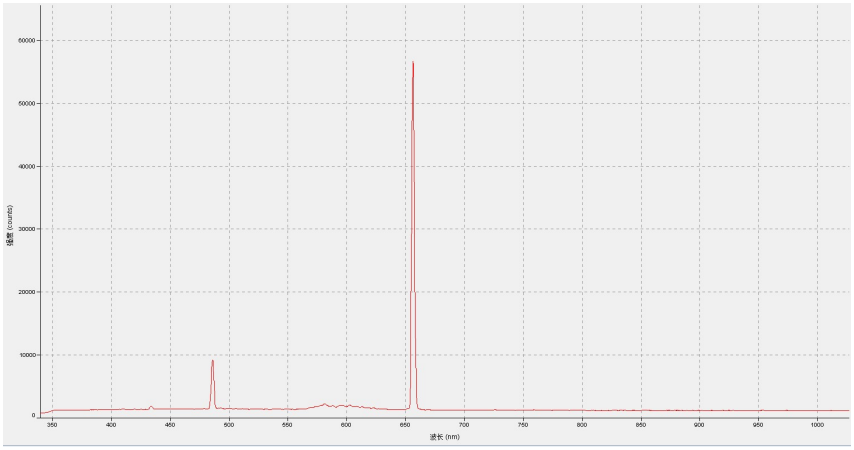


图 12: H

- 溴钨灯

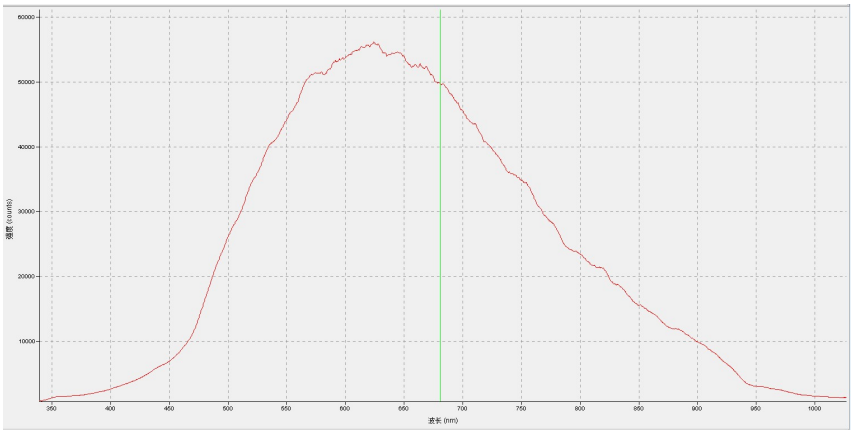


图 13: BrW

- LED
  - 红

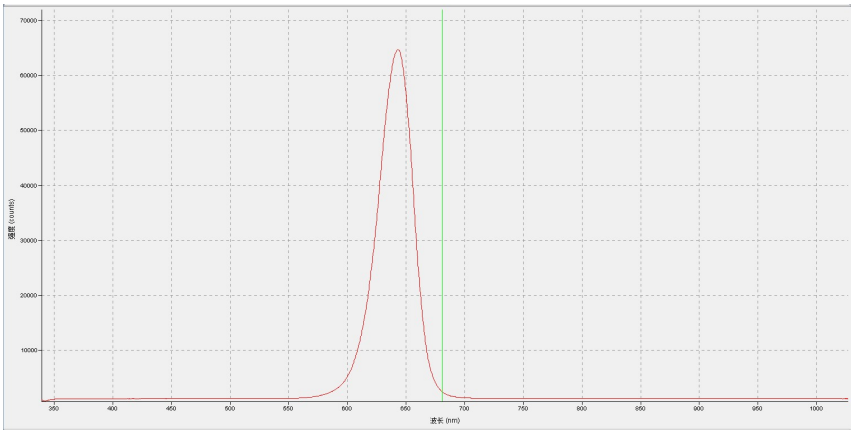


图 14: red

  - 黄

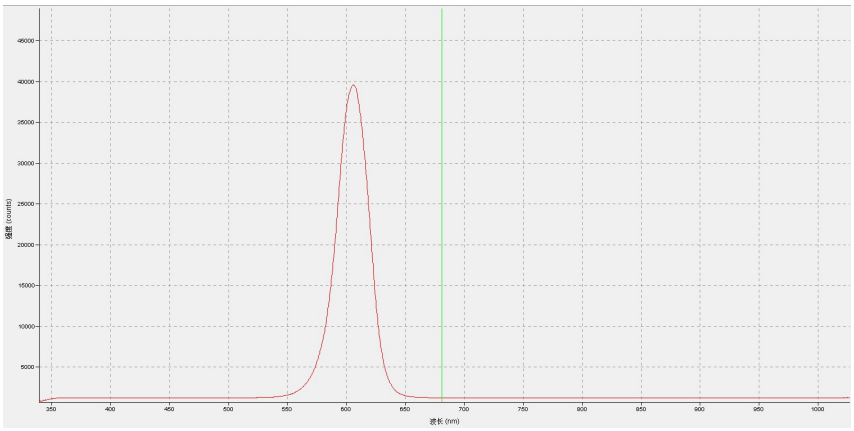


图 15: yellow

  - 蓝

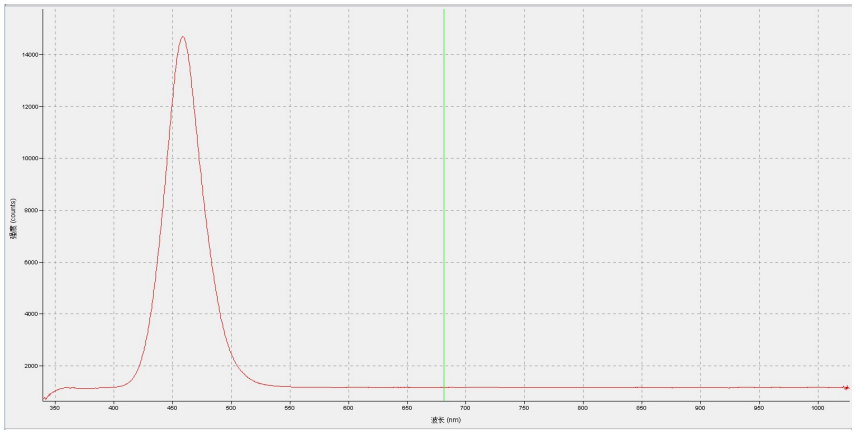


图 16: blue

– 绿

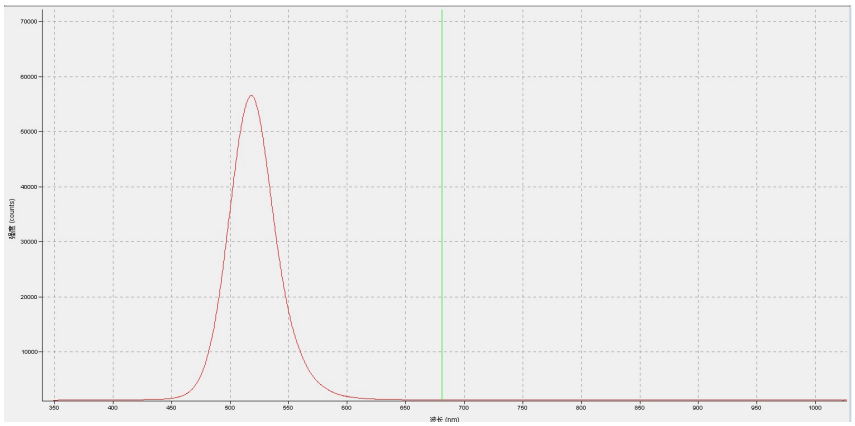


图 17: green

– 白

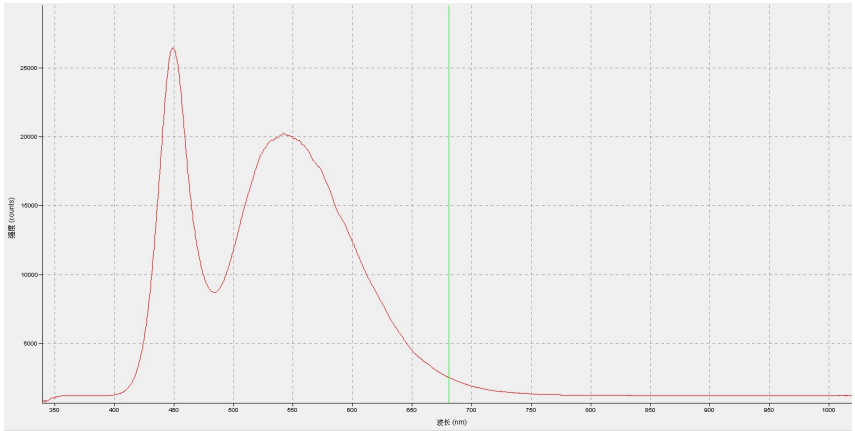


图 18: white

7.2 (2)

- Hg 灯

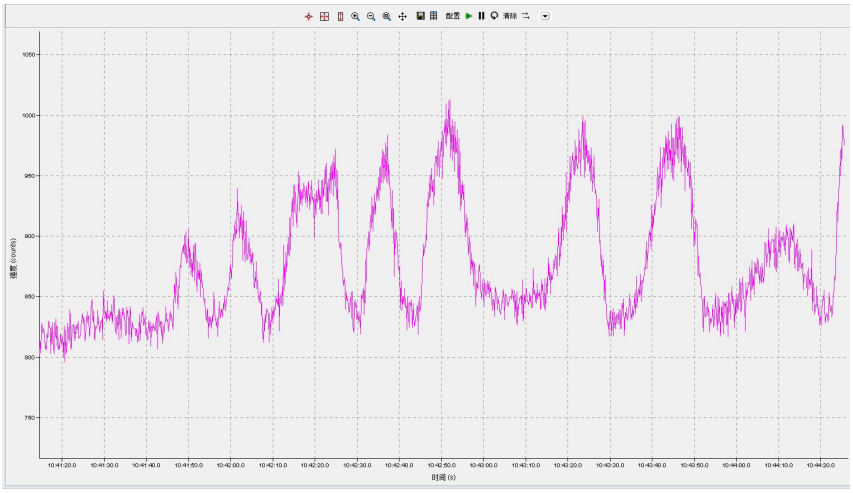


图 19: Hg'

- Na 灯

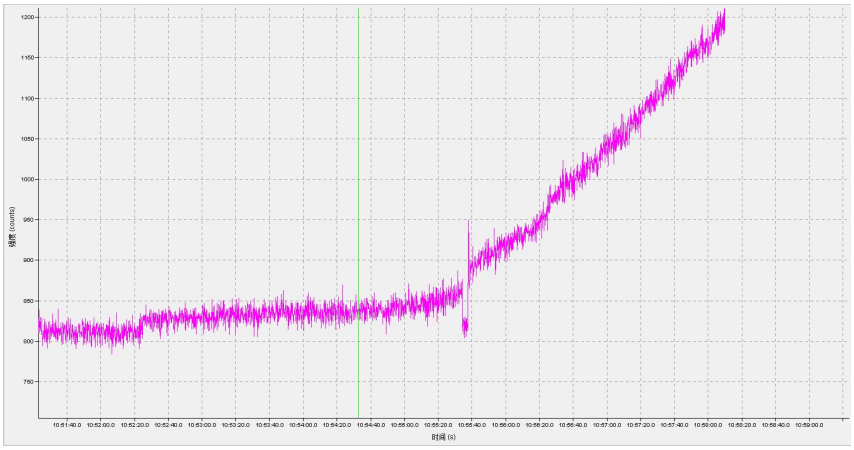


图 20: Na'

7.3 (3)

- 红

I/mA	0	2	4	6	8	10	12	14
光强/counts	0	2745	5032	7501	9768	12248	14584	16814
I/mA	16	18	20	22	24	26	28	30
光强/counts	18564	21514	22814	25224	27024	28484	29364	29664
I/mA	32	34	36	38	40			
光强/counts	30854	31624	36824	38484	40624			

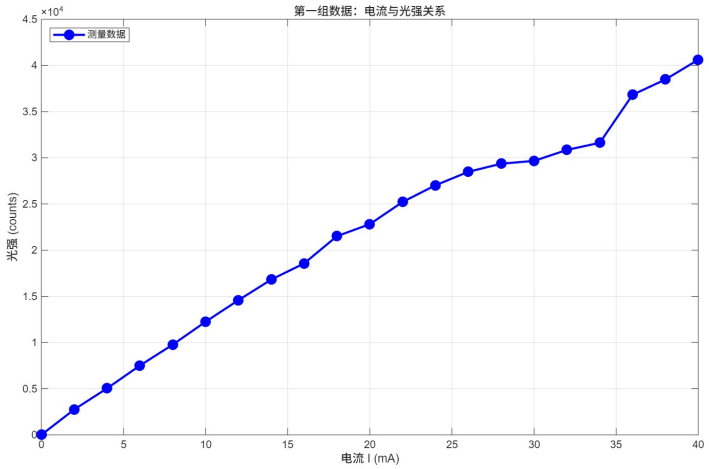


图 21: 红光光强随电流强度变化

- 白

I/mA	0	2	4	6	8	10	12	14
光强/counts	0	1792	2406	3118	3626	4108	4351	5071
I/mA	16	18	20	22	24	26	28	30
光强/counts	5298	5840	6261	6222	6407	6688	6944	7058
I/mA	32	34	36	38	40			
光强/counts	7120	7208	7281	7305	7341			

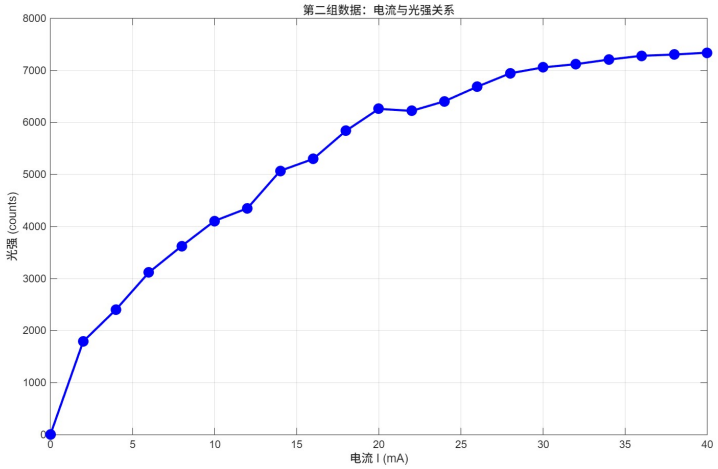


图 22: 白光光强随电流强度变化

8 讨论与分析

本实验利用光纤光谱仪对多种光源（低压汞灯、低压钠灯、氢氘灯、溴钨灯、各色LED）的发射光谱进行了测量，并进一步研究了光谱强度随时间及电流变化的规律。

8.1 光源发射光谱特征分析

- **低压汞灯**：在可见光区观测到多条分立谱线，查阅资料在汞原子理论值附近，说明实验较成功，符合汞原子能级跃迁特征。谱线尖锐，说明原子能级分立性强。
- **低压钠灯**：在 589.0 nm 附近处为峰值，属于主线系的共振线。由于光谱仪分辨率限制，双线未能完全分辨，但仍可见光谱峰形不对称。
- **氢氘灯**：观察到巴尔末线系（如 656 nm、486 nm、434 nm），氢与氘的同位素位移因分辨率限制未能明显区分。
- **溴钨灯**：光谱呈连续分布，符合黑体辐射特征，峰值波长随温度变化，属于热辐射光源。
- **各色 LED**：光谱为连续宽带，峰值波长与发光颜色一致，如红光 LED 峰值在 627 nm 附近，白光 LED 光谱覆盖整个可见光区，具有蓝光激发荧光粉的典型特征。

8.2 光谱强度随时间变化分析

- **汞灯（546 nm）与钠灯（589 nm）**：刚点亮时谱线强度较低，随灯管温度升高、原子激发态布居数增加，强度逐渐上升，约 3 分钟后趋于稳定。说明气体放电光源需预热才能达到稳定发光状态。在钠灯线接近 10:55:40 附近谱线强度突然下降一段，且该段前与该段后上升速率对接有部分偏差，系当时不慎碰到光纤导致变化，但发现并不影响我们发现一般规律，故选择继续完成实验

8.3 LED 光强随电流变化分析

- **红光 LED**：光强随电流近似线性增长，在 0–40 mA 范围内未明显饱和，说明在该范围内 LED 发光效率基本稳定。在 30mA 后曲线波折，是因为系统误差导致，我们所使用的器械，在我们使用螺丝刀调节灵敏度很高，会出现突变状况，同时调完后拿走后，光强会下降，两个效应都在高电流 (接近 40mA) 时更为明显，故出现该波折
- **白光 LED**：光强随电流增长趋势逐渐减缓，尤其在 >30 mA 后增长缓慢，可能由于发热导致效率下降（效率衰减效应），或荧光粉饱和现象。

9 思考题

(1) 钠原子光谱有哪些特征？如何从光谱图判别各谱线所属线系？

①钠原子光谱具有线系结构明确、双线特征显著、光强和分布区域有规律等特点

线系的共同规律:

- 同一线系内，谱线越靠近短波方向（高能级），相邻谱线的波数差越小，最终趋于连续谱（因高能级间距逐渐缩小，直至连续）。
- 同一线系内，越靠近短波方向，谱线光强越小（激发到高能级的概率低）。

各线系的区别特征:

- 主线系：仅两条谱线（钠双黄线，波长 589.0 nm 和 589.6 nm）在可见光区，其余在紫外区；下能级是基态（3s / ），是钠原子的共振线（吸收光谱中最明显的线系）。
- 锐线系：除第一条线在红外区外，其余在可见光区；双线的波数差处处相等（由内 3p 能级与各 s 能级跃迁的特性决定）。
- 漫线系：除第一条线在红外区外，其余在可见光区；谱线较锐线系模糊（复双重线结构）。
- 基线系：全部在红外区；谱线更模糊，是复双重线结构。

②从光谱图判别线系的方法:

可通过光谱区域、双线特征、光强变化三方面综合判断：

- 若谱线在可见光区且有一对强度很高的黄线（钠双黄线），则属于主线系；
- 若谱线在可见光区，且双线的波数差始终相等，则属于锐线系；
- 若谱线在可见光区但结构较模糊（复双重线），则属于漫线系；
- 若谱线在红外区，则属于基线系。

(2) 什么是标准谱线？

标准谱线是指波长已知且精度很高的特征谱线（如低压汞灯、钠灯的特征谱线）。

(3) 光谱仪使用前为什么要进行标定？

光谱仪使用前标定的核心目的是确保波长测量的准确性，具体原因如下：

- 色散的非线性：光栅或棱镜的色散率（波长随衍射角 / 偏向角的变化率）不是严格常数，不同波长的谱线在光谱仪中的“位置 - 波长”关系是非线性的。只有通过标准谱线标定，才能建立准确的“位置 - 波长”校准曲线。
- 机械与光学误差：光谱仪的狭缝宽度、反射镜角度、光栅安装精度等存在微小误差，会导致谱线的实际位置与理论位置偏离。标定可修正这些系统误差。
- 光电元件的响应差异：光电倍增管、CCD 等检测元件对不同波长的光响应效率不同，标定过程中也可辅助修正光强测量的系统误差（若涉及光强定量分析）。

参考文献

[1] 沈韩. 大学物理实验[M]. 北京: 科学出版社, 2024.

附录: 源代码

Listing 1: MATLAB Code

```
% 第一组数据
I1 = [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32,
    34, 36, 38, 40];
counts1 = [0, 2745, 5032, 7501, 9768, 12248, 14584, 16814, 18564,
    21514, 22814, 25224, 27024, 28484, 29364, 29664, 30854, 31624,
    36824, 38484, 40624];

% 线性拟合
p1 = polyfit(I1, counts1, 1);
slope1 = p1(1);
intercept1 = p1(2);

% 计算 R²
fitted_values1 = polyval(p1, I1);
ss_res1 = sum((counts1 - fitted_values1).^2);
ss_tot1 = sum((counts1 - mean(counts1)).^2);
r_squared1 = 1 - (ss_res1 / ss_tot1);

% 绘图
figure;
plot(I1, counts1, 'bo-', 'LineWidth', 2, 'MarkerSize', 8,
    'MarkerFaceColor', 'blue');
hold on;

% 图形设置
grid on;
xlabel('电流 I (mA)');
ylabel('光强 (counts)');
title('第一组数据: 电流与光强关系');

legend('测量数据', '线性拟合', 'Location', 'northwest');

% 输出结果到命令窗口
fprintf('第一组数据线性方程: y = %.2fx + %.2f\n', slope1,
    intercept1);
fprintf('R² = %.6f\n\n', r_squared1);

%% 绘制第二张图

% 第二组数据
I2 = [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32,
    34, 36, 38, 40];
counts2 = [0, 1792, 2406, 3118, 3626, 4108, 4351, 5071, 5298, 5840,
```

```
    6261, 6222, 6407, 6688, 6944, 7058, 7120, 7208, 7281, 7305,
    7341];

% 线性拟合
p2 = polyfit(I2, counts2, 1);
slope2 = p2(1);
intercept2 = p2(2);

% 计算 R²
fitted_values2 = polyval(p2, I2);
ss_res2 = sum((counts2 - fitted_values2).^2);
ss_tot2 = sum((counts2 - mean(counts2)).^2);
r_squared2 = 1 - (ss_res2 / ss_tot2);

% 绘图
figure;
plot(I2, counts2, 'bo-', 'LineWidth', 2, 'MarkerSize', 8,
    'MarkerFaceColor', 'blue');
hold on;

% 图形设置
grid on;
xlabel('电流 I (mA)');
ylabel('光强 (counts)');
title('第二组数据: 电流与光强关系');


% 输出结果到命令窗口
fprintf('第二组数据线性方程: y = %.2fx + %.2f\n', slope2,
    intercept2);
fprintf('R² = %.6f\n', r_squared2);
```

附录: 原始数据

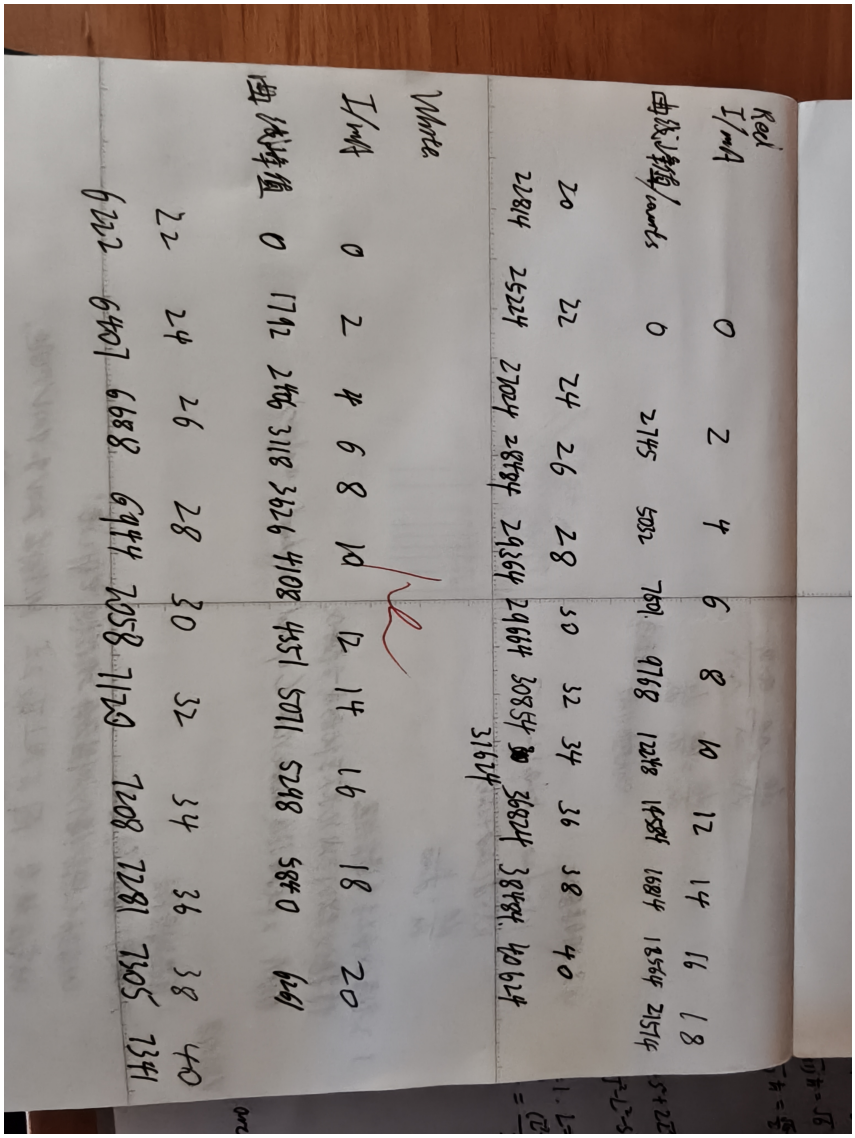


图 23: data

Listing 2: 汞灯

```
暗光谱: 否
参考光谱: 否
光谱数量: 1
光谱仪: USB2+H14105
积分时间（毫秒）: 1 (USB2+H14105)
平均次数: 1 (USB2+H14105)
平滑度: 3 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)
339.35  762.15
339.73  762.15
340.11  762.15
340.49  745.85
340.87  731.29
```

```
14  341.25  731.63
15  341.63  724.01
16  342.01  736.14
17  342.39  733.37
18  342.77  732.33
19  343.15  738.92
20  343.53  762.84
21  343.91  764.23
22  344.29  786.42
23  344.66  802.72
24  345.04  832.88
25  345.42  862.70
26  345.80  876.23
27  346.18  885.94
28  346.56  911.25
29  346.94  921.65
30  347.32  919.57
31  347.70  917.14
32  348.08  922.34
33  348.45  937.25
34  348.83  963.26
35  349.21  974.70
36  349.59  1000.02
37  349.97  1025.67
38  350.35  1036.08
39  350.73  1055.84
40  351.10  1071.10
41  351.48  1072.14
42  351.86  1087.40
43  352.24  1091.56
44  352.62  1095.72
45  353.00  1118.26
46  353.37  1127.62
47  353.75  1128.31
48  354.13  1138.02
49  354.51  1140.10
50  354.89  1144.61
51  355.27  1140.45
52  355.64  1140.10
53  356.02  1140.45
54  356.40  1147.38
55  356.78  1151.20
56  357.16  1158.48
57  357.53  1143.92
58  357.91  1165.41
59  358.29  1177.55
60  358.67  1189.69
```

```
 61  359.04  1206.33
 62  359.42  1220.20
 63  359.80  1226.79
 64  360.18  1254.18
 65  360.55  1268.40
 66  360.93  1295.10
 67  361.31  1335.32
 68  361.69  1417.84
 69  362.06  1548.91
 70  362.44  1712.23
 71  362.82  1929.29
 72  363.20  2156.07
 73  363.57  2368.62
 74  363.95  2499.00
 75  364.33  2552.40
 76  364.71  2520.15
 77  365.08  2425.84
 78  365.46  2246.22
 79  365.84  2033.32
 80  366.21  1812.79
 81  366.59  1640.46
 82  366.97  1507.65
 83  367.34  1391.15
 84  367.72  1306.54
 85  368.10  1261.46
 86  368.47  1220.20
 87  368.85  1190.38
 88  369.23  1178.94
 89  369.60  1144.96
 90  369.98  1136.98
 91  370.36  1142.18
 92  370.73  1131.09
 93  371.11  1128.66
 94  371.49  1117.22
 95  371.86  1115.14
 96  372.24  1125.54
 97  372.62  1131.43
 98  372.99  1120.68
 99  373.37  1124.15
100  373.74  1120.34
101  374.12  1114.44
102  374.50  1107.16
103  374.87  1101.27
104  375.25  1100.23
105  375.62  1103.35
106  376.00  1094.68
107  376.38  1096.06
```

```
108  376.75  1109.59
109  377.13  1115.83
110  377.50  1112.71
111  377.88  1094.33
112  378.26  1102.65
113  378.63  1103.69
114  379.01  1100.92
115  379.38  1086.36
116  379.76  1078.03
117  380.13  1093.64
118  380.51  1108.89
119  380.88  1098.49
120  381.26  1103.35
121  381.64  1110.28
122  382.01  1123.80
123  382.39  1129.35
124  382.76  1120.68
125  383.14  1114.09
126  383.51  1112.01
127  383.89  1103.69
128  384.26  1102.65
129  384.64  1104.39
130  385.01  1104.39
131  385.39  1105.77
132  385.76  1109.24
133  386.14  1108.20
134  386.51  1113.75
135  386.89  1111.67
136  387.26  1110.28
137  387.64  1109.59
138  388.01  1110.63
139  388.39  1120.34
140  388.76  1128.31
141  389.13  1135.94
142  389.51  1144.61
143  389.88  1144.26
144  390.26  1147.04
145  390.63  1149.12
146  391.01  1138.71
147  391.38  1143.92
148  391.76  1134.90
149  392.13  1132.82
150  392.50  1143.22
151  392.88  1143.22
152  393.25  1144.61
153  393.63  1157.78
154  394.00  1150.50
```

```
155  394.37  1164.37
156  394.75  1163.33
157  395.12  1160.91
158  395.50  1170.61
159  395.87  1169.57
160  396.24  1159.87
161  396.62  1166.80
162  396.99  1166.80
163  397.37  1169.57
164  397.74  1179.28
165  398.11  1184.83
166  398.49  1201.82
167  398.86  1212.92
168  399.23  1218.12
169  399.61  1219.16
170  399.98  1240.31
171  400.35  1280.19
172  400.73  1481.30
173  401.10  1908.84
174  401.47  2694.22
175  401.85  4005.96
176  402.22  5770.55
177  402.59  7693.95
178  402.97  9535.17
179  403.34  10773.05
180  403.71  11363.21
181  404.09  11229.37
182  404.46  10256.05
183  404.83  8673.51
184  405.20  6937.00
185  405.58  5274.35
186  405.95  4047.22
187  406.32  3217.11
188  406.70  2703.58
189  407.07  2441.09
190  407.44  2303.43
191  407.81  2108.91
192  408.19  1875.55
193  408.56  1659.18
194  408.93  1453.56
195  409.30  1313.13
196  409.68  1231.30
197  410.05  1184.83
198  410.42  1168.88
199  410.79  1165.41
200  411.17  1164.37
201  411.54  1167.15
```

```
202  411.91  1167.15
203  412.28  1162.99
204  412.65  1151.20
205  413.03  1148.42
206  413.40  1149.81
207  413.77  1141.83
208  414.14  1141.14
209  414.51  1135.25
210  414.89  1134.55
211  415.26  1140.10
212  415.63  1138.71
213  416.00  1141.83
214  416.37  1135.94
215  416.74  1131.78
216  417.12  1139.06
217  417.49  1146.00
218  417.86  1146.00
219  418.23  1142.18
220  418.60  1140.79
221  418.97  1149.81
222  419.34  1151.89
223  419.72  1155.01
224  420.09  1137.67
225  420.46  1148.42
226  420.83  1153.62
227  421.20  1170.27
228  421.57  1170.96
229  421.94  1183.79
230  422.31  1173.39
231  422.69  1190.38
232  423.06  1180.67
233  423.43  1185.87
234  423.80  1175.47
235  424.17  1178.94
236  424.54  1173.39
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1546  873.70  1081.50
1547  874.02  1087.40
1548  874.33  1093.29
1549  874.64  1103.69
1550  874.95  1112.01
1551  875.27  1104.39
1552  875.58  1099.18
1553  875.89  1096.76
1554  876.21  1088.78
1555  876.52  1089.13
1556  876.83  1094.33
1557  877.14  1096.76
1558  877.46  1112.36
1559  877.77  1119.64
1560  878.08  1115.48
1561  878.39  1123.80
1562  878.71  1121.72
1563  879.02  1114.79
1564  879.33  1108.55
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880.27  1096.41
880.58  1092.25
880.89  1095.72
881.20  1094.33
881.52  1106.81
881.83  1104.04
882.14  1106.81
882.45  1107.85
882.76  1100.57
883.07  1103.69
883.39  1100.57
883.70  1096.76
884.01  1110.97
884.32  1114.10
884.63  1118.26
884.94  1124.15
885.25  1116.87
885.57  1119.99
885.88  1122.07
886.19  1115.14
886.50  1123.46
886.81  1129.70
887.12  1132.13
887.43  1124.15
887.74  1133.51
888.05  1119.30
888.37  1129.01
888.68  1123.11
888.99  1117.56
889.30  1122.42
889.61  1130.05
889.92  1127.96
890.23  1134.21
890.54  1123.11
890.85  1118.95
891.16  1125.19
891.47  1123.11
891.78  1131.78
892.09  1124.84
892.40  1116.87
892.71  1123.11
893.02  1118.60
893.33  1112.71
893.64  1107.16
893.95  1097.10
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895.81  1097.45
896.12  1109.59
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896.74  1114.09
897.05  1108.55
897.36  1112.36
897.67  1110.28
897.98  1112.71
898.29  1108.89
898.60  1126.23
898.91  1119.64
899.22  1124.50
899.52  1129.01
899.83  1126.23
900.14  1115.83
900.45  1104.04
900.76  1084.62
901.07  1085.66
901.38  1087.05
901.69  1081.50
902.00  1088.44
902.30  1093.98
902.61  1103.69
902.92  1107.51
903.23  1101.96
903.54  1103.35
903.85  1112.01
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904.46  1106.12
904.77  1109.24
905.08  1112.36
905.39  1112.01
905.70  1101.61
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906.31  1092.25
906.62  1092.60
906.93  1084.97
907.24  1091.56
907.55  1094.68
907.85  1097.45
908.16  1094.33
908.47  1098.49
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1661	909.39	1117.22
1662	909.70	1121.03
1663	910.01	1130.74
1664	910.31	1133.86
1665	910.62	1124.50
1666	910.93	1122.07
1667	911.24	1120.34
1668	911.54	1123.46
1669	911.85	1117.56
1670	912.16	1113.05
1671	912.46	1118.26
1672	912.77	1127.27
1673	913.08	1132.13
1674	913.39	1123.80
1675	913.69	1115.14
1676	914.00	1126.23
1677	914.31	1133.17
1678	914.61	1124.50
1679	914.92	1127.96
1680	915.23	1132.82
1681	915.53	1138.37
1682	915.84	1138.02
1683	916.15	1139.41
1684	916.45	1129.70
1685	916.76	1132.13
1686	917.06	1126.58
1687	917.37	1127.62
1688	917.68	1125.54
1689	917.98	1120.68
1690	918.29	1114.44
1691	918.60	1113.75
1692	918.90	1129.70
1693	919.21	1137.33
1694	919.51	1127.27
1695	919.82	1123.80
1696	920.13	1136.63
1697	920.43	1140.45
1698	920.74	1141.49
1699	921.04	1124.84
1700	921.35	1117.91
1701	921.65	1128.31
1702	921.96	1132.47
1703	922.26	1122.76
1704	922.57	1111.32
1705	922.88	1113.05

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1713  925.32  1116.87
1714  925.62  1121.38
1715  925.93  1127.96
1716  926.23  1116.52
1717  926.54  1121.38
1718  926.84  1129.70
1719  927.15  1122.76
1720  927.45  1122.42
1721  927.76  1128.31
1722  928.06  1132.82
1723  928.36  1138.02
1724  928.67  1132.13
1725  928.97  1121.03
1726  929.28  1125.54
1727  929.58  1132.13
1728  929.89  1130.74
1729  930.19  1116.18
1730  930.50  1115.83
1731  930.80  1121.03
1732  931.10  1117.91
1733  931.41  1113.05
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1736  932.32  1121.03
1737  932.62  1117.56
1738  932.93  1108.20
1739  933.23  1116.18
1740  933.53  1120.34
1741  933.84  1117.56
1742  934.14  1109.24
1743  934.44  1115.14
1744  934.75  1108.20
1745  935.05  1114.79
1746  935.36  1108.89
1747  935.66  1096.41
1748  935.96  1106.47
1749  936.27  1097.45
1750  936.57  1087.05
1751  936.87  1090.52
1752  937.17  1084.97
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1754  937.78  1088.78
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1757  938.69  1114.79
1758  938.99  1120.68
1759  939.29  1121.38
1760  939.60  1124.50
1761  939.90  1137.67
1762  940.20  1122.42
1763  940.51  1112.36
1764  940.81  1113.75
1765  941.11  1106.12
1766  941.41  1106.12
1767  941.71  1105.08
1768  942.02  1093.29
1769  942.32  1096.41
1770  942.62  1096.06
1771  942.92  1090.86
1772  943.23  1091.56
1773  943.53  1092.60
1774  943.83  1093.29
1775  944.13  1091.90
1776  944.43  1094.33
1777  944.74  1095.02
1778  945.04  1104.04
1779  945.34  1104.73
1780  945.64  1114.10
1781  945.94  1113.05
1782  946.24  1117.22
1783  946.55  1117.56
1784  946.85  1116.18
1785  947.15  1099.18
1786  947.45  1101.27
1787  947.75  1097.45
1788  948.05  1100.23
1789  948.35  1104.04
1790  948.66  1109.93
1791  948.96  1112.01
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1793  949.56  1131.09
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1797  950.76  1117.91
1798  951.06  1122.42
1799  951.36  1116.52
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1806  953.47  1115.83
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1808  954.07  1126.58
1809  954.37  1119.64
1810  954.67  1117.56
1811  954.97  1117.22
1812  955.27  1110.63
1813  955.57  1108.89
1814  955.87  1116.18
1815  956.17  1109.93
1816  956.47  1119.64
1817  956.77  1123.46
1818  957.07  1125.88
1819  957.37  1133.17
1820  957.67  1146.34
1821  957.97  1134.90
1822  958.27  1131.09
1823  958.57  1130.74
1824  958.87  1121.38
1825  959.17  1120.34
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1827  959.76  1093.64
1828  960.06  1087.74
1829  960.36  1095.02
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1832  961.26  1101.61
1833  961.56  1112.01
1834  961.86  1119.99
1835  962.16  1124.15
1836  962.46  1121.03
1837  962.76  1105.08
1838  963.05  1099.88
1839  963.35  1109.59
1840  963.65  1096.76
1841  963.95  1093.98
1842  964.25  1090.17
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1844  964.85  1083.58
1845  965.14  1083.23
1846  965.44  1084.97
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1856  968.42  1114.10
1857  968.72  1114.10
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1861  969.91  1127.62
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1864  970.81  1120.68
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1874  973.78  1138.02
1875  974.08  1133.86
1876  974.37  1130.39
1877  974.67  1124.84
1878  974.97  1126.23
1879  975.26  1125.54
1880  975.56  1121.03
1881  975.86  1124.15
1882  976.15  1136.63
1883  976.45  1138.71
1884  976.75  1133.17
1885  977.04  1121.38
1886  977.34  1114.10
1887  977.63  1116.87
1888  977.93  1105.08
1889  978.23  1105.77
1890  978.52  1106.47
1891  978.82  1103.69
1892  979.12  1123.80
1893  979.41  1135.59
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1897  980.60  1115.14
1898  980.89  1119.99
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1904  982.67  1104.73
1905  982.96  1125.19
1906  983.26  1132.13
1907  983.55  1117.22
1908  983.85  1119.64
1909  984.14  1122.07
1910  984.44  1121.03
1911  984.73  1129.01
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1913  985.32  1083.93
1914  985.62  1089.82
1915  985.91  1083.93
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1917  986.50  1106.12
1918  986.80  1098.14
1919  987.09  1108.55
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1921  987.68  1127.27
1922  987.98  1132.82
1923  988.27  1126.92
1924  988.56  1126.23
1925  988.86  1130.05
1926  989.15  1118.60
1927  989.45  1108.20
1928  989.74  1106.81
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1930  990.33  1103.35
1931  990.62  1101.27
1932  990.92  1102.31
1933  991.21  1115.83
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1935  991.80  1107.16
1936  992.09  1106.81
1937  992.39  1103.35
1938  992.68  1104.73
1939  992.98  1110.28
1940  993.27  1108.89
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1942  993.86  1124.15
1943  994.15  1133.17
1944  994.44  1140.45
1945  994.74  1149.81
1946  995.03  1145.65
1947  995.32  1139.41
1948  995.62  1138.37
1949  995.91  1124.84
1950  996.20  1104.73
1951  996.50  1093.98
1952  996.79  1084.97
1953  997.08  1086.01
1954  997.38  1085.32
1955  997.67  1087.74
1956  997.96  1095.37
1957  998.25  1104.04
1958  998.55  1116.87
1959  998.84  1105.43
1960  999.13  1098.84
1961  999.42  1090.17
1962  999.72  1088.09
1963  1000.01 1085.66
1964  1000.30 1089.82
1965  1000.59 1085.66
1966  1000.89 1100.57
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1968  1001.47 1110.28
1969  1001.76 1107.85
1970  1002.06 1103.00
1971  1002.35 1108.55
1972  1002.64 1110.28
1973  1002.93 1108.20
1974  1003.22 1114.79
1975  1003.52 1101.61
1976  1003.81 1100.57
1977  1004.10 1110.97
1978  1004.39 1099.18
1979  1004.68 1096.76
1980  1004.97 1094.68
1981  1005.27 1094.68
1982  1005.56 1108.89
1983  1005.85 1109.24
1984  1006.14 1115.48
1985  1006.43 1125.88
1986  1006.72 1121.38
1987  1007.01 1116.52
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1990  1007.89 1118.60
1991  1008.18 1111.67
1992  1008.47 1110.63
1993  1008.76 1109.93
1994  1009.05 1115.14
1995  1009.34 1123.11
1996  1009.63 1117.91
1997  1009.93 1116.52
1998  1010.22 1124.15
1999  1010.51 1128.31
2000  1010.80 1134.21
2001  1011.09 1140.79
2002  1011.38 1132.47
2003  1011.67 1130.39
2004  1011.96 1126.92
2005  1012.25 1115.83
2006  1012.54 1103.35
2007  1012.83 1106.81
2008  1013.12 1101.27
2009  1013.41 1104.04
2010  1013.70 1122.07
2011  1013.99 1129.70
2012  1014.28 1141.49
2013  1014.57 1149.81
2014  1014.86 1161.60
2015  1015.15 1182.06
2016  1015.44 1197.66
2017  1015.73 1200.78
2018  1016.02 1210.84
2019  1016.31 1203.21
2020  1016.60 1205.98
2021  1016.89 1201.13
2022  1017.18 1194.19
2023  1017.47 1182.40
2024  1017.76 1171.31
2025  1018.05 1162.29
2026  1018.34 1162.29
2027  1018.62 1157.09
2028  1018.91 1146.00
2029  1019.20 1147.73
2030  1019.49 1143.92
2031  1019.78 1132.13
2032  1020.07 1135.59
2033  1020.36 1126.92
2034  1020.65 1122.07
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2036  1021.23 1089.48
2037  1021.51 1096.76
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2039  1022.09 1091.21
2040  1022.38 1098.84
2041  1022.67 1100.23
2042  1022.96 1110.28
2043  1023.25 1116.87
2044  1023.53 1117.56
2045  1023.82 1119.30
2046  1024.11 1134.21
2047  1024.40 1141.14
2048  1024.69 1148.08
2049  1024.97 1140.45
2050  1025.26 1143.57
2051  1025.55 1130.74
2052  1025.84 1119.99
2053  1026.13 1107.51
2054  1026.41 1075.26
2055  1026.70 1075.26
2056  1026.99 1075.26
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Listing 3: 钠灯

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1  暗光谱: 否
2  参考光谱: 否
3  光谱数量: 1
4  光谱仪: USB2+H14105
5  积分时间（毫秒）: 1 (USB2+H14105)
6  平均次数: 1 (USB2+H14105)
7  平滑度: 3 (USB2+H14105)
8  去除暗噪声: 否 (USB2+H14105)
9
10
11  339.35  774.28
12  339.73  774.28
13  340.11  774.28
14  340.49  754.52
15  340.87  737.18
16  341.25  745.50
17  341.63  741.00
18  342.01  735.80
19  342.39  749.32
20  342.77  768.04
21  343.15  781.91
22  343.53  804.80
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23  343.91  803.06
24  344.29  819.01
25  344.66  848.14
26  345.04  860.62
27  345.42  866.17
28  345.80  884.90
29  346.18  908.82
30  346.56  933.09
31  346.94  955.98
32  347.32  962.91
33  347.70  979.56
34  348.08  998.63
35  348.45  1017.70
36  348.83  1034.69
37  349.21  1055.15
38  349.59  1058.62
39  349.97  1077.69
40  350.35  1086.36
41  350.73  1110.97
42  351.10  1108.55
43  351.48  1113.40
44  351.86  1112.36
45  352.24  1122.42
46  352.62  1119.99
47  353.00  1136.63
48  353.37  1125.54
49  353.75  1144.96
50  354.13  1140.79
51  354.51  1148.77
52  354.89  1152.24
53  355.27  1155.36
54  355.64  1143.92
55  356.02  1140.79
56  356.40  1131.09
57  356.78  1125.88
58  357.16  1122.07
59  357.53  1123.46
60  357.91  1125.19
61  358.29  1132.47
62  358.67  1135.59
63  359.04  1133.17
64  359.42  1142.53
65  359.80  1131.78
66  360.18  1123.11
67  360.55  1124.84
68  360.93  1115.48
69  361.31  1113.75
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 73  362.82  1118.95
 74  363.20  1107.16
 75  363.57  1113.05
 76  363.95  1110.97
 77  364.33  1107.16
 78  364.71  1110.28
 79  365.08  1106.12
 80  365.46  1100.23
 81  365.84  1101.27
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 83  366.59  1090.52
 84  366.97  1090.86
 85  367.34  1090.17
 86  367.72  1091.21
 87  368.10  1093.29
 88  368.47  1083.58
 89  368.85  1087.05
 90  369.23  1083.23
 91  369.60  1089.82
 92  369.98  1080.11
 93  370.36  1074.91
 94  370.73  1076.99
 95  371.11  1088.78
 96  371.49  1082.89
 97  371.86  1088.78
 98  372.24  1086.70
 99  372.62  1090.86
100  372.99  1092.60
101  373.37  1086.70
102  373.74  1089.13
103  374.12  1100.57
104  374.50  1102.65
105  374.87  1100.57
106  375.25  1107.16
107  375.62  1118.95
108  376.00  1119.64
109  376.38  1109.59
110  376.75  1103.35
111  377.13  1105.77
112  377.50  1103.35
113  377.88  1101.96
114  378.26  1101.96
115  378.63  1106.47
116  379.01  1108.89
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118  379.76  1122.42
119  380.13  1131.43
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123  381.64  1112.71
124  382.01  1093.64
125  382.39  1092.25
126  382.76  1082.89
127  383.14  1083.93
128  383.51  1097.10
129  383.89  1087.74
130  384.26  1099.18
131  384.64  1108.20
132  385.01  1102.65
133  385.39  1100.23
134  385.76  1117.22
135  386.14  1110.63
136  386.51  1106.81
137  386.89  1115.14
138  387.26  1116.87
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1478	851.67	1200.09
1479	851.98	1221.59

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1482  852.93  1225.05
1483  853.25  1209.45
1484  853.56  1200.09
1485  853.88  1171.31
1486  854.20  1153.97
1487  854.51  1144.61
1488  854.83  1133.86
1489  855.14  1125.88
1490  855.46  1129.01
1491  855.78  1130.05
1492  856.09  1139.41
1493  856.41  1139.75
1494  856.72  1131.43
1495  857.04  1136.98
1496  857.35  1142.18
1497  857.67  1154.32
1498  857.99  1149.12
1499  858.30  1143.92
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1501  858.93  1151.89
1502  859.25  1146.00
1503  859.56  1146.00
1504  859.88  1135.59
1505  860.19  1141.49
1506  860.51  1134.90
1507  860.82  1125.19
1508  861.14  1128.31
1509  861.45  1140.79
1510  861.77  1134.55
1511  862.08  1132.47
1512  862.40  1134.21
1513  862.71  1142.18
1514  863.03  1150.85
1515  863.34  1146.34
1516  863.66  1135.59
1517  863.97  1135.94
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1519  864.60  1124.15
1520  864.92  1120.68
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1523  865.86  1144.96
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1525  866.49  1168.53
1526  866.80  1175.82
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1531  868.37  1168.53
1532  868.69  1153.28
1533  869.00  1147.38
1534  869.31  1130.74
1535  869.63  1138.37
1536  869.94  1143.22
1537  870.25  1138.02
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1539  870.88  1147.73
1540  871.20  1146.00
1541  871.51  1156.74
1542  871.82  1147.04
1543  872.14  1134.21
1544  872.45  1131.78
1545  872.76  1128.66
1546  873.08  1120.34
1547  873.39  1131.09
1548  873.70  1132.82
1549  874.02  1142.53
1550  874.33  1146.00
1551  874.64  1139.75
1552  874.95  1140.45
1553  875.27  1145.30
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1555  875.89  1134.90
1556  876.21  1122.42
1557  876.52  1123.46
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1560  877.46  1141.83
1561  877.77  1144.61
1562  878.08  1145.30
1563  878.39  1146.00
1564  878.71  1148.08
1565  879.02  1149.81
1566  879.33  1142.18
1567  879.64  1139.06
1568  879.95  1145.65
1569  880.27  1147.04
1570  880.58  1142.18
1571  880.89  1141.14
1572  881.20  1135.94
1573  881.52  1142.87
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1576	882.45	1128.66
1577	882.76	1138.02
1578	883.07	1140.10
1579	883.39	1140.79
1580	883.70	1149.12
1581	884.01	1153.28
1582	884.32	1146.69
1583	884.63	1139.06
1584	884.94	1135.59
1585	885.25	1130.05
1586	885.57	1119.64
1587	885.88	1105.43
1588	886.19	1106.12
1589	886.50	1115.14
1590	886.81	1118.60
1591	887.12	1120.34
1592	887.43	1113.05
1593	887.74	1119.99
1594	888.05	1115.83
1595	888.37	1115.14
1596	888.68	1097.80
1597	888.99	1100.92
1598	889.30	1096.76
1599	889.61	1110.97
1600	889.92	1114.09
1601	890.23	1116.18
1602	890.54	1118.60
1603	890.85	1121.72
1604	891.16	1115.48
1605	891.47	1114.10
1606	891.78	1107.16
1607	892.09	1109.59
1608	892.40	1108.89
1609	892.71	1113.05
1610	893.02	1125.88
1611	893.33	1118.95
1612	893.64	1132.13
1613	893.95	1133.17
1614	894.26	1131.09
1615	894.57	1141.14
1616	894.88	1138.71
1617	895.19	1144.26
1618	895.50	1154.32
1619	895.81	1141.14
1620	896.12	1141.49

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1625  897.67  1126.23
1626  897.98  1123.80
1627  898.29  1119.99
1628  898.60  1113.75
1629  898.91  1117.56
1630  899.22  1123.80
1631  899.52  1134.55
1632  899.83  1130.39
1633  900.14  1142.87
1634  900.45  1150.85
1635  900.76  1148.08
1636  901.07  1149.46
1637  901.38  1142.53
1638  901.69  1137.67
1639  902.00  1141.83
1640  902.30  1136.29
1641  902.61  1132.13
1642  902.92  1138.37
1643  903.23  1132.82
1644  903.54  1128.66
1645  903.85  1123.11
1646  904.16  1121.72
1647  904.46  1123.80
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1650  905.39  1144.96
1651  905.70  1157.78
1652  906.01  1166.45
1653  906.31  1169.92
1654  906.62  1172.35
1655  906.93  1162.29
1656  907.24  1150.50
1657  907.55  1133.51
1658  907.85  1134.21
1659  908.16  1130.39
1660  908.47  1124.84
1661  908.78  1123.11
1662  909.08  1119.30
1663  909.39  1121.38
1664  909.70  1135.94
1665  910.01  1128.66
1666  910.31  1132.13
1667  910.62  1151.20
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1673	912.46	1295.79
1674	912.77	1285.73
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1676	913.39	1238.58
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1679	914.31	1146.69
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1684	915.84	1141.14
1685	916.15	1144.26
1686	916.45	1149.12
1687	916.76	1155.36
1688	917.06	1149.81
1689	917.37	1153.62
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1691	917.98	1141.83
1692	918.29	1140.79
1693	918.60	1145.65
1694	918.90	1137.67
1695	919.21	1143.92
1696	919.51	1134.21
1697	919.82	1136.63
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1699	920.43	1141.83
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1702	921.35	1147.04
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1706	922.57	1176.86
1707	922.88	1168.88
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1709	923.49	1159.52
1710	923.79	1138.02
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1712	924.40	1127.96
1713	924.71	1119.99
1714	925.01	1121.03

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1720  926.84  1126.92
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1722  927.45  1134.55
1723  927.76  1130.05
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1726  928.67  1135.94
1727  928.97  1131.43
1728  929.28  1125.88
1729  929.58  1121.38
1730  929.89  1132.13
1731  930.19  1129.70
1732  930.50  1119.64
1733  930.80  1115.48
1734  931.10  1115.83
1735  931.41  1121.38
1736  931.71  1125.19
1737  932.02  1119.99
1738  932.32  1119.30
1739  932.62  1108.55
1740  932.93  1111.67
1741  933.23  1111.32
1742  933.53  1110.97
1743  933.84  1108.55
1744  934.14  1107.85
1745  934.44  1110.97
1746  934.75  1129.35
1747  935.05  1144.96
1748  935.36  1146.00
1749  935.66  1141.83
1750  935.96  1134.55
1751  936.27  1130.05
1752  936.57  1121.38
1753  936.87  1117.91
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1755  937.48  1120.34
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1759  938.69  1153.62
1760  938.99  1143.22
1761  939.29  1143.22
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1766	940.81	1150.85
1767	941.11	1161.60
1768	941.41	1147.04
1769	941.71	1141.49
1770	942.02	1144.61
1771	942.32	1129.01
1772	942.62	1133.17
1773	942.92	1123.80
1774	943.23	1125.19
1775	943.53	1131.78
1776	943.83	1131.09
1777	944.13	1133.86
1778	944.43	1150.16
1779	944.74	1143.57
1780	945.04	1151.54
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1783	945.94	1130.74
1784	946.24	1126.23
1785	946.55	1116.52
1786	946.85	1106.81
1787	947.15	1107.51
1788	947.45	1116.52
1789	947.75	1119.30
1790	948.05	1112.71
1791	948.35	1115.14
1792	948.66	1115.14
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1795	949.56	1116.18
1796	949.86	1116.87
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1799	950.76	1135.59
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1801	951.36	1136.98
1802	951.66	1129.70
1803	951.97	1128.31
1804	952.27	1120.68
1805	952.57	1111.67
1806	952.87	1113.05
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1808	953.47	1117.56

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1814  955.27  1122.07
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1816  955.87  1120.68
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1823  957.97  1162.64
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1828  959.47  1141.14
1829  959.76  1141.14
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1832  960.66  1138.02
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1842  963.65  1120.34
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1844  964.25  1135.94
1845  964.55  1143.22
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1847  965.14  1162.64
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1849  965.74  1168.19
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1855  967.53  1134.55
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1859  968.72  1135.94
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1876  973.78  1123.80
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1878  974.37  1107.51
1879  974.67  1111.67
1880  974.97  1110.63
1881  975.26  1110.63
1882  975.56  1121.72
1883  975.86  1122.07
1884  976.15  1132.47
1885  976.45  1139.06
1886  976.75  1133.51
1887  977.04  1134.55
1888  977.34  1135.59
1889  977.63  1125.88
1890  977.93  1127.96
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1892  978.52  1123.80
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1894  979.12  1123.80
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1898  980.30  1121.38
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1900  980.89  1130.74
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1902  981.48  1119.99
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1907  982.96  1123.46
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1912  984.44  1110.28
1913  984.73  1108.20
1914  985.03  1115.48
1915  985.32  1125.88
1916  985.62  1129.35
1917  985.91  1144.61
1918  986.21  1155.01
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1923  987.68  1131.43
1924  987.98  1128.66
1925  988.27  1133.86
1926  988.56  1123.80
1927  988.86  1124.15
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1939  992.39  1135.59
1940  992.68  1126.92
1941  992.98  1123.80
1942  993.27  1125.54
1943  993.56  1119.30
1944  993.86  1115.48
1945  994.15  1105.77
1946  994.44  1115.83
1947  994.74  1120.68
1948  995.03  1126.23
1949  995.32  1127.27
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1954  996.79  1127.96
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1956  997.38  1122.42
1957  997.67  1129.01
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1962  999.13  1147.73
1963  999.42  1148.08
1964  999.72  1143.92
1965  1000.01 1138.02
1966  1000.30 1133.17
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1968  1000.89 1118.95
1969  1001.18 1111.67
1970  1001.47 1117.22
1971  1001.76 1109.24
1972  1002.06 1117.91
1973  1002.35 1113.75
1974  1002.64 1115.83
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1978  1003.81 1136.98
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1981  1004.68 1140.10
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1983  1005.27 1139.06
1984  1005.56 1136.29
1985  1005.85 1121.72
1986  1006.14 1118.26
1987  1006.43 1123.80
1988  1006.72 1131.09
1989  1007.01 1139.06
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2012 1013.70 1121.38
2013 1013.99 1117.22
2014 1014.28 1125.54
2015 1014.57 1131.09
2016 1014.86 1150.16
2017 1015.15 1147.04
2018 1015.44 1150.85
2019 1015.73 1155.70
2020 1016.02 1163.33
2021 1016.31 1167.15
2022 1016.60 1159.87
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2041 1022.09 1134.90
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2043 1022.67 1116.87

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1025.84 1132.82
1026.13 1140.45
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1026.70 1179.63
1026.99 1179.63
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Listing 4: 氢氘灯

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暗光谱: 否
参考光谱: 否
光谱数量: 1
光谱仪: USB2+H14105
积分时间（毫秒）: 23 (USB2+H14105)
平均次数: 1 (USB2+H14105)
平滑度: 3 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)
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339.73  796.13
340.11  796.13
340.49  781.91
340.87  769.43
341.25  769.78
341.63  767.70
342.01  768.39
342.39  770.12
342.77  771.51
343.15  775.67
343.53  798.90
343.91  804.45
344.29  819.36
344.66  840.86
345.04  851.26
345.42  865.82
345.80  889.75
346.18  904.31
346.56  918.53
346.94  932.40
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32  348.08  1016.66
33  348.45  1033.65
34  348.83  1056.19
35  349.21  1084.97
36  349.59  1096.06
37  349.97  1112.71
38  350.35  1121.38
39  350.73  1126.58
40  351.10  1133.51
41  351.48  1147.04
42  351.86  1159.17
43  352.24  1178.24
44  352.62  1181.71
45  353.00  1187.95
46  353.37  1191.42
47  353.75  1205.64
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1411  830.99  1148.42
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1413  831.63  1141.14
1414  831.95  1141.49
1415  832.27  1140.10
1416  832.59  1136.63
1417  832.91  1140.10
1418  833.23  1155.36
1419  833.55  1168.53
1420  833.87  1175.12
1421  834.19  1168.19
1422  834.51  1170.96
1423  834.82  1168.53
1424  835.14  1167.49
1425  835.46  1159.52
1426  835.78  1155.70
1427  836.10  1149.12
1428  836.42  1145.30
1429  836.74  1141.49
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1431  837.38  1152.24
1432  837.69  1148.08
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1434  838.33  1137.33
1435  838.65  1139.75
1436  838.97  1140.79
1437  839.29  1132.13
1438  839.60  1127.96
1439  839.92  1126.58
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1442	840.88	1133.51
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1444	841.51	1119.99
1445	841.83	1127.96
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1447	842.47	1143.22
1448	842.79	1141.83
1449	843.10	1143.92
1450	843.42	1149.12
1451	843.74	1161.60
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1453	844.37	1160.91
1454	844.69	1152.24
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1458	845.96	1159.87
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1462	847.23	1157.09
1463	847.55	1166.45
1464	847.86	1149.81
1465	848.18	1133.17
1466	848.50	1131.09
1467	848.82	1141.49
1468	849.13	1139.06
1469	849.45	1132.47
1470	849.77	1125.19
1471	850.08	1136.98
1472	850.40	1144.96
1473	850.72	1138.37
1474	851.03	1127.62
1475	851.35	1118.60
1476	851.67	1127.27
1477	851.98	1135.25
1478	852.30	1130.39
1479	852.61	1133.86
1480	852.93	1142.53
1481	853.25	1151.54
1482	853.56	1168.19
1483	853.88	1173.04
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1503  860.19  1156.05
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1505  860.82  1148.08
1506  861.14  1148.08
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1508  861.77  1144.96
1509  862.08  1143.92
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1511  862.71  1150.85
1512  863.03  1150.50
1513  863.34  1158.13
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1518  864.92  1147.38
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1533  869.63  1165.41
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872.14  1144.61
872.45  1141.14
872.76  1141.14
873.08  1144.61
873.39  1144.26
873.70  1141.83
874.02  1144.96
874.33  1142.18
874.64  1145.30
874.95  1141.83
875.27  1139.06
875.58  1136.98
875.89  1141.49
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876.52  1125.19
876.83  1122.07
877.14  1114.79
877.46  1119.30
877.77  1121.38
878.08  1123.80
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878.71  1140.10
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879.33  1152.58
879.64  1154.32
879.95  1146.00
880.27  1148.08
880.58  1143.57
880.89  1147.73
881.20  1147.38
881.52  1135.25
881.83  1133.17
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882.45  1142.53
882.76  1138.71
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883.70  1164.72
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884.32  1153.28
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1591  887.74  1167.15
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1601  890.85  1176.51
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1603  891.47  1171.31
1604  891.78  1171.31
1605  892.09  1173.74
1606  892.40  1156.05
1607  892.71  1153.62
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1609  893.33  1145.30
1610  893.64  1140.10
1611  893.95  1127.96
1612  894.26  1129.70
1613  894.57  1138.71
1614  894.88  1144.61
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1627  898.91  1131.09
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1674  913.39  1142.53
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1685  916.76  1151.89
1686  917.06  1146.00
1687  917.37  1145.65
1688  917.68  1146.00
1689  917.98  1145.30
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1721  927.76  1131.78
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1731	930.80	1117.22
1732	931.10	1124.50
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1736	932.32	1156.40
1737	932.62	1161.25
1738	932.93	1165.07
1739	933.23	1156.05
1740	933.53	1154.66
1741	933.84	1158.13
1742	934.14	1147.38
1743	934.44	1146.34
1744	934.75	1138.02
1745	935.05	1138.71
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1749	936.27	1140.45
1750	936.57	1135.59
1751	936.87	1136.29
1752	937.17	1131.78
1753	937.48	1126.58
1754	937.78	1118.60
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1758	938.99	1127.62
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1764	940.81	1154.66
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1766	941.41	1148.77
1767	941.71	1149.81
1768	942.02	1140.79

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1853  967.53  1132.82
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1856  968.42  1154.66
1857  968.72  1141.14
1858  969.02  1149.81
1859  969.32  1148.42
1860  969.62  1156.40
1861  969.91  1152.93
1862  970.21  1155.01
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973.48 1142.18
973.78 1145.65
974.08 1146.00
974.37 1142.53
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974.97 1138.37
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982.67 1132.82
982.96 1134.90
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983.55 1135.94
983.85 1137.67
984.14 1143.57

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1914  985.62  1153.28
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1917  986.50  1154.66
1918  986.80  1154.66
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1920  987.39  1140.45
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1923  988.27  1130.05
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1928  989.74  1132.13
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1930  990.33  1122.07
1931  990.62  1115.48
1932  990.92  1126.92
1933  991.21  1124.50
1934  991.51  1126.23
1935  991.80  1141.83
1936  992.09  1142.18
1937  992.39  1141.14
1938  992.68  1145.65
1939  992.98  1145.65
1940  993.27  1149.46
1941  993.56  1155.70
1942  993.86  1145.30
1943  994.15  1151.54
1944  994.44  1147.73
1945  994.74  1138.71
1946  995.03  1138.37
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1948  995.62  1121.72
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1951  996.50  1127.27
1952  996.79  1133.86
1953  997.08  1128.31
1954  997.38  1126.58
1955  997.67  1133.86
1956  997.96  1137.67
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1007.60 1136.29
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1011.38 1120.68
1011.67 1123.46
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1013.41 1141.14
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1013.99 1133.17
1014.28 1141.83
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1015.73 1135.25
1016.02 1144.96
1016.31 1124.84
1016.60 1131.78
1016.89 1120.34
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1026.70 1075.26
1026.99 1075.26
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Listing 5: 溴钨灯

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暗光谱: 否
参考光谱: 否
光谱数量: 1
光谱仪: USB2+H14105
积分时间（毫秒）: 2 (USB2+H14105)
平均次数: 1 (USB2+H14105)
平滑度: 10 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)
339.35  839.82
339.73  839.82
340.11  839.82
340.49  742.73
340.87  745.16
341.25  703.89
341.63  808.26
342.01  701.47
342.39  779.14
342.77  800.98
343.15  813.00
343.53  818.09
343.91  827.45
344.29  839.13
344.66  859.47
345.04  880.16
345.42  899.81
345.80  921.77
346.18  945.35
346.56  967.07
346.94  991.69
347.32  1016.08
347.70  1036.66
348.08  1057.46
348.45  1083.93
348.83  1106.81
349.21  1129.24
349.59  1154.20
349.97  1176.05
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38  350.35  1192.92
39  350.73  1215.34
40  351.10  1237.31
41  351.48  1253.60
42  351.86  1272.90
43  352.24  1285.85
44  352.62  1299.26
45  353.00  1308.74
46  353.37  1323.53
47  353.75  1329.66
48  354.13  1338.09
49  354.51  1343.18
50  354.89  1346.07
51  355.27  1350.69
52  355.64  1356.93
53  356.02  1360.75
54  356.40  1366.64
55  356.78  1370.80
56  357.16  1374.96
57  357.53  1371.50
58  357.91  1376.35
59  358.29  1381.44
60  358.67  1382.36
61  359.04  1385.25
62  359.42  1386.52
63  359.80  1385.83
64  360.18  1388.26
65  360.55  1383.86
66  360.93  1388.26
67  361.31  1387.91
68  361.69  1387.91
69  362.06  1392.07
70  362.44  1399.12
71  362.82  1407.67
72  363.20  1413.57
73  363.57  1421.89
74  363.95  1430.33
75  364.33  1436.92
76  364.71  1444.31
77  365.08  1451.48
78  365.46  1459.34
79  365.84  1461.65
80  366.21  1467.78
81  366.59  1471.13
82  366.97  1473.67
83  367.34  1485.00
84  367.72  1490.55
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 87  368.85  1514.36
 88  369.23  1523.95
 89  369.60  1534.93
 90  369.98  1542.21
 91  370.36  1550.42
 92  370.73  1553.42
 93  371.11  1560.36
 94  371.49  1564.40
 95  371.86  1569.95
 96  372.24  1572.38
 97  372.62  1578.16
 98  372.99  1583.36
 99  373.37  1593.99
100  373.74  1601.39
101  374.12  1610.52
102  374.50  1623.58
103  374.87  1634.91
104  375.25  1640.69
105  375.62  1648.08
106  376.00  1652.94
107  376.38  1664.73
108  376.75  1674.90
109  377.13  1680.33
110  377.50  1690.27
111  377.88  1698.82
112  378.26  1706.68
113  378.63  1718.13
114  379.01  1727.26
115  379.38  1736.50
116  379.76  1743.32
117  380.13  1758.46
118  380.51  1771.06
119  380.88  1784.36
120  381.26  1793.95
121  381.64  1803.19
122  382.01  1815.79
123  382.39  1824.69
124  382.76  1839.60
125  383.14  1854.86
126  383.51  1868.96
127  383.89  1883.52
128  384.26  1895.43
129  384.64  1906.29
130  385.01  1923.17
131  385.39  1936.35
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134  386.51  1972.29
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136  387.26  1996.56
137  387.64  2008.58
138  388.01  2020.26
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140  388.76  2042.22
141  389.13  2058.52
142  389.51  2076.66
143  389.88  2090.65
144  390.26  2108.91
145  390.63  2126.36
146  391.01  2140.81
147  391.38  2154.56
148  391.76  2172.13
149  392.13  2193.63
150  392.50  2209.23
151  392.88  2220.68
152  393.25  2234.78
153  393.63  2249.11
154  394.00  2262.98
155  394.37  2280.09
156  394.75  2296.38
157  395.12  2320.08
158  395.50  2341.58
159  395.87  2358.91
160  396.24  2384.46
161  396.62  2404.11
162  396.99  2419.94
163  397.37  2436.01
164  397.74  2453.11
165  398.11  2472.30
166  398.49  2487.90
167  398.86  2509.05
168  399.23  2528.82
169  399.61  2545.00
170  399.98  2558.29
171  400.35  2577.02
172  400.73  2596.09
173  401.10  2618.51
174  401.47  2644.05
175  401.85  2667.98
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177  402.59  2715.02
178  402.97  2740.57
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186  405.95  2921.22
187  406.32  2941.56
188  406.70  2958.90
189  407.07  2983.40
190  407.44  3010.91
191  407.81  3033.10
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193  408.56  3092.74
194  408.93  3120.71
195  409.30  3140.36
196  409.68  3162.79
197  410.05  3186.13
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200  411.17  3262.07
201  411.54  3290.04
202  411.91  3309.00
203  412.28  3331.19
204  412.65  3357.43
205  413.03  3378.58
206  413.40  3404.58
207  413.77  3429.09
208  414.14  3450.93
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211  415.26  3526.87
212  415.63  3554.38
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214  416.37  3589.63
215  416.74  3605.12
216  417.12  3632.16
217  417.49  3660.14
218  417.86  3688.22
219  418.23  3705.33
220  418.60  3732.95
221  418.97  3757.46
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224  420.09  3851.08
225  420.46  3881.36
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237  424.91  4287.05
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258  432.69  5094.51
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266  435.65  5468.65
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287  443.40  6240.74
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293  445.61  6425.20
294  445.98  6449.13
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297  447.08  6575.58
298  447.45  6621.00
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388  480.44  16584.86
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392  481.90  17372.09
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397  483.72  18365.75
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404  486.27  19875.37
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453  504.05  27873.87
454  504.42  28010.95
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1011.09 1346.88
1011.38 1341.91
1011.67 1337.05
1011.96 1333.82
1012.25 1329.54
1012.54 1327.11
1012.83 1324.11
1013.12 1318.91
1013.41 1314.75
1013.70 1310.01
1013.99 1304.92
```

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1014.28 1297.41
1014.57 1293.94
1014.86 1288.28
1015.15 1282.27
1015.44 1280.65
1015.73 1278.57
1016.02 1271.63
1016.31 1270.13
1016.60 1268.28
1016.89 1266.20
1017.18 1261.92
1017.47 1260.31
1017.76 1260.88
1018.05 1255.57
1018.34 1255.11
1018.62 1248.86
1018.91 1245.16
1019.20 1241.70
1019.49 1238.69
1019.78 1234.99
1020.07 1232.10
1020.36 1234.76
1020.65 1230.60
1020.94 1230.72
1021.23 1229.33
1021.51 1227.13
1021.80 1226.67
1022.09 1229.10
1022.38 1229.68
1022.67 1229.79
1022.96 1227.02
1023.25 1228.98
1023.53 1225.86
1023.82 1223.32
1024.11 1223.67
1024.40 1228.17
1024.69 1271.86
1024.97 1240.31
1025.26 1233.03
1025.55 1257.30
1025.84 1196.62
1026.13 1235.46
1026.41 1201.48
1026.70 1201.48
1026.99 1201.48
```

Listing 6: 红灯

```
暗光谱: 否
参考光谱: 否
光谱数量: 1
光谱仪: USB2+H14105
积分时间（毫秒）: 1 (USB2+H14105)
平均次数: 1 (USB2+H14105)
平滑度: 20 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)
339.35  762.15
339.73  762.15
340.11  762.15
340.49  735.45
340.87  750.01
341.25  689.33
341.63  771.86
342.01  779.14
342.39  701.47
342.77  691.76
343.15  791.27
343.53  757.29
343.91  808.26
344.29  844.67
344.66  827.68
345.04  956.33
345.42  837.39
345.80  854.38
346.18  883.51
346.56  951.47
346.94  935.61
347.32  945.02
347.70  953.01
348.08  963.25
348.45  973.49
348.83  984.03
349.21  997.65
349.59  1007.12
349.97  1014.34
350.35  1023.16
350.73  1033.46
351.10  1041.57
351.48  1051.58
351.86  1060.28
352.24  1067.50
352.62  1076.86
353.00  1079.23
```

```
46  353.37  1084.97
47  353.75  1091.66
48  354.13  1096.51
49  354.51  1100.72
50  354.89  1105.04
51  355.27  1109.48
52  355.64  1113.38
53  356.02  1116.64
54  356.40  1119.13
55  356.78  1120.31
56  357.16  1124.46
57  357.53  1127.00
58  357.91  1128.07
59  358.29  1129.31
60  358.67  1129.72
61  359.04  1132.15
62  359.42  1133.10
63  359.80  1133.93
64  360.18  1131.97
65  360.55  1130.20
66  360.93  1130.08
67  361.31  1128.42
68  361.69  1128.42
69  362.06  1125.40
70  362.44  1125.11
71  362.82  1124.34
72  363.20  1124.22
73  363.57  1121.85
74  363.95  1119.60
75  364.33  1119.01
76  364.71  1115.46
77  365.08  1113.80
78  365.46  1115.69
79  365.84  1118.00
80  366.21  1118.12
81  366.59  1118.83
82  366.97  1115.87
83  367.34  1116.23
84  367.72  1115.10
85  368.10  1112.67
86  368.47  1114.21
87  368.85  1115.34
88  369.23  1115.69
89  369.60  1116.17
90  369.98  1114.98
91  370.36  1116.94
92  370.73  1117.77
```

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 93  371.11  1117.53
 94  371.49  1117.53
 95  371.86  1117.65
 96  372.24  1120.37
 97  372.62  1118.48
 98  372.99  1118.42
 99  373.37  1117.94
100  373.74  1117.00
101  374.12  1116.64
102  374.50  1115.40
103  374.87  1117.11
104  375.25  1114.15
105  375.62  1114.86
106  376.00  1117.71
107  376.38  1118.30
108  376.75  1119.72
109  377.13  1118.12
110  377.50  1120.43
111  377.88  1119.72
112  378.26  1118.65
113  378.63  1120.25
114  379.01  1121.38
115  379.38  1121.26
116  379.76  1119.90
117  380.13  1121.50
118  380.51  1121.73
119  380.88  1119.30
120  381.26  1117.41
121  381.64  1116.29
122  382.01  1117.47
123  382.39  1120.78
124  382.76  1120.49
125  383.14  1120.73
126  383.51  1122.03
127  383.89  1121.38
128  384.26  1121.26
129  384.64  1121.20
130  385.01  1121.20
131  385.39  1121.55
132  385.76  1119.36
133  386.14  1117.29
134  386.51  1114.98
135  386.89  1114.45
136  387.26  1115.99
137  387.64  1114.33
138  388.01  1115.16
139  388.39  1116.82
```

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388.76  1118.65
389.13  1120.37
389.51  1121.97
389.88  1120.55
390.26  1118.95
390.63  1119.78
391.01  1119.72
391.38  1118.77
391.76  1116.76
392.13  1116.58
392.50  1116.88
392.88  1117.41
393.25  1117.82
393.63  1119.36
394.00  1119.42
394.37  1118.36
394.75  1119.60
395.12  1119.19
395.50  1119.48
395.87  1120.07
396.24  1121.85
396.62  1122.68
396.99  1123.51
397.37  1121.61
397.74  1121.26
398.11  1120.43
398.49  1120.02
398.86  1117.11
399.23  1119.07
399.61  1118.65
399.98  1118.42
400.35  1118.36
400.73  1119.84
401.10  1122.56
401.47  1125.34
401.85  1128.18
402.22  1129.25
402.59  1129.55
402.97  1130.49
403.34  1131.86
403.71  1129.55
404.09  1128.24
404.46  1129.37
404.83  1128.24
405.20  1129.72
405.58  1130.79
405.95  1133.63
```

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187  406.32  1135.35
188  406.70  1134.40
189  407.07  1135.88
190  407.44  1135.29
191  407.81  1136.53
192  408.19  1134.46
193  408.56  1136.47
194  408.93  1136.71
195  409.30  1136.12
196  409.68  1136.89
197  410.05  1137.95
198  410.42  1139.31
199  410.79  1138.37
200  411.17  1138.37
201  411.54  1137.42
202  411.91  1137.36
203  412.28  1136.53
204  412.65  1136.59
205  413.03  1135.11
206  413.40  1135.53
207  413.77  1137.66
208  414.14  1140.44
209  414.51  1137.54
210  414.89  1138.54
211  415.26  1138.37
212  415.63  1138.96
213  416.00  1138.19
214  416.37  1136.00
215  416.74  1132.03
216  417.12  1129.55
217  417.49  1129.84
218  417.86  1128.48
219  418.23  1126.94
220  418.60  1126.35
221  418.97  1129.37
222  419.34  1130.61
223  419.72  1129.55
224  420.09  1128.90
225  420.46  1129.31
226  420.83  1129.61
227  421.20  1128.78
228  421.57  1128.72
229  421.94  1130.61
230  422.31  1131.20
231  422.69  1132.80
232  423.06  1133.99
233  423.43  1133.99
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234  423.80  1131.86
235  424.17  1131.68
236  424.54  1132.86
237  424.91  1134.22
238  425.28  1132.74
239  425.65  1132.74
240  426.02  1133.04
241  426.39  1133.93
242  426.76  1135.53
243  427.13  1137.66
244  427.51  1139.85
245  427.88  1139.97
246  428.25  1140.79
247  428.62  1140.74
248  428.99  1138.60
249  429.36  1137.42
250  429.73  1140.26
251  430.10  1141.33
252  430.47  1140.56
253  430.84  1139.67
254  431.21  1139.31
255  431.58  1140.32
256  431.95  1144.35
257  432.32  1145.77
258  432.69  1145.41
259  433.06  1145.71
260  433.43  1146.06
261  433.80  1144.11
262  434.17  1142.51
263  434.54  1141.68
264  434.91  1141.33
265  435.28  1142.69
266  435.65  1142.87
267  436.02  1143.70
268  436.39  1143.34
269  436.75  1141.74
270  437.12  1140.08
271  437.49  1140.20
272  437.86  1140.62
273  438.23  1140.02
274  438.60  1140.08
275  438.97  1142.75
276  439.34  1143.81
277  439.71  1141.21
278  440.08  1139.91
279  440.45  1141.45
280  440.82  1140.68
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281  441.19  1138.96
282  441.55  1138.84
283  441.92  1137.06
284  442.29  1134.76
285  442.66  1135.23
286  443.03  1134.87
287  443.40  1135.35
288  443.77  1134.52
289  444.14  1135.82
290  444.51  1136.83
291  444.87  1135.29
292  445.24  1133.57
293  445.61  1134.76
294  445.98  1136.53
295  446.35  1137.48
296  446.72  1139.20
297  447.08  1137.72
298  447.45  1140.14
299  447.82  1141.39
300  448.19  1140.38
301  448.56  1139.31
302  448.93  1140.79
303  449.29  1141.15
304  449.66  1142.27
305  450.03  1142.45
306  450.40  1142.27
307  450.77  1141.03
308  451.13  1140.08
309  451.50  1139.31
310  451.87  1139.49
311  452.24  1138.78
312  452.61  1137.78
313  452.97  1135.64
314  453.34  1137.30
315  453.71  1136.35
316  454.08  1134.58
317  454.44  1133.93
318  454.81  1136.65
319  455.18  1136.59
320  455.55  1137.12
321  455.91  1137.60
322  456.28  1140.14
323  456.65  1140.62
324  457.02  1141.92
325  457.38  1141.21
326  457.75  1140.32
327  458.12  1140.74
```

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328  458.48  1139.67
329  458.85  1139.26
330  459.22  1136.71
331  459.59  1135.70
332  459.95  1134.52
333  460.32  1135.88
334  460.69  1135.47
335  461.05  1134.58
336  461.42  1135.76
337  461.79  1133.87
338  462.15  1135.05
339  462.52  1134.52
340  462.89  1133.57
341  463.25  1134.70
342  463.62  1137.83
343  463.99  1138.43
344  464.35  1136.59
345  464.72  1135.76
346  465.09  1135.17
347  465.45  1134.99
348  465.82  1136.89
349  466.18  1136.59
350  466.55  1137.01
351  466.92  1138.19
352  467.28  1138.90
353  467.65  1138.43
354  468.02  1139.43
355  468.38  1137.30
356  468.75  1137.36
357  469.11  1139.08
358  469.48  1138.31
359  469.85  1137.60
360  470.21  1137.01
361  470.58  1134.34
362  470.94  1136.47
363  471.31  1135.05
364  471.67  1133.69
365  472.04  1133.57
366  472.41  1133.28
367  472.77  1133.34
368  473.14  1132.09
369  473.50  1131.97
370  473.87  1133.87
371  474.23  1136.95
372  474.60  1137.95
373  474.96  1139.14
374  475.33  1137.60
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375  475.69  1138.43
376  476.06  1139.61
377  476.42  1140.14
378  476.79  1138.96
379  477.15  1138.37
380  477.52  1138.72
381  477.88  1138.01
382  478.25  1138.66
383  478.61  1137.42
384  478.98  1134.99
385  479.34  1136.12
386  479.71  1135.17
387  480.07  1134.16
388  480.44  1135.82
389  480.80  1134.70
390  481.17  1133.10
391  481.53  1133.10
392  481.90  1133.45
393  482.26  1131.62
394  482.62  1133.22
395  482.99  1131.86
396  483.35  1132.27
397  483.72  1135.29
398  484.08  1133.10
399  484.45  1134.52
400  484.81  1133.10
401  485.17  1134.22
402  485.54  1136.77
403  485.90  1135.88
404  486.27  1135.88
405  486.63  1136.95
406  487.00  1138.49
407  487.36  1140.26
408  487.72  1138.84
409  488.09  1139.91
410  488.45  1138.25
411  488.81  1138.84
412  489.18  1137.54
413  489.54  1135.76
414  489.91  1136.24
415  490.27  1137.30
416  490.63  1139.26
417  491.00  1138.78
418  491.36  1137.78
419  491.72  1139.85
420  492.09  1138.66
421  492.45  1135.64
```

```
422  492.81  1137.06
423  493.18  1136.59
424  493.54  1137.36
425  493.90  1139.02
426  494.27  1139.55
427  494.63  1141.45
428  494.99  1143.99
429  495.35  1142.04
430  495.72  1143.22
431  496.08  1144.17
432  496.44  1145.06
433  496.81  1143.93
434  497.17  1146.77
435  497.53  1147.13
436  497.89  1148.90
437  498.26  1149.02
438  498.62  1147.60
439  498.98  1147.54
440  499.35  1148.02
441  499.71  1150.33
442  500.07  1150.92
443  500.43  1150.74
444  500.79  1149.79
445  501.16  1149.38
446  501.52  1149.56
447  501.88  1147.96
448  502.24  1149.38
449  502.61  1151.63
450  502.97  1153.34
451  503.33  1156.78
452  503.69  1155.36
453  504.05  1155.95
454  504.42  1155.71
455  504.78  1156.19
456  505.14  1157.37
457  505.50  1155.12
458  505.86  1154.59
459  506.23  1155.59
460  506.59  1156.13
461  506.95  1157.73
462  507.31  1160.39
463  507.67  1159.03
464  508.03  1158.55
465  508.40  1156.54
466  508.76  1156.36
467  509.12  1155.77
468  509.48  1154.71
```

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469  509.84  1154.17
470  510.20  1154.06
471  510.56  1152.28
472  510.93  1153.64
473  511.29  1152.28
474  511.65  1153.17
475  512.01  1152.93
476  512.37  1152.16
477  512.73  1150.74
478  513.09  1152.75
479  513.45  1153.70
480  513.81  1155.89
481  514.17  1153.58
482  514.54  1151.98
483  514.90  1152.10
484  515.26  1150.09
485  515.62  1151.15
486  515.98  1150.92
487  516.34  1151.04
488  516.70  1152.75
489  517.06  1150.74
490  517.42  1149.97
491  517.78  1149.38
492  518.14  1148.25
493  518.50  1150.03
494  518.86  1150.38
495  519.22  1153.40
496  519.58  1154.35
497  519.94  1153.29
498  520.30  1152.58
499  520.66  1153.76
500  521.02  1152.58
501  521.38  1151.98
502  521.74  1152.16
503  522.10  1152.22
504  522.46  1153.46
505  522.82  1153.23
506  523.18  1152.46
507  523.54  1151.39
508  523.90  1152.10
509  524.26  1153.82
510  524.62  1154.35
511  524.98  1153.52
512  525.34  1154.94
513  525.70  1155.95
514  526.06  1156.54
515  526.42  1156.07
```

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516  526.78  1155.83
517  527.14  1155.54
518  527.50  1155.89
519  527.86  1153.52
520  528.22  1153.29
521  528.58  1152.22
522  528.93  1152.10
523  529.29  1151.92
524  529.65  1151.04
525  530.01  1152.58
526  530.37  1151.51
527  530.73  1152.81
528  531.09  1151.21
529  531.45  1152.87
530  531.81  1154.82
531  532.16  1155.18
532  532.52  1155.18
533  532.88  1156.84
534  533.24  1156.19
535  533.60  1154.94
536  533.96  1154.23
537  534.32  1153.82
538  534.68  1155.95
539  535.03  1157.90
540  535.39  1158.55
541  535.75  1158.44
542  536.11  1157.37
543  536.47  1157.43
544  536.83  1157.90
545  537.18  1156.96
546  537.54  1157.67
547  537.90  1162.23
548  538.26  1165.60
549  538.62  1164.89
550  538.97  1164.06
551  539.33  1164.89
552  539.69  1166.13
553  540.05  1166.96
554  540.41  1165.95
555  540.76  1165.90
556  541.12  1166.61
557  541.48  1167.02
558  541.84  1167.79
559  542.19  1168.97
560  542.55  1170.81
561  542.91  1172.53
562  543.27  1172.23
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563  543.62  1173.59
564  543.98  1174.83
565  544.34  1177.79
566  544.70  1179.22
567  545.05  1180.34
568  545.41  1180.81
569  545.77  1183.00
570  546.12  1182.83
571  546.48  1183.00
572  546.84  1183.89
573  547.20  1183.83
574  547.55  1184.66
575  547.91  1185.08
576  548.27  1188.81
577  548.62  1189.69
578  548.98  1189.81
579  549.34  1189.63
580  549.69  1191.00
581  550.05  1190.82
582  550.41  1193.42
583  550.76  1196.38
584  551.12  1197.69
585  551.48  1198.28
586  551.83  1200.23
587  552.19  1201.42
588  552.55  1199.28
589  552.90  1199.40
590  553.26  1203.19
591  553.62  1205.56
592  553.97  1207.81
593  554.33  1210.65
594  554.68  1213.14
595  555.04  1214.50
596  555.40  1216.93
597  555.75  1220.12
598  556.11  1223.50
599  556.46  1227.76
600  556.82  1228.17
601  557.18  1229.42
602  557.53  1232.61
603  557.89  1233.98
604  558.24  1238.95
605  558.60  1241.26
606  558.95  1244.04
607  559.31  1246.65
608  559.67  1250.61
609  560.02  1254.22
```

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560.38  1257.12
560.73  1258.07
561.09  1260.91
561.44  1265.83
561.80  1269.61
562.15  1271.63
562.51  1275.77
562.86  1279.32
563.22  1283.59
563.57  1289.39
563.93  1292.82
564.28  1296.43
564.64  1303.42
564.99  1306.02
565.35  1313.48
565.70  1320.53
566.06  1327.34
566.41  1335.21
566.77  1343.02
567.12  1351.43
567.48  1359.48
567.83  1366.76
568.19  1373.39
568.54  1381.80
568.89  1391.51
569.25  1399.92
569.60  1409.21
569.96  1418.68
570.31  1428.45
570.67  1438.63
571.02  1447.81
571.37  1461.84
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1966  1000.89 1133.57
1967  1001.18 1134.76
1968  1001.47 1136.35
1969  1001.76 1134.82
1970  1002.06 1134.05
1971  1002.35 1135.58
1972  1002.64 1134.93
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1002.93 1136.00
1003.22 1135.64
1003.52 1133.10
1003.81 1132.62
1004.10 1130.08
1004.39 1128.66
1004.68 1128.42
1004.97 1127.30
1005.27 1126.94
1005.56 1125.52
1005.85 1126.76
1006.14 1128.90
1006.43 1127.12
1006.72 1127.18
1007.01 1127.71
1007.31 1129.31
1007.60 1127.30
1007.89 1130.32
1008.18 1129.07
1008.47 1129.19
1008.76 1131.91
1009.05 1132.86
1009.34 1132.09
1009.63 1133.93
1009.93 1134.16
1010.22 1135.94
1010.51 1134.16
1010.80 1135.70
1011.09 1136.65
1011.38 1134.64
1011.67 1134.40
1011.96 1136.35
1012.25 1135.53
1012.54 1135.35
1012.83 1134.52
1013.12 1133.39
1013.41 1133.04
1013.70 1133.51
1013.99 1135.76
1014.28 1132.92
1014.57 1132.98
1014.86 1131.50
1015.15 1132.80
1015.44 1134.10
1015.73 1135.94
1016.02 1137.95
1016.31 1136.95
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2020  1016.60 1136.77
2021  1016.89 1137.30
2022  1017.18 1138.37
2023  1017.47 1138.54
2024  1017.76 1140.50
2025  1018.05 1138.19
2026  1018.34 1138.90
2027  1018.62 1141.21
2028  1018.91 1142.22
2029  1019.20 1141.98
2030  1019.49 1145.53
2031  1019.78 1142.04
2032  1020.07 1141.21
2033  1020.36 1139.31
2034  1020.65 1139.43
2035  1020.94 1140.44
2036  1021.23 1141.68
2037  1021.51 1172.35
2038  1021.80 1116.52
2039  1022.09 1080.11
2040  1022.38 1162.64
2041  1022.67 1157.78
2042  1022.96 1157.78
2043  1023.25 1099.53
2044  1023.53 1233.03
2045  1023.82 1087.40
2046  1024.11 1116.52
2047  1024.40 1220.89
2048  1024.69 1152.93
2049  1024.97 1148.08
2050  1025.26 1186.91
2051  1025.55 1063.12
2052  1025.84 1097.10
2053  1026.13 1094.68
2054  1026.41 1199.05
2055  1026.70 1199.05
2056  1026.99 1199.05
```

Listing 7: 黄灯

```
1  暗光谱: 否
2  参考光谱: 否
3  光谱数量: 1
4  光谱仪: USB2+H14105
5  积分时间（毫秒）: 1 (USB2+H14105)
6  平均次数: 1 (USB2+H14105)
7  平滑度: 20 (USB2+H14105)
```

```
 8  去除暗噪声: 否 (USB2+H14105)
 9  339.35  813.12
10  339.73  813.12
11  340.11  813.12
12  340.49  767.00
13  340.87  759.72
14  341.25  720.88
15  341.63  740.30
16  342.01  796.13
17  342.39  725.74
18  342.77  764.58
19  343.15  771.86
20  343.53  767.00
21  343.91  735.45
22  344.29  771.86
23  344.66  883.51
24  345.04  808.26
25  345.42  859.24
26  345.80  878.65
27  346.18  859.24
28  346.56  997.59
29  346.94  936.49
30  347.32  946.08
31  347.70  953.54
32  348.08  962.72
33  348.45  970.30
34  348.83  978.47
35  349.21  989.71
36  349.59  999.19
37  349.97  1008.36
38  350.35  1018.84
39  350.73  1028.25
40  351.10  1037.25
41  351.48  1047.73
42  351.86  1057.91
43  352.24  1067.68
44  352.62  1075.08
45  353.00  1084.61
46  353.37  1090.65
47  353.75  1097.16
48  354.13  1104.09
49  354.51  1107.52
50  354.89  1113.80
51  355.27  1120.37
52  355.64  1122.56
53  356.02  1125.70
54  356.40  1130.55
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 55  356.78  1134.76
 56  357.16  1135.76
 57  357.53  1138.01
 58  357.91  1137.66
 59  358.29  1139.85
 60  358.67  1139.43
 61  359.04  1139.08
 62  359.42  1138.60
 63  359.80  1139.08
 64  360.18  1137.06
 65  360.55  1138.13
 66  360.93  1136.35
 67  361.31  1134.22
 68  361.69  1133.10
 69  362.06  1131.68
 70  362.44  1132.51
 71  362.82  1129.61
 72  363.20  1131.03
 73  363.57  1129.49
 74  363.95  1129.49
 75  364.33  1129.66
 76  364.71  1127.53
 77  365.08  1125.76
 78  365.46  1125.94
 79  365.84  1123.98
 80  366.21  1122.50
 81  366.59  1121.32
 82  366.97  1120.37
 83  367.34  1119.54
 84  367.72  1117.82
 85  368.10  1114.75
 86  368.47  1112.61
 87  368.85  1112.67
 88  369.23  1110.96
 89  369.60  1109.48
 90  369.98  1108.83
 91  370.36  1105.98
 92  370.73  1103.32
 93  371.11  1103.32
 94  371.49  1105.16
 95  371.86  1103.14
 96  372.24  1101.66
 97  372.62  1102.14
 98  372.99  1101.90
 99  373.37  1103.44
100  373.74  1103.32
101  374.12  1104.74
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102  374.50  1103.68
103  374.87  1103.91
104  375.25  1103.97
105  375.62  1107.46
106  376.00  1104.27
107  376.38  1103.97
108  376.75  1105.98
109  377.13  1108.71
110  377.50  1110.78
111  377.88  1107.52
112  378.26  1107.94
113  378.63  1106.04
114  379.01  1107.46
115  379.38  1108.83
116  379.76  1110.19
117  380.13  1111.31
118  380.51  1112.32
119  380.88  1108.83
120  381.26  1109.36
121  381.64  1109.06
122  382.01  1109.42
123  382.39  1109.36
124  382.76  1108.17
125  383.14  1107.82
126  383.51  1108.12
127  383.89  1108.35
128  384.26  1106.52
129  384.64  1107.05
130  385.01  1107.46
131  385.39  1108.77
132  385.76  1108.12
133  386.14  1109.71
134  386.51  1110.78
135  386.89  1108.65
136  387.26  1110.66
137  387.64  1112.14
138  388.01  1113.38
139  388.39  1114.27
140  388.76  1113.62
141  389.13  1114.45
142  389.51  1113.74
143  389.88  1114.75
144  390.26  1116.05
145  390.63  1115.69
146  391.01  1115.22
147  391.38  1116.34
148  391.76  1116.29
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149  392.13  1115.16
150  392.50  1112.67
151  392.88  1110.78
152  393.25  1114.69
153  393.63  1115.58
154  394.00  1114.33
155  394.37  1111.67
156  394.75  1111.19
157  395.12  1107.88
158  395.50  1106.34
159  395.87  1108.47
160  396.24  1111.43
161  396.62  1112.08
162  396.99  1111.49
163  397.37  1112.73
164  397.74  1111.67
165  398.11  1112.38
166  398.49  1112.26
167  398.86  1113.74
168  399.23  1112.20
169  399.61  1114.86
170  399.98  1114.98
171  400.35  1115.34
172  400.73  1113.44
173  401.10  1114.10
174  401.47  1114.45
175  401.85  1115.10
176  402.22  1115.16
177  402.59  1115.75
178  402.97  1113.62
179  403.34  1113.92
180  403.71  1113.56
181  404.09  1112.73
182  404.46  1114.15
183  404.83  1114.15
184  405.20  1116.46
185  405.58  1116.46
186  405.95  1116.46
187  406.32  1114.86
188  406.70  1115.63
189  407.07  1115.99
190  407.44  1117.11
191  407.81  1118.30
192  408.19  1119.42
193  408.56  1118.30
194  408.93  1119.25
195  409.30  1120.37
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196  409.68  1120.43
197  410.05  1120.49
198  410.42  1122.09
199  410.79  1122.98
200  411.17  1122.38
201  411.54  1120.31
202  411.91  1121.20
203  412.28  1122.68
204  412.65  1121.50
205  413.03  1121.26
206  413.40  1121.67
207  413.77  1122.21
208  414.14  1123.15
209  414.51  1125.11
210  414.89  1124.22
211  415.26  1122.44
212  415.63  1122.92
213  416.00  1123.03
214  416.37  1123.21
215  416.74  1123.86
216  417.12  1123.09
217  417.49  1125.34
218  417.86  1122.74
219  418.23  1123.27
220  418.60  1121.79
221  418.97  1121.44
222  419.34  1122.26
223  419.72  1120.02
224  420.09  1119.07
225  420.46  1117.94
226  420.83  1117.88
227  421.20  1118.83
228  421.57  1119.19
229  421.94  1118.71
230  422.31  1120.55
231  422.69  1120.61
232  423.06  1119.96
233  423.43  1119.48
234  423.80  1119.66
235  424.17  1116.88
236  424.54  1118.00
237  424.91  1118.77
238  425.28  1119.01
239  425.65  1120.37
240  426.02  1119.13
241  426.39  1117.11
242  426.76  1118.36
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243  427.13  1116.52
244  427.51  1116.82
245  427.88  1117.71
246  428.25  1117.06
247  428.62  1116.82
248  428.99  1117.53
249  429.36  1116.17
250  429.73  1115.40
251  430.10  1115.40
252  430.47  1118.77
253  430.84  1118.59
254  431.21  1119.01
255  431.58  1120.02
256  431.95  1121.55
257  432.32  1122.03
258  432.69  1120.13
259  433.06  1121.38
260  433.43  1121.61
261  433.80  1122.09
262  434.17  1120.43
263  434.54  1119.54
264  434.91  1118.36
265  435.28  1119.72
266  435.65  1120.02
267  436.02  1119.96
268  436.39  1118.06
269  436.75  1115.69
270  437.12  1116.88
271  437.49  1116.64
272  437.86  1117.29
273  438.23  1117.17
274  438.60  1119.36
275  438.97  1120.43
276  439.34  1122.21
277  439.71  1120.78
278  440.08  1121.55
279  440.45  1122.38
280  440.82  1122.32
281  441.19  1122.62
282  441.55  1122.15
283  441.92  1122.15
284  442.29  1123.98
285  442.66  1124.69
286  443.03  1123.33
287  443.40  1125.34
288  443.77  1125.28
289  444.14  1126.23
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290  444.51  1125.28
291  444.87  1125.88
292  445.24  1126.17
293  445.61  1126.05
294  445.98  1125.46
295  446.35  1127.18
296  446.72  1125.52
297  447.08  1123.09
298  447.45  1122.86
299  447.82  1122.44
300  448.19  1122.62
301  448.56  1122.92
302  448.93  1123.86
303  449.29  1126.29
304  449.66  1127.83
305  450.03  1129.43
306  450.40  1129.49
307  450.77  1129.55
308  451.13  1127.59
309  451.50  1129.55
310  451.87  1131.09
311  452.24  1131.03
312  452.61  1131.32
313  452.97  1129.55
314  453.34  1128.95
315  453.71  1127.30
316  454.08  1127.36
317  454.44  1126.94
318  454.81  1128.95
319  455.18  1127.71
320  455.55  1127.06
321  455.91  1126.94
322  456.28  1128.84
323  456.65  1131.09
324  457.02  1130.67
325  457.38  1129.84
326  457.75  1128.95
327  458.12  1129.72
328  458.48  1130.26
329  458.85  1130.08
330  459.22  1130.26
331  459.59  1131.20
332  459.95  1131.20
333  460.32  1130.61
334  460.69  1129.96
335  461.05  1131.80
336  461.42  1130.38
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337  461.79  1132.51
338  462.15  1132.45
339  462.52  1133.45
340  462.89  1132.57
341  463.25  1133.69
342  463.62  1133.87
343  463.99  1132.21
344  464.35  1130.61
345  464.72  1130.08
346  465.09  1131.20
347  465.45  1131.20
348  465.82  1130.38
349  466.18  1131.80
350  466.55  1133.04
351  466.92  1133.87
352  467.28  1133.63
353  467.65  1133.16
354  468.02  1134.46
355  468.38  1135.05
356  468.75  1136.00
357  469.11  1135.64
358  469.48  1135.58
359  469.85  1134.16
360  470.21  1134.52
361  470.58  1135.94
362  470.94  1136.95
363  471.31  1136.53
364  471.67  1136.83
365  472.04  1136.71
366  472.41  1136.83
367  472.77  1138.19
368  473.14  1137.36
369  473.50  1136.12
370  473.87  1135.70
371  474.23  1134.22
372  474.60  1135.64
373  474.96  1135.76
374  475.33  1136.77
375  475.69  1138.07
376  476.06  1138.13
377  476.42  1139.43
378  476.79  1140.02
379  477.15  1139.20
380  477.52  1140.56
381  477.88  1141.45
382  478.25  1140.08
383  478.61  1138.84
```

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384  478.98  1137.42
385  479.34  1136.77
386  479.71  1137.95
387  480.07  1136.77
388  480.44  1136.41
389  480.80  1136.24
390  481.17  1136.41
391  481.53  1135.82
392  481.90  1134.52
393  482.26  1134.16
394  482.62  1133.63
395  482.99  1133.87
396  483.35  1134.64
397  483.72  1132.98
398  484.08  1132.74
399  484.45  1132.33
400  484.81  1133.34
401  485.17  1135.35
402  485.54  1134.52
403  485.90  1134.34
404  486.27  1134.82
405  486.63  1135.05
406  487.00  1136.30
407  487.36  1137.12
408  487.72  1135.64
409  488.09  1135.94
410  488.45  1135.47
411  488.81  1137.78
412  489.18  1137.89
413  489.54  1136.77
414  489.91  1137.42
415  490.27  1136.77
416  490.63  1137.01
417  491.00  1135.76
418  491.36  1135.47
419  491.72  1134.82
420  492.09  1136.47
421  492.45  1134.05
422  492.81  1134.93
423  493.18  1134.87
424  493.54  1136.35
425  493.90  1139.67
426  494.27  1142.04
427  494.63  1142.10
428  494.99  1142.10
429  495.35  1142.45
430  495.72  1144.29
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431  496.08  1145.12
432  496.44  1143.81
433  496.81  1143.99
434  497.17  1143.40
435  497.53  1143.93
436  497.89  1145.59
437  498.26  1147.01
438  498.62  1148.14
439  498.98  1148.19
440  499.35  1148.55
441  499.71  1147.78
442  500.07  1147.66
443  500.43  1146.54
444  500.79  1144.64
445  501.16  1143.70
446  501.52  1144.88
447  501.88  1145.06
448  502.24  1144.88
449  502.61  1146.36
450  502.97  1146.71
451  503.33  1147.84
452  503.69  1147.48
453  504.05  1147.66
454  504.42  1147.78
455  504.78  1148.31
456  505.14  1150.92
457  505.50  1148.67
458  505.86  1149.14
459  506.23  1148.85
460  506.59  1149.02
461  506.95  1148.25
462  507.31  1147.31
463  507.67  1147.37
464  508.03  1147.96
465  508.40  1147.01
466  508.76  1146.83
467  509.12  1146.42
468  509.48  1146.12
469  509.84  1146.60
470  510.20  1147.13
471  510.56  1146.83
472  510.93  1147.13
473  511.29  1147.13
474  511.65  1148.31
475  512.01  1149.73
476  512.37  1150.21
477  512.73  1149.26
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478  513.09  1148.31
479  513.45  1147.25
480  513.81  1146.12
481  514.17  1146.77
482  514.54  1147.07
483  514.90  1147.66
484  515.26  1149.50
485  515.62  1151.15
486  515.98  1150.38
487  516.34  1149.50
488  516.70  1150.03
489  517.06  1151.63
490  517.42  1151.92
491  517.78  1154.94
492  518.14  1155.12
493  518.50  1154.59
494  518.86  1155.71
495  519.22  1157.13
496  519.58  1156.13
497  519.94  1154.88
498  520.30  1156.19
499  520.66  1155.36
500  521.02  1155.95
501  521.38  1156.07
502  521.74  1155.12
503  522.10  1157.49
504  522.46  1157.43
505  522.82  1158.73
506  523.18  1159.56
507  523.54  1159.15
508  523.90  1159.98
509  524.26  1159.38
510  524.62  1160.15
511  524.98  1161.22
512  525.34  1160.39
513  525.70  1160.57
514  526.06  1161.99
515  526.42  1163.41
516  526.78  1164.00
517  527.14  1165.19
518  527.50  1165.66
519  527.86  1166.43
520  528.22  1167.49
521  528.58  1171.34
522  528.93  1172.53
523  529.29  1174.54
524  529.65  1174.36
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525  530.01  1176.26
526  530.37  1178.33
527  530.73  1182.06
528  531.09  1183.36
529  531.45  1184.66
530  531.81  1184.66
531  532.16  1184.54
532  532.52  1184.31
533  532.88  1185.61
534  533.24  1190.17
535  533.60  1192.48
536  533.96  1194.49
537  534.32  1195.32
538  534.68  1197.03
539  535.03  1201.06
540  535.39  1204.08
541  535.75  1206.98
542  536.11  1209.05
543  536.47  1213.37
544  536.83  1215.51
545  537.18  1218.41
546  537.54  1220.12
547  537.90  1223.79
548  538.26  1230.07
549  538.62  1234.33
550  538.97  1238.83
551  539.33  1243.86
552  539.69  1249.49
553  540.05  1254.22
554  540.41  1258.90
555  540.76  1264.23
556  541.12  1268.85
557  541.48  1274.29
558  541.84  1279.38
559  542.19  1286.19
560  542.55  1293.65
561  542.91  1301.05
562  543.27  1306.97
563  543.62  1313.42
564  543.98  1320.29
565  544.34  1326.74
566  544.70  1333.37
567  545.05  1339.77
568  545.41  1347.88
569  545.77  1356.05
570  546.12  1364.57
571  546.48  1373.33
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572  546.84  1383.46
573  547.20  1392.57
574  547.55  1401.10
575  547.91  1410.28
576  548.27  1419.45
577  548.62  1427.98
578  548.98  1439.99
579  549.34  1451.72
580  549.69  1463.56
581  550.05  1478.18
582  550.41  1490.67
583  550.76  1505.65
584  551.12  1519.50
585  551.48  1534.42
586  551.83  1550.64
587  552.19  1568.28
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971.70  1135.05
972.00  1134.93
972.29  1132.86
972.59  1134.05
972.89  1133.69
973.18  1133.10
973.48  1134.22
973.78  1135.47
974.08  1135.29
974.37  1135.35
974.67  1135.53
974.97  1137.30
975.26  1138.49
975.56  1137.01
975.86  1137.36
976.15  1137.36
976.45  1137.66
976.75  1135.94
977.04  1136.24
977.34  1136.47
977.63  1135.29
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1889  978.23  1133.45
1890  978.52  1133.75
1891  978.82  1132.74
1892  979.12  1132.27
1893  979.41  1133.28
1894  979.71  1134.93
1895  980.00  1133.34
1896  980.30  1133.69
1897  980.60  1134.46
1898  980.89  1134.28
1899  981.19  1133.93
1900  981.48  1134.58
1901  981.78  1136.35
1902  982.07  1134.93
1903  982.37  1136.06
1904  982.67  1137.24
1905  982.96  1135.64
1906  983.26  1135.23
1907  983.55  1135.76
1908  983.85  1137.48
1909  984.14  1137.12
1910  984.44  1138.96
1911  984.73  1138.19
1912  985.03  1136.35
1913  985.32  1135.94
1914  985.62  1134.58
1915  985.91  1133.28
1916  986.21  1134.87
1917  986.50  1135.88
1918  986.80  1135.23
1919  987.09  1131.74
1920  987.39  1129.31
1921  987.68  1132.15
1922  987.98  1132.33
1923  988.27  1131.97
1924  988.56  1132.33
1925  988.86  1133.34
1926  989.15  1132.74
1927  989.45  1132.51
1928  989.74  1131.38
1929  990.04  1132.92
1930  990.33  1132.33
1931  990.62  1132.62
1932  990.92  1133.93
1933  991.21  1132.74
1934  991.51  1129.72
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1937  992.39  1131.32
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1939  992.98  1130.14
1940  993.27  1130.20
1941  993.56  1132.74
1942  993.86  1133.16
1943  994.15  1133.87
1944  994.44  1134.05
1945  994.74  1132.39
1946  995.03  1132.39
1947  995.32  1132.74
1948  995.62  1133.10
1949  995.91  1131.38
1950  996.20  1129.25
1951  996.50  1127.30
1952  996.79  1128.90
1953  997.08  1132.03
1954  997.38  1133.34
1955  997.67  1133.69
1956  997.96  1134.52
1957  998.25  1132.33
1958  998.55  1130.61
1959  998.84  1131.20
1960  999.13  1133.39
1961  999.42  1134.22
1962  999.72  1133.51
1963  1000.01 1134.10
1964  1000.30 1133.69
1965  1000.59 1134.76
1966  1000.89 1136.00
1967  1001.18 1134.82
1968  1001.47 1134.82
1969  1001.76 1135.05
1970  1002.06 1137.36
1971  1002.35 1136.83
1972  1002.64 1137.24
1973  1002.93 1134.70
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1975  1003.52 1135.70
1976  1003.81 1133.99
1977  1004.10 1133.81
1978  1004.39 1134.22
1979  1004.68 1134.82
1980  1004.97 1135.53
1981  1005.27 1134.93
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1983  1005.85 1130.43
1984  1006.14 1130.79
1985  1006.43 1129.78
1986  1006.72 1130.02
1987  1007.01 1131.62
1988  1007.31 1132.68
1989  1007.60 1130.91
1990  1007.89 1130.08
1991  1008.18 1131.14
1992  1008.47 1134.46
1993  1008.76 1133.87
1994  1009.05 1130.55
1995  1009.34 1131.09
1996  1009.63 1131.62
1997  1009.93 1131.68
1998  1010.22 1132.62
1999  1010.51 1132.80
2000  1010.80 1133.04
2001  1011.09 1131.74
2002  1011.38 1131.14
2003  1011.67 1130.97
2004  1011.96 1130.43
2005  1012.25 1132.33
2006  1012.54 1131.32
2007  1012.83 1129.61
2008  1013.12 1129.55
2009  1013.41 1130.97
2010  1013.70 1131.56
2011  1013.99 1130.73
2012  1014.28 1130.55
2013  1014.57 1131.03
2014  1014.86 1133.34
2015  1015.15 1135.05
2016  1015.44 1135.53
2017  1015.73 1136.89
2018  1016.02 1135.76
2019  1016.31 1133.93
2020  1016.60 1132.45
2021  1016.89 1131.91
2022  1017.18 1132.62
2023  1017.47 1132.98
2024  1017.76 1133.63
2025  1018.05 1131.91
2026  1018.34 1132.57
2027  1018.62 1131.56
2028  1018.91 1130.32
```

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2029  1019.20 1129.37
2030  1019.49 1131.14
2031  1019.78 1132.74
2032  1020.07 1133.63
2033  1020.36 1132.86
2034  1020.65 1132.57
2035  1020.94 1134.87
2036  1021.23 1133.93
2037  1021.51 1148.08
2038  1021.80 1104.39
2039  1022.09 1075.26
2040  1022.38 1104.39
2041  1022.67 1131.09
2042  1022.96 1157.78
2043  1023.25 1174.78
2044  1023.53 1121.38
2045  1023.82 1067.98
2046  1024.11 1118.95
2047  1024.40 1097.10
2048  1024.69 1128.66
2049  1024.97 1123.80
2050  1025.26 1167.49
2051  1025.55 1177.20
2052  1025.84 1121.38
2053  1026.13 1172.35
2054  1026.41 1150.50
2055  1026.70 1150.50
2056  1026.99 1150.50
```

Listing 8: 蓝灯

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1
2  暗光谱: 否
3  参考光谱: 否
4  光谱数量: 1
5  光谱仪: USB2+H14105
6  积分时间（毫秒）: 74 (USB2+H14105)
7  平均次数: 1 (USB2+H14105)
8  平滑度: 20 (USB2+H14105)
9  去除暗噪声: 否 (USB2+H14105)
10 339.35  745.16
11 339.73  745.16
12 340.11  745.16
13 340.49  745.16
14 340.87  767.00
15 341.25  783.99
16 341.63  759.72
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17  342.01  703.89
18  342.39  733.02
19  342.77  759.72
20  343.15  747.58
21  343.53  793.70
22  343.91  813.12
23  344.29  820.40
24  344.66  917.49
25  345.04  900.50
26  345.42  844.67
27  345.80  883.51
28  346.18  929.63
29  346.56  934.48
30  346.94  942.12
31  347.32  951.35
32  347.70  960.88
33  348.08  970.18
34  348.45  979.59
35  348.83  987.05
36  349.21  995.40
37  349.59  1004.69
38  349.97  1015.70
39  350.35  1026.24
40  350.73  1034.71
41  351.10  1046.13
42  351.48  1054.72
43  351.86  1062.83
44  352.24  1070.29
45  352.62  1074.67
46  353.00  1080.11
47  353.37  1089.53
48  353.75  1096.28
49  354.13  1101.96
50  354.51  1107.29
51  354.89  1111.19
52  355.27  1116.34
53  355.64  1121.61
54  356.02  1123.63
55  356.40  1127.00
56  356.78  1130.79
57  357.16  1134.05
58  357.53  1137.42
59  357.91  1136.77
60  358.29  1137.18
61  358.67  1137.30
62  359.04  1138.78
63  359.42  1139.31
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 64  359.80  1140.02
 65  360.18  1138.13
 66  360.55  1137.24
 67  360.93  1137.54
 68  361.31  1137.83
 69  361.69  1138.43
 70  362.06  1138.49
 71  362.44  1134.99
 72  362.82  1136.35
 73  363.20  1136.30
 74  363.57  1137.06
 75  363.95  1138.72
 76  364.33  1140.44
 77  364.71  1140.08
 78  365.08  1140.74
 79  365.46  1138.43
 80  365.84  1137.24
 81  366.21  1138.01
 82  366.59  1137.78
 83  366.97  1136.41
 84  367.34  1136.53
 85  367.72  1135.82
 86  368.10  1135.23
 87  368.47  1134.70
 88  368.85  1132.03
 89  369.23  1133.10
 90  369.60  1132.57
 91  369.98  1132.98
 92  370.36  1134.34
 93  370.73  1135.53
 94  371.11  1134.70
 95  371.49  1135.41
 96  371.86  1134.76
 97  372.24  1133.22
 98  372.62  1131.26
 99  372.99  1130.79
100  373.37  1132.21
101  373.74  1131.91
102  374.12  1131.32
103  374.50  1132.57
104  374.87  1131.74
105  375.25  1132.68
106  375.62  1133.04
107  376.00  1133.87
108  376.38  1134.46
109  376.75  1134.10
110  377.13  1133.93
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112  377.88  1135.29
113  378.26  1133.51
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115  379.01  1133.99
116  379.38  1132.45
117  379.76  1131.38
118  380.13  1131.74
119  380.51  1131.62
120  380.88  1132.98
121  381.26  1134.93
122  381.64  1134.28
123  382.01  1134.40
124  382.39  1134.93
125  382.76  1132.27
126  383.14  1132.98
127  383.51  1134.87
128  383.89  1135.05
129  384.26  1136.95
130  384.64  1134.93
131  385.01  1134.93
132  385.39  1135.76
133  385.76  1135.35
134  386.14  1134.87
135  386.51  1134.76
136  386.89  1135.29
137  387.26  1136.77
138  387.64  1138.54
139  388.01  1138.96
140  388.39  1140.32
141  388.76  1141.50
142  389.13  1142.27
143  389.51  1143.99
144  389.88  1144.88
145  390.26  1147.19
146  390.63  1147.19
147  391.01  1149.38
148  391.38  1149.26
149  391.76  1149.79
150  392.13  1150.62
151  392.50  1150.86
152  392.88  1149.56
153  393.25  1151.27
154  393.63  1152.10
155  394.00  1153.88
156  394.37  1152.93
157  394.75  1153.76
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159  395.50  1153.46
160  395.87  1153.05
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162  396.62  1154.41
163  396.99  1155.89
164  397.37  1157.73
165  397.74  1158.91
166  398.11  1162.82
167  398.49  1162.70
168  398.86  1164.36
169  399.23  1165.78
170  399.61  1164.83
171  399.98  1167.38
172  400.35  1169.21
173  400.73  1169.03
174  401.10  1170.39
175  401.47  1173.06
176  401.85  1174.89
177  402.22  1176.85
178  402.59  1178.03
179  402.97  1180.81
180  403.34  1183.95
181  403.71  1186.79
182  404.09  1186.97
183  404.46  1190.94
184  404.83  1192.77
185  405.20  1193.90
186  405.58  1198.63
187  405.95  1203.19
188  406.32  1206.15
189  406.70  1210.95
190  407.07  1215.15
191  407.44  1220.36
192  407.81  1224.86
193  408.19  1231.84
194  408.56  1236.52
195  408.93  1245.11
196  409.30  1250.79
197  409.68  1256.71
198  410.05  1266.18
199  410.42  1277.67
200  410.79  1288.50
201  411.17  1299.51
202  411.54  1308.57
203  411.91  1319.99
204  412.28  1331.78
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413.40  1367.53
413.77  1383.52
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414.89  1433.66
415.26  1452.49
415.63  1472.61
416.00  1494.46
416.37  1519.50
416.74  1544.36
417.12  1568.81
417.49  1595.04
417.86  1622.09
418.23  1649.39
418.60  1678.75
418.97  1708.76
419.34  1743.93
419.72  1780.34
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420.46  1856.11
420.83  1891.28
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421.57  1974.57
421.94  2019.69
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422.69  2115.77
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423.43  2218.84
423.80  2276.14
424.17  2335.34
424.54  2395.55
424.91  2462.33
425.28  2526.92
425.65  2593.87
426.02  2660.77
426.39  2732.82
426.76  2809.01
427.13  2883.72
427.51  2967.55
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428.62  3218.38
428.99  3307.06
429.36  3399.47
429.73  3499.58
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256  431.58  4019.66
257  431.95  4130.66
258  432.32  4246.04
259  432.69  4359.82
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262  433.80  4739.24
263  434.17  4873.86
264  434.54  5007.95
265  434.91  5144.11
266  435.28  5283.06
267  435.65  5428.75
268  436.02  5569.17
269  436.39  5716.82
270  436.75  5864.70
271  437.12  6010.51
272  437.49  6163.72
273  437.86  6321.67
274  438.23  6482.46
275  438.60  6644.25
276  438.97  6807.77
277  439.34  6971.87
278  439.71  7139.35
279  440.08  7297.77
280  440.45  7461.81
281  440.82  7630.12
282  441.19  7805.12
283  441.55  7980.83
284  441.92  8153.16
285  442.29  8341.89
286  442.66  8518.84
287  443.03  8697.21
288  443.40  8885.65
289  443.77  9065.85
290  444.14  9250.50
291  444.51  9437.69
292  444.87  9617.31
293  445.24  9808.52
294  445.61  10006.55
295  445.98  10204.81
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297  446.72  10602.17
298  447.08  10793.44
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301  448.19  11366.68
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303  448.93  11723.31
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305  449.66  12086.32
306  450.03  12266.59
307  450.40  12436.97
308  450.77  12602.43
309  451.13  12781.40
310  451.50  12939.76
311  451.87  13098.24
312  452.24  13257.19
313  452.61  13395.31
314  452.97  13536.20
315  453.34  13670.12
316  453.71  13797.75
317  454.08  13909.40
318  454.44  14016.08
319  454.81  14123.59
320  455.18  14221.15
321  455.55  14304.27
322  455.91  14383.54
323  456.28  14449.14
324  456.65  14510.59
325  457.02  14565.17
326  457.38  14604.95
327  457.75  14637.69
328  458.12  14668.77
329  458.48  14677.77
330  458.85  14680.14
331  459.22  14678.00
332  459.59  14661.55
333  459.95  14637.45
334  460.32  14603.18
335  460.69  14548.65
336  461.05  14482.52
337  461.42  14416.69
338  461.79  14334.70
339  462.15  14257.80
340  462.52  14171.37
341  462.89  14076.05
342  463.25  13962.57
343  463.62  13862.28
344  463.99  13755.31
345  464.35  13632.82
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348  465.45  13225.40
349  465.82  13077.28
350  466.18  12919.87
351  466.55  12771.92
352  466.92  12614.75
353  467.28  12451.29
354  467.65  12294.47
355  468.02  12126.40
356  468.38  11964.78
357  468.75  11792.51
358  469.11  11637.29
359  469.48  11477.03
360  469.85  11302.92
361  470.21  11140.89
362  470.58  10980.99
363  470.94  10816.41
364  471.31  10653.32
365  471.67  10485.84
366  472.04  10319.60
367  472.41  10145.37
368  472.77  9982.28
369  473.14  9824.45
370  473.50  9667.45
371  473.87  9516.07
372  474.23  9360.26
373  474.60  9207.76
374  474.96  9057.68
375  475.33  8900.09
376  475.69  8750.37
377  476.06  8609.06
378  476.42  8464.02
379  476.79  8321.17
380  477.15  8174.65
381  477.52  8032.80
382  477.88  7885.22
383  478.25  7754.86
384  478.61  7610.94
385  478.98  7477.98
386  479.34  7345.01
387  479.71  7210.74
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389  480.44  6948.90
390  480.80  6829.43
391  481.17  6701.97
392  481.53  6574.58
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395  482.62  6215.05
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398  483.72  5867.07
399  484.08  5749.68
400  484.45  5631.51
401  484.81  5523.82
402  485.17  5413.83
403  485.54  5307.21
404  485.90  5201.77
405  486.27  5098.94
406  486.63  4997.24
407  487.00  4901.63
408  487.36  4809.81
409  487.72  4715.62
410  488.09  4616.87
411  488.45  4522.92
412  488.81  4430.98
413  489.18  4342.48
414  489.54  4253.20
415  489.91  4166.06
416  490.27  4082.05
417  490.63  3997.40
418  491.00  3916.53
419  491.36  3833.65
420  491.72  3757.28
421  492.09  3683.87
422  492.45  3609.10
423  492.81  3542.74
424  493.18  3472.76
425  493.54  3406.40
426  493.90  3338.44
427  494.27  3270.83
428  494.63  3202.99
429  494.99  3142.01
430  495.35  3084.29
431  495.72  3023.73
432  496.08  2971.39
433  496.44  2918.65
434  496.81  2865.54
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436  497.53  2769.76
437  497.89  2720.62
438  498.26  2676.75
439  498.62  2625.43
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444  500.43  2413.19
445  500.79  2377.32
446  501.16  2341.50
447  501.52  2304.97
448  501.88  2265.01
449  502.24  2228.31
450  502.61  2192.85
451  502.97  2160.94
452  503.33  2130.27
453  503.69  2100.91
454  504.05  2071.60
455  504.42  2045.38
456  504.78  2019.92
457  505.14  1996.36
458  505.50  1973.04
459  505.86  1948.88
460  506.23  1927.69
461  506.59  1903.95
462  506.95  1879.03
463  507.31  1856.94
464  507.67  1837.58
465  508.03  1818.70
466  508.40  1800.53
467  508.76  1781.11
468  509.12  1765.95
469  509.48  1749.79
470  509.84  1731.68
471  510.20  1717.59
472  510.56  1699.71
473  510.93  1682.42
474  511.29  1664.36
475  511.65  1650.63
476  512.01  1636.42
477  512.37  1616.41
478  512.73  1600.61
479  513.09  1586.75
480  513.45  1574.85
481  513.81  1560.76
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1751  936.57  1170.22
1752  936.87  1172.29
1753  937.17  1171.99
1754  937.48  1170.10
1755  937.78  1170.81
```

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1756  938.08  1170.16
1757  938.39  1169.33
1758  938.69  1171.28
1759  938.99  1170.81
1760  939.29  1168.44
1761  939.60  1166.07
1762  939.90  1167.20
1763  940.20  1165.72
1764  940.51  1166.37
1765  940.81  1166.01
1766  941.11  1166.84
1767  941.41  1166.90
1768  941.71  1166.49
1769  942.02  1164.18
1770  942.32  1163.23
1771  942.62  1165.48
1772  942.92  1164.24
1773  943.23  1164.36
1774  943.53  1165.36
1775  943.83  1165.07
1776  944.13  1164.06
1777  944.43  1160.86
1778  944.74  1161.57
1779  945.04  1160.09
1780  945.34  1161.51
1781  945.64  1159.80
1782  945.94  1157.61
1783  946.24  1157.25
1784  946.55  1155.89
1785  946.85  1155.48
1786  947.15  1156.60
1787  947.45  1155.89
1788  947.75  1155.42
1789  948.05  1153.05
1790  948.35  1152.40
1791  948.66  1150.33
1792  948.96  1149.44
1793  949.26  1146.77
1794  949.56  1146.06
1795  949.86  1147.01
1796  950.16  1148.02
1797  950.46  1147.54
1798  950.76  1149.02
1799  951.06  1147.84
1800  951.36  1149.20
1801  951.66  1150.98
1802  951.97  1151.15
```

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1803  952.27  1150.56
1804  952.57  1151.33
1805  952.87  1150.44
1806  953.17  1149.79
1807  953.47  1149.20
1808  953.77  1150.27
1809  954.07  1150.98
1810  954.37  1150.92
1811  954.67  1151.98
1812  954.97  1149.79
1813  955.27  1150.92
1814  955.57  1151.21
1815  955.87  1149.08
1816  956.17  1150.09
1817  956.47  1152.22
1818  956.77  1154.11
1819  957.07  1153.58
1820  957.37  1153.88
1821  957.67  1152.93
1822  957.97  1154.11
1823  958.27  1155.12
1824  958.57  1156.66
1825  958.87  1155.77
1826  959.17  1156.19
1827  959.47  1155.18
1828  959.76  1157.25
1829  960.06  1158.61
1830  960.36  1157.78
1831  960.66  1159.27
1832  960.96  1161.63
1833  961.26  1161.99
1834  961.56  1164.06
1835  961.86  1164.71
1836  962.16  1163.88
1837  962.46  1162.46
1838  962.76  1163.71
1839  963.05  1161.40
1840  963.35  1161.16
1841  963.65  1162.05
1842  963.95  1162.11
1843  964.25  1163.35
1844  964.55  1163.41
1845  964.85  1163.88
1846  965.14  1162.99
1847  965.44  1162.52
1848  965.74  1162.76
1849  966.04  1161.16
```

```
966.34  1160.39
966.64  1161.69
966.93  1158.67
967.23  1158.85
967.53  1157.55
967.83  1158.14
968.13  1159.80
968.42  1157.31
968.72  1156.42
969.02  1155.71
969.32  1155.95
969.62  1156.13
969.91  1154.17
970.21  1152.28
970.51  1151.69
970.81  1148.90
971.10  1148.43
971.40  1150.03
971.70  1150.03
972.00  1147.84
972.29  1147.19
972.59  1147.90
972.89  1146.83
973.18  1145.29
973.48  1143.75
973.78  1141.03
974.08  1140.44
974.37  1142.57
974.67  1142.16
974.97  1140.74
975.26  1141.80
975.56  1141.62
975.86  1139.85
976.15  1138.66
976.45  1139.43
976.75  1141.21
977.04  1141.92
977.34  1143.22
977.63  1143.70
977.93  1141.39
978.23  1142.63
978.52  1143.81
978.82  1143.34
979.12  1146.95
979.41  1147.48
979.71  1148.67
980.00  1148.67
```

```
1897  980.30  1146.30
1898  980.60  1147.25
1899  980.89  1147.31
1900  981.19  1149.08
1901  981.48  1149.08
1902  981.78  1148.85
1903  982.07  1151.33
1904  982.37  1150.38
1905  982.67  1151.75
1906  982.96  1151.63
1907  983.26  1152.28
1908  983.55  1149.67
1909  983.85  1148.55
1910  984.14  1149.97
1911  984.44  1150.68
1912  984.73  1151.10
1913  985.03  1150.56
1914  985.32  1151.15
1915  985.62  1153.64
1916  985.91  1154.94
1917  986.21  1154.53
1918  986.50  1153.64
1919  986.80  1152.63
1920  987.09  1154.41
1921  987.39  1154.35
1922  987.68  1155.83
1923  987.98  1155.77
1924  988.27  1154.59
1925  988.56  1154.82
1926  988.86  1155.24
1927  989.15  1156.13
1928  989.45  1156.48
1929  989.74  1157.96
1930  990.04  1159.44
1931  990.33  1159.38
1932  990.62  1158.91
1933  990.92  1158.79
1934  991.21  1156.60
1935  991.51  1157.49
1936  991.80  1159.74
1937  992.09  1158.44
1938  992.39  1160.80
1939  992.68  1161.28
1940  992.98  1160.51
1941  993.27  1159.92
1942  993.56  1158.67
1943  993.86  1158.85
```

```
994.15  1158.73
994.44  1160.98
994.74  1159.32
995.03  1163.53
995.32  1165.42
995.62  1168.38
995.91  1169.63
996.20  1169.15
996.50  1168.56
996.79  1169.27
997.08  1168.50
997.38  1165.78
997.67  1166.01
997.96  1166.78
998.25  1168.74
998.55  1171.22
998.84  1170.99
999.13  1170.57
999.42  1170.81
999.72  1169.21
1000.01 1170.57
1000.30 1172.17
1000.59 1172.59
1000.89 1168.86
1001.18 1167.14
1001.47 1167.97
1001.76 1166.49
1002.06 1167.55
1002.35 1167.49
1002.64 1166.19
1002.93 1165.84
1003.22 1165.19
1003.52 1162.46
1003.81 1160.15
1004.10 1162.46
1004.39 1161.99
1004.68 1161.81
1004.97 1160.27
1005.27 1159.98
1005.56 1159.86
1005.85 1159.44
1006.14 1157.19
1006.43 1158.08
1006.72 1159.03
1007.01 1156.01
1007.31 1155.42
1007.60 1153.05
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1991  1007.89 1154.77
1992  1008.18 1155.30
1993  1008.47 1155.42
1994  1008.76 1154.53
1995  1009.05 1156.42
1996  1009.34 1158.32
1997  1009.63 1157.02
1998  1009.93 1156.36
1999  1010.22 1155.42
2000  1010.51 1152.93
2001  1010.80 1155.24
2002  1011.09 1153.23
2003  1011.38 1152.87
2004  1011.67 1153.88
2005  1011.96 1152.99
2006  1012.25 1151.81
2007  1012.54 1148.49
2008  1012.83 1149.44
2009  1013.12 1148.25
2010  1013.41 1146.42
2011  1013.70 1146.71
2012  1013.99 1144.94
2013  1014.28 1145.00
2014  1014.57 1146.66
2015  1014.86 1146.60
2016  1015.15 1148.55
2017  1015.44 1148.61
2018  1015.73 1150.03
2019  1016.02 1149.44
2020  1016.31 1149.62
2021  1016.60 1149.50
2022  1016.89 1151.27
2023  1017.18 1149.38
2024  1017.47 1150.50
2025  1017.76 1152.22
2026  1018.05 1153.70
2027  1018.34 1150.56
2028  1018.62 1150.74
2029  1018.91 1153.40
2030  1019.20 1151.63
2031  1019.49 1150.74
2032  1019.78 1149.02
2033  1020.07 1148.14
2034  1020.36 1147.54
2035  1020.65 1147.31
2036  1020.94 1145.65
2037  1021.23 1145.23
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2038  1021.51 1211.18
2039  1021.80 1179.63
2040  1022.09 1157.78
2041  1022.38 1148.08
2042  1022.67 1157.78
2043  1022.96 1082.54
2044  1023.25 1169.92
2045  1023.53 1169.92
2046  1023.82 1177.20
2047  1024.11 1063.12
2048  1024.40 1143.22
2049  1024.69 1225.75
2050  1024.97 1133.51
2051  1025.26 1099.53
2052  1025.55 1169.92
2053  1025.84 1169.92
2054  1026.13 1138.37
2055  1026.41 1123.80
2056  1026.70 1123.80
2057  1026.99 1123.80
```

Listing 9: 绿灯

```
1  暗光谱: 否
2  参考光谱: 否
3  光谱数量: 1
4  光谱仪: USB2+H14105
5  积分时间（毫秒）: 1 (USB2+H14105)
6  平均次数: 1 (USB2+H14105)
7  平滑度: 20 (USB2+H14105)
8  去除暗噪声: 否 (USB2+H14105)
9  339.35  820.40
10 339.73  820.40
11 340.11  820.40
12 340.49  750.01
13 340.87  771.86
14 341.25  762.15
15 341.63  713.60
16 342.01  759.72
17 342.39  725.74
18 342.77  786.42
19 343.15  742.73
20 343.53  759.72
21 343.91  830.11
22 344.29  839.82
23 344.66  805.84
24 345.04  827.68
```

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25  345.42  864.09
26  345.80  873.80
27  346.18  888.36
28  346.56  932.05
29  346.94  954.67
30  347.32  962.25
31  347.70  971.13
32  348.08  979.65
33  348.45  990.01
34  348.83  1000.90
35  349.21  1010.08
36  349.59  1022.27
37  349.97  1031.16
38  350.35  1041.52
39  350.73  1051.16
40  351.10  1060.70
41  351.48  1071.41
42  351.86  1080.41
43  352.24  1086.69
44  352.62  1094.32
45  353.00  1101.90
46  353.37  1109.00
47  353.75  1115.52
48  354.13  1120.73
49  354.51  1125.58
50  354.89  1131.09
51  355.27  1136.65
52  355.64  1140.85
53  356.02  1142.87
54  356.40  1145.77
55  356.78  1146.54
56  357.16  1148.96
57  357.53  1147.31
58  357.91  1149.50
59  358.29  1150.56
60  358.67  1151.10
61  359.04  1151.21
62  359.42  1151.15
63  359.80  1150.98
64  360.18  1150.74
65  360.55  1151.27
66  360.93  1150.86
67  361.31  1148.14
68  361.69  1147.78
69  362.06  1146.00
70  362.44  1145.59
71  362.82  1146.60
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 72  363.20  1145.59
 73  363.57  1144.76
 74  363.95  1143.46
 75  364.33  1141.15
 76  364.71  1139.55
 77  365.08  1136.65
 78  365.46  1137.24
 79  365.84  1136.89
 80  366.21  1134.05
 81  366.59  1134.93
 82  366.97  1134.52
 83  367.34  1132.80
 84  367.72  1133.87
 85  368.10  1133.22
 86  368.47  1134.52
 87  368.85  1132.27
 88  369.23  1133.16
 89  369.60  1133.34
 90  369.98  1133.39
 91  370.36  1132.62
 92  370.73  1131.26
 93  371.11  1130.32
 94  371.49  1131.62
 95  371.86  1131.38
 96  372.24  1132.68
 97  372.62  1133.28
 98  372.99  1135.94
 99  373.37  1137.42
100  373.74  1136.47
101  374.12  1136.59
102  374.50  1137.60
103  374.87  1136.71
104  375.25  1137.60
105  375.62  1137.60
106  376.00  1138.49
107  376.38  1138.49
108  376.75  1140.20
109  377.13  1140.32
110  377.50  1141.33
111  377.88  1141.27
112  378.26  1139.37
113  378.63  1137.89
114  379.01  1139.08
115  379.38  1138.96
116  379.76  1139.43
117  380.13  1141.45
118  380.51  1142.10
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119  380.88  1142.16
120  381.26  1139.43
121  381.64  1140.91
122  382.01  1138.90
123  382.39  1137.54
124  382.76  1138.66
125  383.14  1138.13
126  383.51  1139.08
127  383.89  1137.72
128  384.26  1139.02
129  384.64  1138.13
130  385.01  1138.49
131  385.39  1139.26
132  385.76  1139.67
133  386.14  1139.02
134  386.51  1138.96
135  386.89  1137.95
136  387.26  1136.59
137  387.64  1135.47
138  388.01  1135.76
139  388.39  1135.23
140  388.76  1135.41
141  389.13  1136.06
142  389.51  1137.42
143  389.88  1137.18
144  390.26  1137.18
145  390.63  1136.77
146  391.01  1137.95
147  391.38  1137.42
148  391.76  1137.89
149  392.13  1137.72
150  392.50  1136.89
151  392.88  1136.77
152  393.25  1139.31
153  393.63  1140.50
154  394.00  1143.70
155  394.37  1143.81
156  394.75  1145.06
157  395.12  1144.58
158  395.50  1145.35
159  395.87  1146.77
160  396.24  1146.89
161  396.62  1149.02
162  396.99  1148.37
163  397.37  1150.98
164  397.74  1150.44
165  398.11  1149.67
```

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166  398.49  1151.15
167  398.86  1151.57
168  399.23  1152.40
169  399.61  1153.94
170  399.98  1155.54
171  400.35  1156.42
172  400.73  1157.02
173  401.10  1159.27
174  401.47  1161.57
175  401.85  1161.57
176  402.22  1161.57
177  402.59  1164.06
178  402.97  1164.77
179  403.34  1166.13
180  403.71  1167.20
181  404.09  1167.20
182  404.46  1168.03
183  404.83  1168.86
184  405.20  1167.85
185  405.58  1168.20
186  405.95  1168.32
187  406.32  1167.97
188  406.70  1167.97
189  407.07  1166.43
190  407.44  1169.21
191  407.81  1170.87
192  408.19  1170.51
193  408.56  1170.75
194  408.93  1172.64
195  409.30  1171.93
196  409.68  1170.51
197  410.05  1170.99
198  410.42  1172.53
199  410.79  1173.00
200  411.17  1172.94
201  411.54  1174.36
202  411.91  1177.68
203  412.28  1180.04
204  412.65  1179.33
205  413.03  1180.81
206  413.40  1182.89
207  413.77  1181.47
208  414.14  1183.00
209  414.51  1184.01
210  414.89  1185.79
211  415.26  1187.74
212  415.63  1189.34
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213  416.00  1191.11
214  416.37  1191.29
215  416.74  1191.06
216  417.12  1194.43
217  417.49  1194.13
218  417.86  1194.07
219  418.23  1197.39
220  418.60  1196.92
221  418.97  1196.92
222  419.34  1194.55
223  419.72  1195.79
224  420.09  1194.13
225  420.46  1195.79
226  420.83  1195.73
227  421.20  1196.92
228  421.57  1198.99
229  421.94  1200.65
230  422.31  1203.49
231  422.69  1201.30
232  423.06  1202.30
233  423.43  1204.97
234  423.80  1204.73
235  424.17  1205.26
236  424.54  1205.86
237  424.91  1208.76
238  425.28  1211.36
239  425.65  1212.07
240  426.02  1211.72
241  426.39  1213.08
242  426.76  1213.61
243  427.13  1213.20
244  427.51  1213.55
245  427.88  1215.21
246  428.25  1215.33
247  428.62  1214.20
248  428.99  1216.28
249  429.36  1215.92
250  429.73  1217.40
251  430.10  1216.39
252  430.47  1215.86
253  430.84  1216.75
254  431.21  1216.81
255  431.58  1217.76
256  431.95  1220.66
257  432.32  1220.72
258  432.69  1226.10
259  433.06  1226.52
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260  433.43  1226.99
261  433.80  1229.89
262  434.17  1232.67
263  434.54  1237.05
264  434.91  1238.53
265  435.28  1241.67
266  435.65  1243.80
267  436.02  1246.53
268  436.39  1247.95
269  436.75  1247.71
270  437.12  1250.08
271  437.49  1251.97
272  437.86  1255.64
273  438.23  1258.07
274  438.60  1259.85
275  438.97  1263.40
276  439.34  1265.83
277  439.71  1270.68
278  440.08  1273.05
279  440.45  1274.88
280  440.82  1279.50
281  441.19  1283.47
282  441.55  1286.49
283  441.92  1288.56
284  442.29  1290.93
285  442.66  1296.73
286  443.03  1301.29
287  443.40  1307.44
288  443.77  1314.07
289  444.14  1321.30
290  444.51  1326.45
291  444.87  1331.30
292  445.24  1338.70
293  445.61  1345.15
294  445.98  1351.61
295  446.35  1357.35
296  446.72  1362.80
297  447.08  1370.08
298  447.45  1377.36
299  447.82  1385.17
300  448.19  1395.30
301  448.56  1403.35
302  448.93  1411.46
303  449.29  1420.04
304  449.66  1427.80
305  450.03  1438.16
306  450.40  1448.52
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307  450.77  1459.00
308  451.13  1471.49
309  451.50  1483.80
310  451.87  1500.14
311  452.24  1514.59
312  452.61  1529.62
313  452.97  1545.02
314  453.34  1557.63
315  453.71  1573.20
316  454.08  1588.17
317  454.44  1605.10
318  454.81  1623.87
319  455.18  1644.89
320  455.55  1665.61
321  455.91  1684.97
322  456.28  1705.57
323  456.65  1727.71
324  457.02  1753.11
325  457.38  1779.27
326  457.75  1804.08
327  458.12  1830.90
328  458.48  1859.90
329  458.85  1888.62
330  459.22  1918.45
331  459.59  1950.89
332  459.95  1986.12
333  460.32  2023.65
334  460.69  2058.40
335  461.05  2094.69
336  461.42  2134.95
337  461.79  2177.28
338  462.15  2219.07
339  462.52  2260.75
340  462.89  2302.78
341  463.25  2351.92
342  463.62  2400.64
343  463.99  2454.04
344  464.35  2506.37
345  464.72  2566.58
346  465.09  2628.56
347  465.45  2691.26
348  465.82  2756.26
349  466.18  2822.98
350  466.55  2893.49
351  466.92  2967.25
352  467.28  3042.91
353  467.65  3122.65
```

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354  468.02  3208.73
355  468.38  3298.65
356  468.75  3391.13
357  469.11  3489.28
358  469.48  3584.30
359  469.85  3684.70
360  470.21  3790.73
361  470.58  3903.45
362  470.94  4019.01
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1674  913.39  1147.84
1675  913.69  1148.85
1676  914.00  1150.27
1677  914.31  1149.32
1678  914.61  1149.62
1679  914.92  1152.40
1680  915.23  1152.16
1681  915.53  1152.46
1682  915.84  1154.06
1683  916.15  1155.65
1684  916.45  1156.07
1685  916.76  1155.36
1686  917.06  1153.82
1687  917.37  1154.65
1688  917.68  1153.29
1689  917.98  1157.07
1690  918.29  1156.84
1691  918.60  1157.37
1692  918.90  1159.09
1693  919.21  1158.73
1694  919.51  1158.38
1695  919.82  1157.25
1696  920.13  1157.49
1697  920.43  1157.25
1698  920.74  1154.59
1699  921.04  1152.52
1700  921.35  1154.41
1701  921.65  1157.49
1702  921.96  1160.63
1703  922.26  1160.69
1704  922.57  1160.39
1705  922.88  1160.80
1706  923.18  1160.92
1707  923.49  1159.92
1708  923.79  1159.50
1709  924.10  1159.09
1710  924.40  1159.03
1711  924.71  1159.32
1712  925.01  1159.15
1713  925.32  1159.80
1714  925.62  1158.32
1715  925.93  1158.38
1716  926.23  1158.38
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1717  926.54  1157.61
1718  926.84  1158.26
1719  927.15  1156.72
1720  927.45  1155.12
1721  927.76  1154.17
1722  928.06  1154.00
1723  928.36  1151.86
1724  928.67  1151.81
1725  928.97  1153.34
1726  929.28  1151.98
1727  929.58  1153.64
1728  929.89  1153.88
1729  930.19  1155.12
1730  930.50  1151.81
1731  930.80  1152.40
1732  931.10  1151.98
1733  931.41  1150.38
1734  931.71  1149.73
1735  932.02  1149.91
1736  932.32  1149.56
1737  932.62  1149.26
1738  932.93  1149.44
1739  933.23  1151.10
1740  933.53  1151.21
1741  933.84  1149.38
1742  934.14  1148.79
1743  934.44  1148.67
1744  934.75  1150.68
1745  935.05  1151.81
1746  935.36  1152.93
1747  935.66  1154.71
1748  935.96  1154.11
1749  936.27  1154.35
1750  936.57  1154.06
1751  936.87  1153.94
1752  937.17  1152.75
1753  937.48  1152.58
1754  937.78  1153.11
1755  938.08  1155.36
1756  938.39  1155.12
1757  938.69  1153.58
1758  938.99  1154.06
1759  939.29  1153.11
1760  939.60  1155.12
1761  939.90  1155.42
1762  940.20  1155.89
1763  940.51  1155.59
```

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1764  940.81  1155.95
1765  941.11  1154.23
1766  941.41  1151.45
1767  941.71  1152.52
1768  942.02  1151.51
1769  942.32  1151.04
1770  942.62  1151.75
1771  942.92  1150.15
1772  943.23  1149.79
1773  943.53  1150.03
1774  943.83  1151.63
1775  944.13  1152.63
1776  944.43  1153.34
1777  944.74  1154.41
1778  945.04  1154.41
1779  945.34  1154.29
1780  945.64  1154.77
1781  945.94  1155.30
1782  946.24  1155.89
1783  946.55  1154.17
1784  946.85  1152.63
1785  947.15  1150.80
1786  947.45  1150.86
1787  947.75  1149.97
1788  948.05  1149.14
1789  948.35  1150.33
1790  948.66  1150.68
1791  948.96  1150.68
1792  949.26  1149.91
1793  949.56  1149.56
1794  949.86  1148.37
1795  950.16  1149.02
1796  950.46  1146.77
1797  950.76  1147.78
1798  951.06  1149.79
1799  951.36  1151.69
1800  951.66  1153.88
1801  951.97  1153.17
1802  952.27  1153.82
1803  952.57  1152.58
1804  952.87  1151.81
1805  953.17  1151.57
1806  953.47  1152.93
1807  953.77  1152.28
1808  954.07  1152.40
1809  954.37  1154.77
1810  954.67  1156.30
```

```
954.97  1156.60
955.27  1157.61
955.57  1159.32
955.87  1161.04
956.17  1161.16
956.47  1161.40
956.77  1160.80
957.07  1162.17
957.37  1164.24
957.67  1165.60
957.97  1166.07
958.27  1166.19
958.57  1166.72
958.87  1165.60
959.17  1166.84
959.47  1167.20
959.76  1166.90
960.06  1169.39
960.36  1169.57
960.66  1167.97
960.96  1167.32
961.26  1166.84
961.56  1167.43
961.86  1167.49
962.16  1168.03
962.46  1166.67
962.76  1166.67
963.05  1165.60
963.35  1161.99
963.65  1159.38
963.95  1157.78
964.25  1157.02
964.55  1157.55
964.85  1159.62
965.14  1161.34
965.44  1161.28
965.74  1162.52
966.04  1163.47
966.34  1165.13
966.64  1161.16
966.93  1159.68
967.23  1159.38
967.53  1160.57
967.83  1158.73
968.13  1156.90
968.42  1157.19
968.72  1156.90
```

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1858  969.02  1157.19
1859  969.32  1155.18
1860  969.62  1153.34
1861  969.91  1152.99
1862  970.21  1152.40
1863  970.51  1151.10
1864  970.81  1149.67
1865  971.10  1152.46
1866  971.40  1151.51
1867  971.70  1151.98
1868  972.00  1150.86
1869  972.29  1147.19
1870  972.59  1147.37
1871  972.89  1148.96
1872  973.18  1150.33
1873  973.48  1153.17
1874  973.78  1154.41
1875  974.08  1154.53
1876  974.37  1156.54
1877  974.67  1157.43
1878  974.97  1155.00
1879  975.26  1156.36
1880  975.56  1160.75
1881  975.86  1161.40
1882  976.15  1160.51
1883  976.45  1161.99
1884  976.75  1161.10
1885  977.04  1161.40
1886  977.34  1161.46
1887  977.63  1161.16
1888  977.93  1160.98
1889  978.23  1160.98
1890  978.52  1160.45
1891  978.82  1160.98
1892  979.12  1161.87
1893  979.41  1160.39
1894  979.71  1158.91
1895  980.00  1158.32
1896  980.30  1157.78
1897  980.60  1157.49
1898  980.89  1157.31
1899  981.19  1160.33
1900  981.48  1161.99
1901  981.78  1161.75
1902  982.07  1161.16
1903  982.37  1163.29
1904  982.67  1162.88
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1905  982.96  1163.29
1906  983.26  1161.93
1907  983.55  1162.23
1908  983.85  1161.40
1909  984.14  1163.17
1910  984.44  1163.71
1911  984.73  1162.64
1912  985.03  1163.17
1913  985.32  1163.17
1914  985.62  1160.75
1915  985.91  1161.28
1916  986.21  1161.87
1917  986.50  1160.92
1918  986.80  1161.04
1919  987.09  1162.11
1920  987.39  1159.44
1921  987.68  1156.54
1922  987.98  1155.48
1923  988.27  1155.36
1924  988.56  1155.06
1925  988.86  1155.89
1926  989.15  1154.59
1927  989.45  1152.52
1928  989.74  1154.41
1929  990.04  1153.52
1930  990.33  1154.59
1931  990.62  1152.52
1932  990.92  1153.76
1933  991.21  1153.40
1934  991.51  1154.88
1935  991.80  1154.77
1936  992.09  1157.13
1937  992.39  1158.08
1938  992.68  1157.02
1939  992.98  1157.07
1940  993.27  1154.71
1941  993.56  1154.71
1942  993.86  1155.71
1943  994.15  1156.01
1944  994.44  1152.58
1945  994.74  1152.87
1946  995.03  1151.86
1947  995.32  1153.34
1948  995.62  1152.58
1949  995.91  1151.10
1950  996.20  1149.26
1951  996.50  1149.32
```

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996.79  1149.79
997.08  1149.50
997.38  1148.49
997.67  1148.73
997.96  1146.89
998.25  1146.42
998.55  1145.77
998.84  1145.65
999.13  1146.54
999.42  1149.38
999.72  1149.91
1000.01 1152.75
1000.30 1154.17
1000.59 1153.34
1000.89 1151.92
1001.18 1151.15
1001.47 1151.57
1001.76 1150.38
1002.06 1150.74
1002.35 1150.98
1002.64 1151.51
1002.93 1151.15
1003.22 1152.52
1003.52 1153.88
1003.81 1155.89
1004.10 1154.47
1004.39 1153.23
1004.68 1152.63
1004.97 1151.69
1005.27 1149.56
1005.56 1149.20
1005.85 1148.79
1006.14 1149.08
1006.43 1149.73
1006.72 1151.39
1007.01 1152.16
1007.31 1151.33
1007.60 1151.63
1007.89 1152.63
1008.18 1153.17
1008.47 1154.23
1008.76 1154.71
1009.05 1154.71
1009.34 1155.42
1009.63 1155.30
1009.93 1155.06
1010.22 1155.89
```

1999 1010.51 1156.30
2000 1010.80 1157.43
2001 1011.09 1158.02
2002 1011.38 1156.60
2003 1011.67 1155.89
2004 1011.96 1153.82
2005 1012.25 1154.00
2006 1012.54 1154.77
2007 1012.83 1156.25
2008 1013.12 1158.44
2009 1013.41 1159.44
2010 1013.70 1158.44
2011 1013.99 1158.55
2012 1014.28 1158.44
2013 1014.57 1158.73
2014 1014.86 1159.44
2015 1015.15 1157.67
2016 1015.44 1155.00
2017 1015.73 1153.94
2018 1016.02 1153.94
2019 1016.31 1153.76
2020 1016.60 1155.65
2021 1016.89 1155.59
2022 1017.18 1155.89
2023 1017.47 1154.59
2024 1017.76 1153.94
2025 1018.05 1152.16
2026 1018.34 1152.81
2027 1018.62 1152.28
2028 1018.91 1152.58
2029 1019.20 1152.34
2030 1019.49 1151.98
2031 1019.78 1152.04
2032 1020.07 1153.46
2033 1020.36 1152.52
2034 1020.65 1152.81
2035 1020.94 1153.23
2036 1021.23 1153.46
2037 1021.51 1162.64
2038 1021.80 1152.93
2039 1022.09 1111.67
2040 1022.38 1191.77
2041 1022.67 1143.22
2042 1022.96 1106.81
2043 1023.25 1128.66
2044 1023.53 1126.23
2045 1023.82 1092.25

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1024.11 1172.35
1024.40 1150.50
1024.69 1150.50
1024.97 1143.22
1025.26 1131.09
1025.55 1148.08
1025.84 1206.33
1026.13 1138.37
1026.41 1172.35
1026.70 1172.35
1026.99 1172.35
```

Listing 10: 白灯

```
暗光谱: 否
参考光谱: 否
光谱数量: 1
光谱仪: USB2+H14105
积分时间（毫秒）: 50 (USB2+H14105)
平均次数: 1 (USB2+H14105)
平滑度: 20 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)

339.35  783.99
339.73  783.99
340.11  783.99
340.49  757.29
340.87  757.29
341.25  825.26
341.63  713.60
342.01  691.76
342.39  827.68
342.77  820.40
343.15  783.99
343.53  750.01
343.91  783.99
344.29  822.83
344.66  827.68
345.04  837.39
345.42  871.37
345.80  842.25
346.18  912.64
346.56  975.74
346.94  952.30
347.32  962.19
347.70  971.60
348.08  981.66
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34  348.45  991.19
35  348.83  1000.02
36  349.21  1007.47
37  349.59  1019.73
38  349.97  1030.44
39  350.35  1039.62
40  350.73  1048.15
41  351.10  1056.55
42  351.48  1066.38
43  351.86  1075.91
44  352.24  1082.25
45  352.62  1090.00
46  353.00  1098.64
47  353.37  1105.27
48  353.75  1112.85
49  354.13  1118.54
50  354.51  1121.50
51  354.89  1125.58
52  355.27  1130.38
53  355.64  1134.40
54  356.02  1137.18
55  356.40  1138.54
56  356.78  1141.56
57  357.16  1144.46
58  357.53  1145.35
59  357.91  1147.42
60  358.29  1148.55
61  358.67  1149.56
62  359.04  1150.09
63  359.42  1150.33
64  359.80  1151.51
65  360.18  1151.33
66  360.55  1148.73
67  360.93  1150.38
68  361.31  1149.20
69  361.69  1149.62
70  362.06  1148.02
71  362.44  1147.96
72  362.82  1145.94
73  363.20  1143.04
74  363.57  1142.27
75  363.95  1142.57
76  364.33  1144.29
77  364.71  1143.28
78  365.08  1140.08
79  365.46  1140.32
80  365.84  1136.83
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 81  366.21  1135.76
 82  366.59  1134.40
 83  366.97  1133.51
 84  367.34  1133.22
 85  367.72  1136.41
 86  368.10  1137.78
 87  368.47  1136.77
 88  368.85  1138.37
 89  369.23  1138.66
 90  369.60  1139.73
 91  369.98  1141.15
 92  370.36  1142.16
 93  370.73  1141.68
 94  371.11  1141.62
 95  371.49  1141.56
 96  371.86  1141.56
 97  372.24  1140.32
 98  372.62  1140.62
 99  372.99  1138.72
100  373.37  1139.31
101  373.74  1139.02
102  374.12  1139.97
103  374.50  1139.20
104  374.87  1140.02
105  375.25  1139.91
106  375.62  1142.10
107  376.00  1141.86
108  376.38  1141.62
109  376.75  1141.62
110  377.13  1140.74
111  377.50  1141.21
112  377.88  1140.50
113  378.26  1141.74
114  378.63  1143.28
115  379.01  1142.87
116  379.38  1143.99
117  379.76  1142.10
118  380.13  1142.93
119  380.51  1145.47
120  380.88  1147.54
121  381.26  1150.74
122  381.64  1152.10
123  382.01  1154.29
124  382.39  1156.07
125  382.76  1154.65
126  383.14  1151.92
127  383.51  1150.21
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128  383.89  1148.96
129  384.26  1147.42
130  384.64  1146.42
131  385.01  1147.01
132  385.39  1147.01
133  385.76  1147.31
134  386.14  1147.66
135  386.51  1147.60
136  386.89  1148.19
137  387.26  1149.14
138  387.64  1149.97
139  388.01  1149.50
140  388.39  1152.58
141  388.76  1153.46
142  389.13  1153.46
143  389.51  1152.46
144  389.88  1153.82
145  390.26  1155.95
146  390.63  1154.94
147  391.01  1155.42
148  391.38  1157.43
149  391.76  1158.14
150  392.13  1158.67
151  392.50  1161.46
152  392.88  1162.23
153  393.25  1164.18
154  393.63  1164.00
155  394.00  1165.01
156  394.37  1166.78
157  394.75  1168.38
158  395.12  1170.93
159  395.50  1173.47
160  395.87  1175.78
161  396.24  1178.33
162  396.62  1179.16
163  396.99  1182.00
164  397.37  1182.47
165  397.74  1184.25
166  398.11  1188.57
167  398.49  1194.73
168  398.86  1197.92
169  399.23  1200.88
170  399.61  1206.21
171  399.98  1211.72
172  400.35  1214.97
173  400.73  1218.41
174  401.10  1225.63
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401.47  1231.73
401.85  1239.66
402.22  1247.83
402.59  1256.35
402.97  1265.53
403.34  1274.35
403.71  1283.23
404.09  1292.82
404.46  1305.02
404.83  1317.86
405.20  1330.59
405.58  1343.02
405.95  1358.59
406.32  1374.75
406.70  1391.09
407.07  1409.51
407.44  1428.57
407.81  1447.51
408.19  1469.00
408.56  1491.56
408.93  1515.47
409.30  1542.29
409.68  1566.45
410.05  1592.44
410.42  1622.04
410.79  1654.00
411.17  1683.07
411.54  1714.45
411.91  1751.74
412.28  1789.04
412.65  1832.32
413.03  1876.18
413.40  1921.89
413.77  1966.29
414.14  2017.44
414.51  2070.95
414.89  2123.70
415.26  2183.14
415.63  2244.00
416.00  2310.18
416.37  2378.74
416.74  2451.32
417.12  2522.42
417.49  2602.46
417.86  2691.91
418.23  2778.28
418.60  2870.40
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222  418.97  2967.84
223  419.34  3067.24
224  419.72  3175.16
225  420.09  3287.35
226  420.46  3403.20
227  420.83  3520.60
228  421.20  3642.85
229  421.57  3779.84
230  421.94  3915.46
231  422.31  4058.14
232  422.69  4212.71
233  423.06  4360.59
234  423.43  4513.33
235  423.80  4676.84
236  424.17  4861.67
237  424.54  5042.70
238  424.91  5238.89
239  425.28  5433.43
240  425.65  5634.06
241  426.02  5837.29
242  426.39  6048.76
243  426.76  6275.91
244  427.13  6510.99
245  427.51  6764.08
246  427.88  7013.01
247  428.25  7284.86
248  428.62  7560.44
249  428.99  7836.43
250  429.36  8124.27
251  429.73  8423.65
252  430.10  8720.89
253  430.47  9042.94
254  430.84  9371.09
255  431.21  9696.99
256  431.58  10049.35
257  431.95  10416.99
258  432.32  10803.45
259  432.69  11170.67
260  433.06  11540.20
261  433.43  11930.86
262  433.80  12323.36
263  434.17  12696.92
264  434.54  13114.64
265  434.91  13532.77
266  435.28  13935.39
267  435.65  14371.11
268  436.02  14811.03
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269  436.39  15238.40
270  436.75  15657.54
271  437.12  16082.54
272  437.49  16544.48
273  437.86  16976.88
274  438.23  17418.52
275  438.60  17849.61
276  438.97  18276.39
277  439.34  18704.71
278  439.71  19150.84
279  440.08  19580.58
280  440.45  20004.46
281  440.82  20457.99
282  441.19  20914.43
283  441.55  21379.27
284  441.92  21824.75
285  442.29  22231.40
286  442.66  22621.59
287  443.03  23002.61
288  443.40  23374.45
289  443.77  23699.16
290  444.14  24023.82
291  444.51  24339.77
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293  445.24  24946.99
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1669  911.54  1142.57
1670  911.85  1141.45
1671  912.16  1141.09
1672  912.46  1141.74
1673  912.77  1142.16
1674  913.08  1143.70
1675  913.39  1143.46
1676  913.69  1143.28
1677  914.00  1141.92
1678  914.31  1141.39
```

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914.61  1142.39
914.92  1143.75
915.23  1145.12
915.53  1146.18
915.84  1145.29
916.15  1145.00
916.45  1146.12
916.76  1145.47
917.06  1144.05
917.37  1144.82
917.68  1146.89
917.98  1146.83
918.29  1145.23
918.60  1145.06
918.90  1145.00
919.21  1147.42
919.51  1148.14
919.82  1148.85
920.13  1147.13
920.43  1148.37
920.74  1150.15
921.04  1149.67
921.35  1149.26
921.65  1149.50
921.96  1151.27
922.26  1151.75
922.57  1150.86
922.88  1150.68
923.18  1151.92
923.49  1152.63
923.79  1153.76
924.10  1154.23
924.40  1155.71
924.71  1156.07
925.01  1155.71
925.32  1155.77
925.62  1156.01
925.93  1156.42
926.23  1156.07
926.54  1158.02
926.84  1159.38
927.15  1159.98
927.45  1160.39
927.76  1158.61
928.06  1155.95
928.36  1155.65
928.67  1156.13
```

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1726  928.97  1153.76
1727  929.28  1154.23
1728  929.58  1155.42
1729  929.89  1155.42
1730  930.19  1153.11
1731  930.50  1151.69
1732  930.80  1150.03
1733  931.10  1148.96
1734  931.41  1151.81
1735  931.71  1150.33
1736  932.02  1150.92
1737  932.32  1151.86
1738  932.62  1152.28
1739  932.93  1150.38
1740  933.23  1150.56
1741  933.53  1149.08
1742  933.84  1150.09
1743  934.14  1149.79
1744  934.44  1149.50
1745  934.75  1148.61
1746  935.05  1149.50
1747  935.36  1149.97
1748  935.66  1148.96
1749  935.96  1149.91
1750  936.27  1149.85
1751  936.57  1148.96
1752  936.87  1148.73
1753  937.17  1148.85
1754  937.48  1148.25
1755  937.78  1147.78
1756  938.08  1147.84
1757  938.39  1147.72
1758  938.69  1147.19
1759  938.99  1146.71
1760  939.29  1145.29
1761  939.60  1143.70
1762  939.90  1143.46
1763  940.20  1145.06
1764  940.51  1148.02
1765  940.81  1149.26
1766  941.11  1148.85
1767  941.41  1151.04
1768  941.71  1151.33
1769  942.02  1150.44
1770  942.32  1150.09
1771  942.62  1150.86
1772  942.92  1150.03
```

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1773  943.23  1150.50
1774  943.53  1150.27
1775  943.83  1150.21
1776  944.13  1150.80
1777  944.43  1150.15
1778  944.74  1150.33
1779  945.04  1151.33
1780  945.34  1152.87
1781  945.64  1152.22
1782  945.94  1153.88
1783  946.24  1153.58
1784  946.55  1154.00
1785  946.85  1154.41
1786  947.15  1155.42
1787  947.45  1154.06
1788  947.75  1152.58
1789  948.05  1151.15
1790  948.35  1149.73
1791  948.66  1149.38
1792  948.96  1150.27
1793  949.26  1147.54
1794  949.56  1147.25
1795  949.86  1147.72
1796  950.16  1150.09
1797  950.46  1151.33
1798  950.76  1150.38
1799  951.06  1149.73
1800  951.36  1149.38
1801  951.66  1150.38
1802  951.97  1153.05
1803  952.27  1151.21
1804  952.57  1152.10
1805  952.87  1150.68
1806  953.17  1150.68
1807  953.47  1151.57
1808  953.77  1151.39
1809  954.07  1151.92
1810  954.37  1152.46
1811  954.67  1154.17
1812  954.97  1154.47
1813  955.27  1156.13
1814  955.57  1156.96
1815  955.87  1157.73
1816  956.17  1157.19
1817  956.47  1157.02
1818  956.77  1158.55
1819  957.07  1158.32
```

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1820  957.37  1158.14
1821  957.67  1155.71
1822  957.97  1154.53
1823  958.27  1154.88
1824  958.57  1153.11
1825  958.87  1153.64
1826  959.17  1152.16
1827  959.47  1153.05
1828  959.76  1153.82
1829  960.06  1155.06
1830  960.36  1157.07
1831  960.66  1156.36
1832  960.96  1156.84
1833  961.26  1156.07
1834  961.56  1156.30
1835  961.86  1156.72
1836  962.16  1158.20
1837  962.46  1157.61
1838  962.76  1157.19
1839  963.05  1158.08
1840  963.35  1157.49
1841  963.65  1157.25
1842  963.95  1155.95
1843  964.25  1153.76
1844  964.55  1156.13
1845  964.85  1154.94
1846  965.14  1154.65
1847  965.44  1153.88
1848  965.74  1153.82
1849  966.04  1155.48
1850  966.34  1154.06
1851  966.64  1155.42
1852  966.93  1152.81
1853  967.23  1153.52
1854  967.53  1155.06
1855  967.83  1155.42
1856  968.13  1155.30
1857  968.42  1152.93
1858  968.72  1153.17
1859  969.02  1151.27
1860  969.32  1152.46
1861  969.62  1154.06
1862  969.91  1156.36
1863  970.21  1157.07
1864  970.51  1154.82
1865  970.81  1155.54
1866  971.10  1153.11
```

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1867  971.40  1153.70
1868  971.70  1152.52
1869  972.00  1152.40
1870  972.29  1151.86
1871  972.59  1152.04
1872  972.89  1154.06
1873  973.18  1153.58
1874  973.48  1153.11
1875  973.78  1155.30
1876  974.08  1155.89
1877  974.37  1155.59
1878  974.67  1154.29
1879  974.97  1153.70
1880  975.26  1154.41
1881  975.56  1156.48
1882  975.86  1156.36
1883  976.15  1157.13
1884  976.45  1158.08
1885  976.75  1158.26
1886  977.04  1158.73
1887  977.34  1159.80
1888  977.63  1160.98
1889  977.93  1161.87
1890  978.23  1159.98
1891  978.52  1160.63
1892  978.82  1160.03
1893  979.12  1161.93
1894  979.41  1161.22
1895  979.71  1160.45
1896  980.00  1161.63
1897  980.30  1160.33
1898  980.60  1161.69
1899  980.89  1162.46
1900  981.19  1164.42
1901  981.48  1162.40
1902  981.78  1160.92
1903  982.07  1160.63
1904  982.37  1160.09
1905  982.67  1159.62
1906  982.96  1161.46
1907  983.26  1161.40
1908  983.55  1162.40
1909  983.85  1161.81
1910  984.14  1161.16
1911  984.44  1161.63
1912  984.73  1163.11
1913  985.03  1161.99
```

1914	985.32	1162.05
1915	985.62	1163.29
1916	985.91	1160.09
1917	986.21	1160.21
1918	986.50	1159.62
1919	986.80	1160.69
1920	987.09	1161.51
1921	987.39	1162.17
1922	987.68	1161.51
1923	987.98	1161.57
1924	988.27	1161.22
1925	988.56	1162.76
1926	988.86	1161.16
1927	989.15	1160.86
1928	989.45	1161.63
1929	989.74	1160.92
1930	990.04	1160.09
1931	990.33	1161.40
1932	990.62	1161.69
1933	990.92	1161.69
1934	991.21	1162.11
1935	991.51	1161.81
1936	991.80	1160.98
1937	992.09	1160.15
1938	992.39	1162.34
1939	992.68	1161.51
1940	992.98	1161.22
1941	993.27	1159.32
1942	993.56	1159.68
1943	993.86	1157.96
1944	994.15	1157.37
1945	994.44	1159.86
1946	994.74	1161.57
1947	995.03	1159.21
1948	995.32	1159.68
1949	995.62	1158.79
1950	995.91	1160.27
1951	996.20	1161.04
1952	996.50	1160.51
1953	996.79	1158.38
1954	997.08	1158.50
1955	997.38	1158.73
1956	997.67	1158.44
1957	997.96	1161.75
1958	998.25	1161.87
1959	998.55	1160.98
1960	998.84	1161.10

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1961  999.13  1160.27
1962  999.42  1158.97
1963  999.72  1158.44
1964  1000.01 1157.43
1965  1000.30 1158.91
1966  1000.59 1157.37
1967  1000.89 1158.97
1968  1001.18 1158.50
1969  1001.47 1156.60
1970  1001.76 1156.30
1971  1002.06 1155.30
1972  1002.35 1155.48
1973  1002.64 1156.96
1974  1002.93 1157.37
1975  1003.22 1156.96
1976  1003.52 1156.54
1977  1003.81 1157.43
1978  1004.10 1157.25
1979  1004.39 1156.25
1980  1004.68 1156.54
1981  1004.97 1157.02
1982  1005.27 1158.50
1983  1005.56 1157.73
1984  1005.85 1159.44
1985  1006.14 1160.98
1986  1006.43 1159.50
1987  1006.72 1160.45
1988  1007.01 1162.17
1989  1007.31 1164.47
1990  1007.60 1164.77
1991  1007.89 1165.13
1992  1008.18 1165.36
1993  1008.47 1166.55
1994  1008.76 1167.14
1995  1009.05 1167.67
1996  1009.34 1169.51
1997  1009.63 1168.68
1998  1009.93 1167.20
1999  1010.22 1167.79
2000  1010.51 1170.22
2001  1010.80 1171.52
2002  1011.09 1172.17
2003  1011.38 1170.34
2004  1011.67 1170.39
2005  1011.96 1170.51
2006  1012.25 1168.74
2007  1012.54 1167.32
```

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2008  1012.83 1165.78
2009  1013.12 1165.13
2010  1013.41 1165.42
2011  1013.70 1166.61
2012  1013.99 1168.68
2013  1014.28 1167.43
2014  1014.57 1166.55
2015  1014.86 1166.37
2016  1015.15 1166.49
2017  1015.44 1167.26
2018  1015.73 1167.08
2019  1016.02 1165.54
2020  1016.31 1164.53
2021  1016.60 1163.05
2022  1016.89 1162.40
2023  1017.18 1160.63
2024  1017.47 1163.41
2025  1017.76 1164.42
2026  1018.05 1162.52
2027  1018.34 1162.17
2028  1018.62 1161.63
2029  1018.91 1160.86
2030  1019.20 1159.56
2031  1019.49 1158.79
2032  1019.78 1157.49
2033  1020.07 1158.26
2034  1020.36 1157.02
2035  1020.65 1156.36
2036  1020.94 1155.48
2037  1021.23 1152.99
2038  1021.51 1172.35
2039  1021.80 1092.25
2040  1022.09 1097.10
2041  1022.38 1070.40
2042  1022.67 1169.92
2043  1022.96 1109.24
2044  1023.25 1220.89
2045  1023.53 1194.19
2046  1023.82 1157.78
2047  1024.11 1155.36
2048  1024.40 1152.93
2049  1024.69 1143.22
2050  1024.97 1157.78
2051  1025.26 1145.65
2052  1025.55 1162.64
2053  1025.84 1201.48
2054  1026.13 1116.52
```

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2055  1026.41 1131.09
2056  1026.70 1131.09
2057  1026.99 1131.09
```

Listing 11: 汞灯时序

```
 1  Trend type: Single 波长 at 339.35
 2  积分时间（毫秒）: 10 (USB2+H14105)
 3  平均次数: 15 (USB2+H14105)
 4  平滑度: 0 (USB2+H14105)
 5  去除暗噪声: 否 (USB2+H14105)
 6  10:41:14.680    801.31
 7  10:41:14.830    811.99
 8  10:41:14.980    802.60
 9  10:41:15.130    824.45
10  10:41:15.280    827.84
11  10:41:15.430    822.18
12  10:41:15.580    820.08
13  10:41:15.730    823.80
14  10:41:15.880    817.33
15  10:41:16.030    806.32
16  10:41:16.180    808.26
17  10:41:16.329    805.51
18  10:41:16.479    819.43
19  10:41:16.629    815.87
20  10:41:16.779    817.65
21  10:41:16.929    829.79
22  10:41:17.079    826.23
23  10:41:17.229    821.05
24  10:41:17.379    802.60
25  10:41:17.529    824.77
26  10:41:17.679    810.37
27  10:41:17.829    828.33
28  10:41:17.979    813.77
29  10:41:18.129    802.44
30  10:41:18.279    828.33
31  10:41:18.429    816.84
32  10:41:18.579    831.57
33  10:41:18.729    819.27
34  10:41:18.879    809.40
35  10:41:19.029    825.26
36  10:41:19.179    806.00
37  10:41:19.329    809.07
38  10:41:19.479    806.16
39  10:41:19.629    817.97
40  10:41:19.778    818.94
41  10:41:19.928    811.34
```

```
42  10:41:20.078    809.72
43  10:41:20.228    817.33
44  10:41:20.378    799.69
45  10:41:20.528    825.42
46  10:41:20.678    795.81
47  10:41:20.828    822.67
48  10:41:20.978    802.76
49  10:41:21.128    804.71
50  10:41:21.278    812.63
51  10:41:21.428    827.52
52  10:41:21.577    815.38
53  10:41:21.727    812.96
54  10:41:21.877    838.69
55  10:41:22.027    832.21
56  10:41:22.177    806.49
57  10:41:22.327    808.26
58  10:41:22.477    826.06
59  10:41:22.627    825.74
60  10:41:22.777    826.87
61  10:41:22.927    813.60
62  10:41:23.077    825.26
63  10:41:23.227    815.38
64  10:41:23.377    823.48
65  10:41:23.527    826.23
66  10:41:23.677    822.34
67  10:41:23.827    829.79
68  10:41:23.977    815.38
69  10:41:24.127    816.03
70  10:41:24.276    816.68
71  10:41:24.436    817.81
72  10:41:24.576    825.42
73  10:41:24.726    828.98
74  10:41:24.886    836.10
75  10:41:25.036    827.20
76  10:41:25.186    833.02
77  10:41:25.336    828.49
78  10:41:25.486    812.96
79  10:41:25.636    817.81
80  10:41:25.786    823.48
81  10:41:25.936    837.23
82  10:41:26.086    828.49
83  10:41:26.235    833.99
84  10:41:26.385    832.05
85  10:41:26.535    817.33
86  10:41:26.685    818.62
87  10:41:26.835    828.98
88  10:41:26.985    840.79
```

89	10:41:27.135	847.42
90	10:41:27.285	831.57
91	10:41:27.435	829.30
92	10:41:27.585	838.04
93	10:41:27.735	815.71
94	10:41:27.885	813.93
95	10:41:28.035	824.93
96	10:41:28.185	823.15
97	10:41:28.335	826.23
98	10:41:28.485	838.52
99	10:41:28.635	828.01
100	10:41:28.784	828.49
101	10:41:28.934	822.50
102	10:41:29.084	820.24
103	10:41:29.234	829.30
104	10:41:29.384	829.30
105	10:41:29.534	831.40
106	10:41:29.684	813.77
107	10:41:29.834	854.87
108	10:41:29.984	842.89
109	10:41:30.134	840.30
110	10:41:30.284	832.54
111	10:41:30.434	838.04
112	10:41:30.584	829.30
113	10:41:30.734	846.78
114	10:41:30.883	831.40
115	10:41:31.033	834.80
116	10:41:31.183	832.21
117	10:41:31.333	830.27
118	10:41:31.484	837.88
119	10:41:31.634	832.05
120	10:41:31.784	843.06
121	10:41:31.933	827.52
122	10:41:32.083	832.05
123	10:41:32.233	844.03
124	10:41:32.383	850.66
125	10:41:32.533	846.45
126	10:41:32.683	824.77
127	10:41:32.834	818.30
128	10:41:32.983	832.86
129	10:41:33.133	822.99
130	10:41:33.283	829.95
131	10:41:33.433	839.01
132	10:41:33.583	814.41
133	10:41:33.733	839.66
134	10:41:33.883	829.30
135	10:41:34.033	815.55

```
136  10:41:34.183    830.11
137  10:41:34.333    840.30
138  10:41:34.483    833.99
139  10:41:34.633    840.14
140  10:41:34.782    832.54
141  10:41:34.932    833.35
142  10:41:35.082    817.49
143  10:41:35.232    820.56
144  10:41:35.382    841.28
145  10:41:35.532    836.26
146  10:41:35.682    838.20
147  10:41:35.832    827.52
148  10:41:35.982    810.85
149  10:41:36.132    841.44
150  10:41:36.282    836.58
151  10:41:36.432    813.12
152  10:41:36.582    820.24
153  10:41:36.732    828.65
154  10:41:36.882    826.06
155  10:41:37.032    823.80
156  10:41:37.182    823.31
157  10:41:37.332    823.80
158  10:41:37.482    817.65
159  10:41:37.632    823.31
160  10:41:37.782    822.02
161  10:41:37.932    811.82
162  10:41:38.084    820.89
163  10:41:38.232    813.12
164  10:41:38.382    835.29
165  10:41:38.532    814.09
166  10:41:38.682    825.09
167  10:41:38.832    830.11
168  10:41:38.980    836.74
169  10:41:39.130    835.29
170  10:41:39.280    840.47
171  10:41:39.430    825.74
172  10:41:39.580    828.98
173  10:41:39.730    834.48
174  10:41:39.880    818.94
175  10:41:40.030    824.12
176  10:41:40.180    817.49
177  10:41:40.330    814.25
178  10:41:40.480    815.06
179  10:41:40.630    830.27
180  10:41:40.779    821.05
181  10:41:40.929    828.98
182  10:41:41.079    808.75
```

183	10:41:41.229	823.48
184	10:41:41.379	825.74
185	10:41:41.529	821.21
186	10:41:41.679	827.52
187	10:41:41.829	821.86
188	10:41:41.979	824.77
189	10:41:42.129	821.86
190	10:41:42.279	825.90
191	10:41:42.429	833.99
192	10:41:42.578	830.92
193	10:41:42.728	806.32
194	10:41:42.879	837.39
195	10:41:43.029	830.76
196	10:41:43.179	825.09
197	10:41:43.328	829.79
198	10:41:43.478	828.17
199	10:41:43.628	817.97
200	10:41:43.778	829.14
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202	10:41:44.078	824.77
203	10:41:44.228	819.75
204	10:41:44.378	828.33
205	10:41:44.528	828.49
206	10:41:44.678	841.44
207	10:41:44.828	830.92
208	10:41:44.978	829.62
209	10:41:45.128	830.92
210	10:41:45.277	820.89
211	10:41:45.427	829.79
212	10:41:45.577	835.13
213	10:41:45.727	838.04
214	10:41:45.877	830.11
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216	10:41:46.177	830.60
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218	10:41:46.477	837.07
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220	10:41:46.777	860.21
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223	10:41:47.227	862.15
224	10:41:47.377	855.68
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318	10:42:01.483	902.76
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323	10:42:02.232	917.49

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726	10:43:02.682	833.35
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1243	10:44:20.229	836.91
1244	10:44:20.379	847.10
1245	10:44:20.529	847.75
1246	10:44:20.679	846.78
1247	10:44:20.829	855.03
1248	10:44:20.979	828.01
1249	10:44:21.129	831.73
1250	10:44:21.279	834.64
1251	10:44:21.429	835.94
1252	10:44:21.579	833.67
1253	10:44:21.729	828.49
1254	10:44:21.878	832.54
1255	10:44:22.028	853.09
1256	10:44:22.178	839.17
1257	10:44:22.328	837.39
1258	10:44:22.478	842.25
1259	10:44:22.628	832.70
1260	10:44:22.778	837.88
1261	10:44:22.928	846.13
1262	10:44:23.081	850.34
1263	10:44:23.226	866.68

```
1264  10:44:23.376    863.28
1265  10:44:23.526    876.87
1266  10:44:23.676    892.09
1267  10:44:23.826    913.44
1268  10:44:23.976    928.01
1269  10:44:24.126    911.66
1270  10:44:24.276    936.26
1271  10:44:24.426    944.67
1272  10:44:24.576    971.05
1273  10:44:24.726    949.85
1274  10:44:24.876    972.83
1275  10:44:25.026    964.09
1276  10:44:25.176    991.76
1277  10:44:25.325    988.85
1278  10:44:25.475    983.02
1279  10:44:25.625    972.35
```

Listing 12: 钠灯时序

```
 1  10:51:22.735    808.91
 2  10:51:22.884    809.07
 3  10:51:23.034    817.33
 4  10:51:23.184    827.68
 5  10:51:23.334    816.36
 6  10:51:23.484    837.72
 7  10:51:23.634    810.69
 8  10:51:23.784    818.46
 9  10:51:23.934    809.40
10  10:51:24.084    820.56
11  10:51:24.234    828.65
12  10:51:24.384    818.30
13  10:51:24.534    815.71
14  10:51:24.684    807.46
15  10:51:24.833    810.04
16  10:51:24.983    816.52
17  10:51:25.133    828.17
18  10:51:25.283    793.54
19  10:51:25.433    808.10
20  10:51:25.585    802.12
21  10:51:25.735    803.90
22  10:51:25.885    801.15
23  10:51:26.035    808.91
24  10:51:26.185    812.47
25  10:51:26.335    803.41
26  10:51:26.484    814.41
27  10:51:26.634    826.39
28  10:51:26.784    798.23
```

29	10:51:26.934	813.28
30	10:51:27.084	801.47
31	10:51:27.234	812.63
32	10:51:27.384	808.43
33	10:51:27.534	823.31
34	10:51:27.684	797.42
35	10:51:27.834	818.14
36	10:51:27.984	807.94
37	10:51:28.134	798.72
38	10:51:28.284	815.71
39	10:51:28.433	790.63
40	10:51:28.583	812.96
41	10:51:28.733	803.73
42	10:51:28.883	811.18
43	10:51:29.033	809.56
44	10:51:29.183	822.83
45	10:51:29.333	823.64
46	10:51:29.483	818.30
47	10:51:29.633	805.68
48	10:51:29.783	797.10
49	10:51:29.932	806.49
50	10:51:30.082	819.27
51	10:51:30.232	815.55
52	10:51:30.382	815.71
53	10:51:30.532	814.74
54	10:51:30.682	805.51
55	10:51:30.832	816.36
56	10:51:30.982	812.80
57	10:51:31.132	807.13
58	10:51:31.282	833.18
59	10:51:31.432	809.07
60	10:51:31.582	823.80
61	10:51:31.732	811.02
62	10:51:31.881	813.60
63	10:51:32.031	805.35
64	10:51:32.181	808.91
65	10:51:32.331	821.37
66	10:51:32.481	804.38
67	10:51:32.631	807.78
68	10:51:32.781	811.18
69	10:51:32.931	802.93
70	10:51:33.081	821.70
71	10:51:33.231	801.47
72	10:51:33.381	815.55
73	10:51:33.531	826.23
74	10:51:33.680	809.56
75	10:51:33.830	811.82

76	10:51:33.980	814.74
77	10:51:34.130	801.63
78	10:51:34.280	813.93
79	10:51:34.431	799.53
80	10:51:34.581	802.60
81	10:51:34.731	839.33
82	10:51:34.881	808.91
83	10:51:35.030	811.18
84	10:51:35.180	815.87
85	10:51:35.330	812.47
86	10:51:35.480	806.16
87	10:51:35.630	810.04
88	10:51:35.780	814.90
89	10:51:35.930	815.55
90	10:51:36.080	807.62
91	10:51:36.230	817.65
92	10:51:36.380	815.87
93	10:51:36.530	821.86
94	10:51:36.680	799.53
95	10:51:36.829	806.00
96	10:51:36.979	795.00
97	10:51:37.129	811.99
98	10:51:37.279	822.83
99	10:51:37.429	821.05
100	10:51:37.579	810.69
101	10:51:37.729	796.45
102	10:51:37.879	820.24
103	10:51:38.029	812.80
104	10:51:38.179	812.96
105	10:51:38.329	815.87
106	10:51:38.479	818.46
107	10:51:38.629	819.43
108	10:51:38.778	812.63
109	10:51:38.928	802.28
110	10:51:39.078	803.90
111	10:51:39.236	796.94
112	10:51:39.378	817.65
113	10:51:39.528	818.14
114	10:51:39.678	809.40
115	10:51:39.828	821.70
116	10:51:39.978	825.42
117	10:51:40.128	799.37
118	10:51:40.278	811.99
119	10:51:40.428	805.03
120	10:51:40.578	813.12
121	10:51:40.728	808.10
122	10:51:40.877	817.16

123	10:51:41.027	827.84
124	10:51:41.177	828.01
125	10:51:41.327	801.63
126	10:51:41.477	800.17
127	10:51:41.637	803.25
128	10:51:41.787	816.52
129	10:51:41.937	824.12
130	10:51:42.087	826.39
131	10:51:42.237	803.73
132	10:51:42.387	804.22
133	10:51:42.537	807.94
134	10:51:42.687	795.64
135	10:51:42.836	798.39
136	10:51:42.986	809.40
137	10:51:43.136	819.11
138	10:51:43.286	810.04
139	10:51:43.436	819.92
140	10:51:43.586	801.31
141	10:51:43.736	806.65
142	10:51:43.886	810.69
143	10:51:44.036	813.77
144	10:51:44.186	796.13
145	10:51:44.336	802.60
146	10:51:44.485	809.88
147	10:51:44.635	807.94
148	10:51:44.785	817.97
149	10:51:44.935	794.83
150	10:51:45.085	812.96
151	10:51:45.235	804.22
152	10:51:45.385	790.14
153	10:51:45.535	817.16
154	10:51:45.685	812.31
155	10:51:45.835	811.50
156	10:51:45.985	801.47
157	10:51:46.135	797.10
158	10:51:46.285	801.15
159	10:51:46.434	806.97
160	10:51:46.584	810.85
161	10:51:46.734	817.49
162	10:51:46.884	820.89
163	10:51:47.034	818.30
164	10:51:47.184	815.38
165	10:51:47.334	804.22
166	10:51:47.484	815.71
167	10:51:47.634	820.40
168	10:51:47.784	810.69
169	10:51:47.934	811.02

170	10:51:48.084	808.43
171	10:51:48.233	809.72
172	10:51:48.383	803.90
173	10:51:48.533	808.59
174	10:51:48.683	808.75
175	10:51:48.833	814.09
176	10:51:48.983	805.35
177	10:51:49.133	826.39
178	10:51:49.283	821.05
179	10:51:49.433	819.59
180	10:51:49.583	822.67
181	10:51:49.733	794.35
182	10:51:49.883	812.63
183	10:51:50.033	816.36
184	10:51:50.183	815.55
185	10:51:50.333	796.61
186	10:51:50.483	821.21
187	10:51:50.633	802.12
188	10:51:50.783	809.24
189	10:51:50.933	808.26
190	10:51:51.083	827.20
191	10:51:51.233	795.32
192	10:51:51.383	809.07
193	10:51:51.533	819.59
194	10:51:51.683	812.80
195	10:51:51.833	822.02
196	10:51:51.983	809.24
197	10:51:52.133	802.76
198	10:51:52.283	798.56
199	10:51:52.433	823.96
200	10:51:52.582	815.55
201	10:51:52.732	812.80
202	10:51:52.882	819.11
203	10:51:53.032	813.28
204	10:51:53.182	810.04
205	10:51:53.332	797.42
206	10:51:53.482	816.36
207	10:51:53.632	816.03
208	10:51:53.782	824.61
209	10:51:53.932	813.28
210	10:51:54.082	817.97
211	10:51:54.232	793.70
212	10:51:54.382	809.72
213	10:51:54.531	791.76
214	10:51:54.681	798.56
215	10:51:54.832	815.22
216	10:51:54.982	808.59

217	10:51:55.132	808.59
218	10:51:55.281	819.43
219	10:51:55.431	806.97
220	10:51:55.581	810.37
221	10:51:55.732	808.26
222	10:51:55.882	828.17
223	10:51:56.032	800.66
224	10:51:56.182	807.62
225	10:51:56.331	816.36
226	10:51:56.481	806.81
227	10:51:56.631	810.04
228	10:51:56.781	817.81
229	10:51:56.931	791.44
230	10:51:57.081	811.18
231	10:51:57.231	817.33
232	10:51:57.381	820.89
233	10:51:57.531	810.37
234	10:51:57.681	820.24
235	10:51:57.831	794.35
236	10:51:57.980	804.54
237	10:51:58.130	816.84
238	10:51:58.280	807.46
239	10:51:58.430	822.18
240	10:51:58.580	812.31
241	10:51:58.730	815.06
242	10:51:58.880	805.03
243	10:51:59.030	811.34
244	10:51:59.180	819.92
245	10:51:59.330	797.10
246	10:51:59.480	819.59
247	10:51:59.630	806.32
248	10:51:59.784	804.87
249	10:51:59.930	791.44
250	10:52:00.080	806.81
251	10:52:00.230	810.21
252	10:52:00.380	816.84
253	10:52:00.530	807.94
254	10:52:00.680	821.37
255	10:52:00.830	794.35
256	10:52:00.980	802.28
257	10:52:01.130	811.02
258	10:52:01.279	813.12
259	10:52:01.429	821.21
260	10:52:01.579	814.90
261	10:52:01.729	817.81
262	10:52:01.879	809.24
263	10:52:02.029	806.97

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264  10:52:02.179    810.85
265  10:52:02.329    801.79
266  10:52:02.479    814.58
267  10:52:02.629    796.45
268  10:52:02.779    812.80
269  10:52:02.929    805.19
270  10:52:03.078    801.95
271  10:52:03.228    807.94
272  10:52:03.378    818.62
273  10:52:03.528    820.40
274  10:52:03.678    812.47
275  10:52:03.828    818.14
276  10:52:03.978    798.88
277  10:52:04.128    815.06
278  10:52:04.278    818.46
279  10:52:04.428    806.97
280  10:52:04.578    815.38
281  10:52:04.728    816.68
282  10:52:04.878    790.63
283  10:52:05.028    821.21
284  10:52:05.178    828.82
285  10:52:05.328    808.75
286  10:52:05.478    813.60
287  10:52:05.628    802.12
288  10:52:05.778    810.21
289  10:52:05.928    802.60
290  10:52:06.078    799.85
291  10:52:06.228    816.68
292  10:52:06.378    808.91
293  10:52:06.527    815.55
294  10:52:06.677    782.70
295  10:52:06.827    809.40
296  10:52:06.977    818.14
297  10:52:07.127    792.57
298  10:52:07.277    822.83
299  10:52:07.427    812.15
300  10:52:07.577    802.12
301  10:52:07.727    808.91
302  10:52:07.877    813.60
303  10:52:08.027    811.02
304  10:52:08.177    817.00
305  10:52:08.327    814.41
306  10:52:08.477    791.76
307  10:52:08.626    803.73
308  10:52:08.776    799.20
309  10:52:08.926    799.69
310  10:52:09.076    833.99
```

```
311  10:52:09.226    802.60
312  10:52:09.376    795.64
313  10:52:09.526    803.25
314  10:52:09.676    802.28
315  10:52:09.826    814.58
316  10:52:09.976    808.43
317  10:52:10.126    804.54
318  10:52:10.276    810.37
319  10:52:10.426    819.59
320  10:52:10.576    798.23
321  10:52:10.726    807.94
322  10:52:10.876    812.47
323  10:52:11.026    797.42
324  10:52:11.176    803.41
325  10:52:11.326    805.03
326  10:52:11.476    808.43
327  10:52:11.633    801.47
328  10:52:11.783    795.48
329  10:52:11.933    799.53
330  10:52:12.083    819.59
331  10:52:12.232    817.33
332  10:52:12.382    804.54
333  10:52:12.532    806.00
334  10:52:12.682    802.44
335  10:52:12.832    817.97
336  10:52:12.982    813.28
337  10:52:13.132    789.98
338  10:52:13.282    808.43
339  10:52:13.432    821.05
340  10:52:13.581    795.16
341  10:52:13.731    813.60
342  10:52:13.881    821.05
343  10:52:14.031    804.22
344  10:52:14.181    807.78
345  10:52:14.331    817.49
346  10:52:14.481    806.49
347  10:52:14.627    814.25
348  10:52:14.777    816.84
349  10:52:14.927    811.66
350  10:52:15.074    826.06
351  10:52:15.224    801.15
352  10:52:15.374    810.69
353  10:52:15.524    811.18
354  10:52:15.684    803.41
355  10:52:15.833    797.10
356  10:52:15.978    810.21
357  10:52:16.128    813.28
```

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10:52:16.277    809.24
10:52:16.428    824.12
10:52:16.577    819.11
10:52:16.727    804.38
10:52:16.877    807.46
10:52:17.027    824.77
10:52:17.177    799.85
10:52:17.327    800.82
10:52:17.477    819.59
10:52:17.627    819.27
10:52:17.777    816.68
10:52:17.927    810.37
10:52:18.077    814.09
10:52:18.227    815.55
10:52:18.377    811.34
10:52:18.527    813.28
10:52:18.677    797.26
10:52:18.826    831.73
10:52:18.976    819.92
10:52:19.126    801.63
10:52:19.276    812.31
10:52:19.426    795.16
10:52:19.576    803.57
10:52:19.726    814.58
10:52:19.876    814.25
10:52:20.026    820.89
10:52:20.176    803.90
10:52:20.326    819.75
10:52:20.476    803.25
10:52:20.626    814.74
10:52:20.776    810.69
10:52:20.925    802.44
10:52:21.075    802.93
10:52:21.225    810.37
10:52:21.375    816.84
10:52:21.526    813.28
10:52:21.675    811.34
10:52:21.825    817.97
10:52:21.975    810.21
10:52:22.125    796.61
10:52:22.275    808.10
10:52:22.425    815.22
10:52:22.575    817.81
10:52:22.725    790.14
10:52:22.875    826.55
10:52:23.025    803.41
10:52:23.175    819.59
```

405	10:52:23.325	833.83
406	10:52:23.475	809.24
407	10:52:23.625	796.13
408	10:52:23.775	818.46
409	10:52:23.925	808.75
410	10:52:24.075	816.52
411	10:52:24.224	829.30
412	10:52:24.374	816.03
413	10:52:24.524	810.85
414	10:52:24.674	802.60
415	10:52:24.824	811.34
416	10:52:24.975	828.17
417	10:52:25.132	825.09
418	10:52:25.282	830.27
419	10:52:25.432	823.48
420	10:52:25.582	831.24
421	10:52:25.732	830.92
422	10:52:25.882	825.42
423	10:52:26.032	830.43
424	10:52:26.182	818.46
425	10:52:26.332	824.45
426	10:52:26.482	831.57
427	10:52:26.632	821.21
428	10:52:26.781	844.19
429	10:52:26.931	825.42
430	10:52:27.081	825.42
431	10:52:27.231	817.00
432	10:52:27.381	821.21
433	10:52:27.531	830.92
434	10:52:27.681	826.71
435	10:52:27.831	823.64
436	10:52:27.981	829.46
437	10:52:28.131	823.48
438	10:52:28.281	822.18
439	10:52:28.431	824.12
440	10:52:28.581	820.08
441	10:52:28.731	824.61
442	10:52:28.881	833.35
443	10:52:29.031	816.68
444	10:52:29.181	838.52
445	10:52:29.331	818.78
446	10:52:29.481	813.93
447	10:52:29.631	838.04
448	10:52:29.781	823.64
449	10:52:29.931	829.14
450	10:52:30.081	818.62
451	10:52:30.231	818.62

452	10:52:30.381	810.85
453	10:52:30.530	816.03
454	10:52:30.680	829.14
455	10:52:30.830	833.51
456	10:52:30.980	832.86
457	10:52:31.130	822.67
458	10:52:31.279	824.77
459	10:52:31.429	835.45
460	10:52:31.579	811.34
461	10:52:31.729	822.02
462	10:52:31.879	824.28
463	10:52:32.029	834.16
464	10:52:32.179	825.42
465	10:52:32.329	833.83
466	10:52:32.479	822.02
467	10:52:32.628	823.80
468	10:52:32.778	834.48
469	10:52:32.928	823.64
470	10:52:33.078	822.18
471	10:52:33.228	817.97
472	10:52:33.378	814.09
473	10:52:33.528	832.86
474	10:52:33.678	828.49
475	10:52:33.828	827.36
476	10:52:33.978	815.55
477	10:52:34.128	816.03
478	10:52:34.278	838.20
479	10:52:34.427	821.53
480	10:52:34.577	833.83
481	10:52:34.727	811.50
482	10:52:34.881	832.05
483	10:52:35.028	830.60
484	10:52:35.174	810.04
485	10:52:35.324	802.44
486	10:52:35.474	805.35
487	10:52:35.624	825.74
488	10:52:35.774	817.00
489	10:52:35.924	828.17
490	10:52:36.074	836.42
491	10:52:36.224	813.93
492	10:52:36.373	819.75
493	10:52:36.523	833.83
494	10:52:36.673	837.88
495	10:52:36.823	822.34
496	10:52:36.973	822.02
497	10:52:37.123	828.17
498	10:52:37.273	829.14

499	10:52:37.423	827.04
500	10:52:37.573	836.26
501	10:52:37.722	830.27
502	10:52:37.882	822.83
503	10:52:38.032	832.70
504	10:52:38.182	825.90
505	10:52:38.332	824.93
506	10:52:38.482	839.01
507	10:52:38.632	827.52
508	10:52:38.781	832.38
509	10:52:38.931	810.21
510	10:52:39.081	811.82
511	10:52:39.231	834.16
512	10:52:39.381	817.00
513	10:52:39.531	823.96
514	10:52:39.681	825.74
515	10:52:39.831	805.84
516	10:52:39.981	828.65
517	10:52:40.131	833.35
518	10:52:40.281	841.11
519	10:52:40.431	833.99
520	10:52:40.581	836.91
521	10:52:40.731	820.24
522	10:52:40.881	828.49
523	10:52:41.031	830.92
524	10:52:41.181	831.08
525	10:52:41.331	837.07
526	10:52:41.480	828.17
527	10:52:41.630	821.70
528	10:52:41.780	834.64
529	10:52:41.930	822.50
530	10:52:42.080	828.01
531	10:52:42.230	826.06
532	10:52:42.380	830.76
533	10:52:42.530	831.24
534	10:52:42.680	830.76
535	10:52:42.830	826.87
536	10:52:42.980	828.49
537	10:52:43.130	814.74
538	10:52:43.280	818.30
539	10:52:43.422	817.00
540	10:52:43.572	808.75
541	10:52:43.722	822.99
542	10:52:43.882	835.13
543	10:52:44.022	825.26
544	10:52:44.172	828.33
545	10:52:44.332	836.58

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548  10:52:44.772    829.79
549  10:52:44.922    831.73
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551  10:52:45.228    819.27
552  10:52:45.378    811.66
553  10:52:45.528    823.80
554  10:52:45.677    830.27
555  10:52:45.827    820.56
556  10:52:45.974    827.68
557  10:52:46.124    816.68
558  10:52:46.273    828.65
559  10:52:46.423    827.68
560  10:52:46.573    840.63
561  10:52:46.723    825.74
562  10:52:46.873    829.62
563  10:52:47.023    831.24
564  10:52:47.174    836.10
565  10:52:47.324    822.67
566  10:52:47.475    831.40
567  10:52:47.625    820.72
568  10:52:47.775    833.51
569  10:52:47.925    832.38
570  10:52:48.074    837.88
571  10:52:48.224    827.36
572  10:52:48.374    837.55
573  10:52:48.524    838.04
574  10:52:48.674    829.14
575  10:52:48.824    820.89
576  10:52:48.974    831.89
577  10:52:49.124    822.18
578  10:52:49.274    829.30
579  10:52:49.424    823.31
580  10:52:49.574    834.48
581  10:52:49.724    833.67
582  10:52:49.873    846.78
583  10:52:50.024    820.56
584  10:52:50.174    815.38
585  10:52:50.324    811.82
586  10:52:50.473    831.08
587  10:52:50.623    839.50
588  10:52:50.773    824.61
589  10:52:50.923    828.33
590  10:52:51.073    824.45
591  10:52:51.223    837.39
592  10:52:51.373    818.94
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593	10:52:51.523	825.74
594	10:52:51.673	825.74
595	10:52:51.823	832.54
596	10:52:51.973	824.61
597	10:52:52.123	816.52
598	10:52:52.272	830.92
599	10:52:52.422	825.74
600	10:52:52.572	821.70
601	10:52:52.722	833.02
602	10:52:52.872	829.30
603	10:52:53.022	827.04
604	10:52:53.172	830.11
605	10:52:53.322	827.68
606	10:52:53.472	838.04
607	10:52:53.622	827.20
608	10:52:53.772	829.95
609	10:52:53.922	831.57
610	10:52:54.072	814.09
611	10:52:54.222	828.98
612	10:52:54.372	821.53
613	10:52:54.522	824.12
614	10:52:54.672	821.37
615	10:52:54.822	839.01
616	10:52:54.972	833.67
617	10:52:55.122	814.74
618	10:52:55.271	817.97
619	10:52:55.421	835.94
620	10:52:55.571	817.00
621	10:52:55.721	839.66
622	10:52:55.871	818.46
623	10:52:56.031	820.08
624	10:52:56.171	814.58
625	10:52:56.321	840.63
626	10:52:56.471	833.99
627	10:52:56.631	808.26
628	10:52:56.770	831.73
629	10:52:56.930	827.68
630	10:52:57.070	829.14
631	10:52:57.230	830.43
632	10:52:57.380	821.37
633	10:52:57.530	833.67
634	10:52:57.680	828.01
635	10:52:57.830	827.68
636	10:52:57.980	833.51
637	10:52:58.130	832.21
638	10:52:58.280	823.96
639	10:52:58.430	828.65

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10:52:58.729    827.04
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10:52:59.179    833.02
10:52:59.329    818.62
10:52:59.479    826.06
10:52:59.629    830.43
10:52:59.779    829.95
10:52:59.929    821.21
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10:53:00.230    842.25
10:53:00.380    843.86
10:53:00.530    821.53
10:53:00.680    841.60
10:53:00.829    808.59
10:53:00.979    821.37
10:53:01.130    834.48
10:53:01.279    838.69
10:53:01.429    845.97
10:53:01.579    814.09
10:53:01.729    839.66
10:53:01.879    830.60
10:53:02.029    831.57
10:53:02.179    822.18
10:53:02.329    832.21
10:53:02.479    831.24
10:53:02.629    822.99
10:53:02.779    838.69
10:53:02.929    823.31
10:53:03.079    836.91
10:53:03.229    819.59
10:53:03.378    814.58
10:53:03.528    827.20
10:53:03.678    826.39
10:53:03.828    812.31
10:53:03.978    833.51
10:53:04.128    815.06
10:53:04.278    831.40
10:53:04.428    811.82
10:53:04.578    834.64
10:53:04.728    828.82
10:53:04.880    826.23
10:53:05.020    828.17
10:53:05.179    825.09
10:53:05.329    828.65
10:53:05.479    820.56
```

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688  10:53:05.779    820.56
689  10:53:05.929    850.82
690  10:53:06.079    823.96
691  10:53:06.229    832.54
692  10:53:06.379    829.30
693  10:53:06.529    841.44
694  10:53:06.679    835.77
695  10:53:06.829    810.37
696  10:53:06.979    828.33
697  10:53:07.128    843.54
698  10:53:07.278    815.06
699  10:53:07.428    822.02
700  10:53:07.578    828.49
701  10:53:07.725    828.65
702  10:53:07.875    826.87
703  10:53:08.025    816.84
704  10:53:08.175    822.18
705  10:53:08.322    830.11
706  10:53:08.472    822.50
707  10:53:08.622    826.87
708  10:53:08.772    829.30
709  10:53:08.922    845.97
710  10:53:09.072    812.80
711  10:53:09.222    835.94
712  10:53:09.372    833.51
713  10:53:09.522    816.84
714  10:53:09.672    819.59
715  10:53:09.822    834.64
716  10:53:09.979    844.35
717  10:53:10.122    827.20
718  10:53:10.272    832.21
719  10:53:10.422    843.22
720  10:53:10.572    827.04
721  10:53:10.722    823.80
722  10:53:10.872    840.63
723  10:53:11.022    828.65
724  10:53:11.172    833.51
725  10:53:11.322    848.56
726  10:53:11.472    827.36
727  10:53:11.622    824.28
728  10:53:11.771    829.62
729  10:53:11.921    827.20
730  10:53:12.071    833.51
731  10:53:12.221    840.63
732  10:53:12.371    828.17
733  10:53:12.521    849.85
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735	10:53:12.821	848.88
736	10:53:12.971	831.73
737	10:53:13.121	844.67
738	10:53:13.271	844.84
739	10:53:13.421	828.01
740	10:53:13.571	820.89
741	10:53:13.720	830.27
742	10:53:13.870	811.82
743	10:53:14.020	834.16
744	10:53:14.170	828.01
745	10:53:14.320	838.85
746	10:53:14.470	832.70
747	10:53:14.620	830.60
748	10:53:14.770	836.10
749	10:53:14.920	831.24
750	10:53:15.070	831.40
751	10:53:15.220	838.20
752	10:53:15.370	829.62
753	10:53:15.520	815.87
754	10:53:15.670	825.58
755	10:53:15.820	823.96
756	10:53:15.970	839.82
757	10:53:16.120	807.46
758	10:53:16.270	827.68
759	10:53:16.419	837.23
760	10:53:16.569	840.95
761	10:53:16.719	817.81
762	10:53:16.869	841.11
763	10:53:17.019	830.76
764	10:53:17.170	823.31
765	10:53:17.319	840.79
766	10:53:17.470	839.33
767	10:53:17.620	828.65
768	10:53:17.770	833.51
769	10:53:17.920	851.63
770	10:53:18.070	833.35
771	10:53:18.220	831.24
772	10:53:18.370	842.25
773	10:53:18.520	822.34
774	10:53:18.669	838.85
775	10:53:18.820	831.40
776	10:53:18.970	833.99
777	10:53:19.121	835.61
778	10:53:19.271	832.86
779	10:53:19.421	822.34
780	10:53:19.571	827.52

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782  10:53:19.871    834.80
783  10:53:20.021    807.29
784  10:53:20.171    828.01
785  10:53:20.321    831.57
786  10:53:20.471    825.90
787  10:53:20.621    851.15
788  10:53:20.771    842.25
789  10:53:20.918    837.07
790  10:53:21.068    816.36
791  10:53:21.228    829.95
792  10:53:21.378    836.58
793  10:53:21.519    841.92
794  10:53:21.679    824.45
795  10:53:21.819    852.60
796  10:53:21.969    828.33
797  10:53:22.119    823.80
798  10:53:22.279    821.86
799  10:53:22.418    832.54
800  10:53:22.578    824.28
801  10:53:22.728    821.53
802  10:53:22.869    838.69
803  10:53:23.019    827.84
804  10:53:23.169    832.54
805  10:53:23.319    840.79
806  10:53:23.469    832.70
807  10:53:23.619    839.98
808  10:53:23.769    832.21
809  10:53:23.918    826.71
810  10:53:24.068    834.48
811  10:53:24.219    841.76
812  10:53:24.369    839.66
813  10:53:24.519    821.37
814  10:53:24.669    824.61
815  10:53:24.819    825.58
816  10:53:24.969    833.35
817  10:53:25.119    830.43
818  10:53:25.269    830.92
819  10:53:25.419    834.48
820  10:53:25.569    825.90
821  10:53:25.719    838.85
822  10:53:25.878    822.18
823  10:53:26.018    843.06
824  10:53:26.168    823.96
825  10:53:26.318    817.33
826  10:53:26.478    826.39
827  10:53:26.618    838.69
```

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10:53:27.527    825.58
10:53:27.677    823.31
10:53:27.827    843.86
10:53:27.977    828.82
10:53:28.127    829.46
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10:53:28.577    826.39
10:53:28.727    848.72
10:53:28.877    837.88
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10:53:29.176    829.79
10:53:29.326    821.21
10:53:29.476    835.29
10:53:29.626    831.73
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10:53:29.926    829.79
10:53:30.077    835.13
10:53:30.227    836.26
10:53:30.377    841.11
10:53:30.527    826.23
10:53:30.676    804.87
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10:53:30.977    852.12
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10:53:31.427    835.29
10:53:31.576    834.48
10:53:31.726    838.36
10:53:31.876    831.73
10:53:32.026    832.05
10:53:32.176    825.26
10:53:32.322    840.79
10:53:32.472    832.21
10:53:32.622    822.83
10:53:32.772    833.51
10:53:32.922    804.38
10:53:33.072    830.76
10:53:33.221    826.23
10:53:33.371    814.58
10:53:33.522    838.52
10:53:33.671    839.33
```

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876  10:53:33.972    825.09
877  10:53:34.118    831.73
878  10:53:34.268    841.44
879  10:53:34.418    821.37
880  10:53:34.568    831.08
881  10:53:34.718    827.04
882  10:53:34.868    828.82
883  10:53:35.018    833.51
884  10:53:35.169    825.42
885  10:53:35.319    821.21
886  10:53:35.469    828.65
887  10:53:35.618    833.99
888  10:53:35.768    844.84
889  10:53:35.918    832.38
890  10:53:36.068    839.17
891  10:53:36.218    837.55
892  10:53:36.368    843.70
893  10:53:36.518    830.60
894  10:53:36.668    823.48
895  10:53:36.818    850.82
896  10:53:36.968    838.69
897  10:53:37.128    828.65
898  10:53:37.278    834.16
899  10:53:37.417    847.10
900  10:53:37.567    845.00
901  10:53:37.717    828.01
902  10:53:37.877    839.82
903  10:53:38.027    836.10
904  10:53:38.177    838.52
905  10:53:38.327    831.89
906  10:53:38.477    841.28
907  10:53:38.627    821.53
908  10:53:38.777    822.02
909  10:53:38.927    829.62
910  10:53:39.068    826.06
911  10:53:39.218    823.48
912  10:53:39.368    831.89
913  10:53:39.517    842.73
914  10:53:39.667    841.92
915  10:53:39.817    834.48
916  10:53:39.967    846.78
917  10:53:40.118    828.17
918  10:53:40.267    837.07
919  10:53:40.417    833.35
920  10:53:40.567    821.21
921  10:53:40.717    836.91
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925	10:53:41.317	849.53
926	10:53:41.477	842.73
927	10:53:41.627	830.43
928	10:53:41.777	834.80
929	10:53:41.927	843.70
930	10:53:42.077	838.52
931	10:53:42.226	827.36
932	10:53:42.376	837.55
933	10:53:42.526	837.07
934	10:53:42.676	831.57
935	10:53:42.826	823.15
936	10:53:42.976	827.04
937	10:53:43.126	833.51
938	10:53:43.276	828.17
939	10:53:43.426	828.17
940	10:53:43.576	833.51
941	10:53:43.726	835.45
942	10:53:43.876	852.93
943	10:53:44.026	843.54
944	10:53:44.175	837.88
945	10:53:44.325	831.24
946	10:53:44.475	843.86
947	10:53:44.625	823.31
948	10:53:44.775	818.30
949	10:53:44.925	854.38
950	10:53:45.075	825.09
951	10:53:45.225	822.99
952	10:53:45.375	824.12
953	10:53:45.525	827.36
954	10:53:45.675	843.38
955	10:53:45.825	834.48
956	10:53:45.975	827.20
957	10:53:46.125	838.69
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959	10:53:46.425	828.65
960	10:53:46.575	832.54
961	10:53:46.725	846.78
962	10:53:46.875	830.43
963	10:53:47.025	845.16
964	10:53:47.174	857.13
965	10:53:47.324	843.70
966	10:53:47.474	849.69
967	10:53:47.624	837.39
968	10:53:47.774	841.44

969	10:53:47.924	828.98
970	10:53:48.074	841.76
971	10:53:48.224	830.76
972	10:53:48.374	862.31
973	10:53:48.524	835.13
974	10:53:48.674	831.89
975	10:53:48.824	839.33
976	10:53:48.973	831.40
977	10:53:49.123	828.17
978	10:53:49.273	828.01
979	10:53:49.423	836.91
980	10:53:49.573	841.92
981	10:53:49.723	845.97
982	10:53:49.873	849.37
983	10:53:50.023	824.45
984	10:53:50.173	828.49
985	10:53:50.323	842.41
986	10:53:50.473	829.62
987	10:53:50.623	843.22
988	10:53:50.773	833.18
989	10:53:50.923	842.25
990	10:53:51.073	838.04
991	10:53:51.223	813.60
992	10:53:51.373	822.18
993	10:53:51.523	841.44
994	10:53:51.673	837.39
995	10:53:51.823	823.15
996	10:53:51.973	833.67
997	10:53:52.123	829.14
998	10:53:52.273	825.58
999	10:53:52.423	830.11
1000	10:53:52.573	844.51
1001	10:53:52.723	820.40
1002	10:53:52.872	846.62
1003	10:53:53.022	845.64
1004	10:53:53.172	833.18
1005	10:53:53.322	835.29
1006	10:53:53.472	841.92
1007	10:53:53.622	845.81
1008	10:53:53.772	832.70
1009	10:53:53.922	826.39
1010	10:53:54.072	824.28
1011	10:53:54.222	832.86
1012	10:53:54.372	820.89
1013	10:53:54.522	830.11
1014	10:53:54.672	834.32
1015	10:53:54.821	833.35

1016	10:53:54.971	841.28
1017	10:53:55.122	832.38
1018	10:53:55.272	831.57
1019	10:53:55.422	823.96
1020	10:53:55.572	842.41
1021	10:53:55.722	844.51
1022	10:53:55.872	846.45
1023	10:53:56.022	830.76
1024	10:53:56.171	821.21
1025	10:53:56.321	828.98
1026	10:53:56.471	846.29
1027	10:53:56.621	812.96
1028	10:53:56.771	847.10
1029	10:53:56.921	831.57
1030	10:53:57.071	845.97
1031	10:53:57.221	830.60
1032	10:53:57.371	833.99
1033	10:53:57.521	825.42
1034	10:53:57.671	827.68
1035	10:53:57.821	824.61
1036	10:53:57.971	841.92
1037	10:53:58.120	826.71
1038	10:53:58.270	842.73
1039	10:53:58.420	839.17
1040	10:53:58.570	844.84
1041	10:53:58.720	826.39
1042	10:53:58.870	846.94
1043	10:53:59.020	814.74
1044	10:53:59.170	823.64
1045	10:53:59.320	835.61
1046	10:53:59.470	842.08
1047	10:53:59.620	820.72
1048	10:53:59.770	826.39
1049	10:53:59.919	828.17
1050	10:54:00.069	838.20
1051	10:54:00.219	811.82
1052	10:54:00.369	832.38
1053	10:54:00.519	840.14
1054	10:54:00.669	826.87
1055	10:54:00.819	837.55
1056	10:54:00.969	830.76
1057	10:54:01.119	838.04
1058	10:54:01.269	849.20
1059	10:54:01.419	829.79
1060	10:54:01.569	842.57
1061	10:54:01.719	834.96
1062	10:54:01.868	839.33

1063	10:54:02.018	833.51
1064	10:54:02.168	831.40
1065	10:54:02.318	855.03
1066	10:54:02.468	832.38
1067	10:54:02.618	826.39
1068	10:54:02.768	832.54
1069	10:54:02.918	859.40
1070	10:54:03.068	822.83
1071	10:54:03.218	820.56
1072	10:54:03.368	848.56
1073	10:54:03.518	835.61
1074	10:54:03.668	832.70
1075	10:54:03.817	820.40
1076	10:54:03.967	840.95
1077	10:54:04.117	841.92
1078	10:54:04.267	853.41
1079	10:54:04.417	838.36
1080	10:54:04.567	849.37
1081	10:54:04.717	824.45
1082	10:54:04.867	839.01
1083	10:54:05.017	828.65
1084	10:54:05.167	816.36
1085	10:54:05.317	849.37
1086	10:54:05.467	830.92
1087	10:54:05.617	846.94
1088	10:54:05.767	847.26
1089	10:54:05.917	839.33
1090	10:54:06.067	836.58
1091	10:54:06.217	830.27
1092	10:54:06.367	853.09
1093	10:54:06.517	823.80
1094	10:54:06.667	844.35
1095	10:54:06.817	835.29
1096	10:54:06.967	837.72
1097	10:54:07.117	828.65
1098	10:54:07.267	836.74
1099	10:54:07.417	837.88
1100	10:54:07.566	822.67
1101	10:54:07.716	839.50
1102	10:54:07.866	845.16
1103	10:54:08.016	837.23
1104	10:54:08.166	827.68
1105	10:54:08.316	828.33
1106	10:54:08.466	832.86
1107	10:54:08.616	833.18
1108	10:54:08.766	844.51
1109	10:54:08.916	831.24

1110	10:54:09.066	830.60
1111	10:54:09.216	827.84
1112	10:54:09.366	827.68
1113	10:54:09.516	828.01
1114	10:54:09.666	834.80
1115	10:54:09.816	826.39
1116	10:54:09.966	830.60
1117	10:54:10.116	844.35
1118	10:54:10.266	830.60
1119	10:54:10.416	830.60
1120	10:54:10.566	817.49
1121	10:54:10.716	815.87
1122	10:54:10.865	831.89
1123	10:54:11.015	839.33
1124	10:54:11.165	828.82
1125	10:54:11.315	822.34
1126	10:54:11.465	838.52
1127	10:54:11.615	834.16
1128	10:54:11.765	834.48
1129	10:54:11.916	819.92
1130	10:54:12.066	829.95
1131	10:54:12.216	822.99
1132	10:54:12.366	842.89
1133	10:54:12.515	850.01
1134	10:54:12.665	839.33
1135	10:54:12.815	831.73
1136	10:54:12.965	833.02
1137	10:54:13.115	841.44
1138	10:54:13.265	849.04
1139	10:54:13.415	815.22
1140	10:54:13.565	826.55
1141	10:54:13.715	834.96
1142	10:54:13.865	830.43
1143	10:54:14.015	833.35
1144	10:54:14.164	845.16
1145	10:54:14.314	827.36
1146	10:54:14.464	837.72
1147	10:54:14.614	834.48
1148	10:54:14.764	839.50
1149	10:54:14.914	839.33
1150	10:54:15.074	828.01
1151	10:54:15.214	818.30
1152	10:54:15.364	831.08
1153	10:54:15.524	847.91
1154	10:54:15.674	836.26
1155	10:54:15.824	837.88
1156	10:54:15.974	825.26

1157	10:54:16.124	831.57
1158	10:54:16.274	833.83
1159	10:54:16.424	840.95
1160	10:54:16.574	824.45
1161	10:54:16.724	850.82
1162	10:54:16.873	818.78
1163	10:54:17.023	839.17
1164	10:54:17.173	846.45
1165	10:54:17.323	833.35
1166	10:54:17.473	829.62
1167	10:54:17.623	831.57
1168	10:54:17.773	823.80
1169	10:54:17.923	846.13
1170	10:54:18.073	831.57
1171	10:54:18.223	830.76
1172	10:54:18.373	844.19
1173	10:54:18.523	845.64
1174	10:54:18.672	852.60
1175	10:54:18.822	825.42
1176	10:54:18.973	836.42
1177	10:54:19.123	844.84
1178	10:54:19.273	813.44
1179	10:54:19.423	840.79
1180	10:54:19.573	837.07
1181	10:54:19.723	838.36
1182	10:54:19.873	825.74
1183	10:54:20.022	830.11
1184	10:54:20.172	832.21
1185	10:54:20.323	842.57
1186	10:54:20.473	840.79
1187	10:54:20.623	842.25
1188	10:54:20.773	853.57
1189	10:54:20.923	834.16
1190	10:54:21.073	820.40
1191	10:54:21.222	845.16
1192	10:54:21.372	827.36
1193	10:54:21.522	828.65
1194	10:54:21.672	814.90
1195	10:54:21.822	834.32
1196	10:54:21.972	848.40
1197	10:54:22.122	836.91
1198	10:54:22.272	838.20
1199	10:54:22.422	829.95
1200	10:54:22.572	835.13
1201	10:54:22.722	833.18
1202	10:54:22.872	849.20
1203	10:54:23.022	822.02

1204	10:54:23.172	869.11
1205	10:54:23.322	838.85
1206	10:54:23.472	820.89
1207	10:54:23.622	828.65
1208	10:54:23.772	846.29
1209	10:54:23.921	830.60
1210	10:54:24.071	833.35
1211	10:54:24.221	834.64
1212	10:54:24.371	835.94
1213	10:54:24.521	818.14
1214	10:54:24.671	829.95
1215	10:54:24.821	830.92
1216	10:54:24.971	843.86
1217	10:54:25.121	832.86
1218	10:54:25.271	834.16
1219	10:54:25.421	825.26
1220	10:54:25.571	840.14
1221	10:54:25.721	844.35
1222	10:54:25.871	839.66
1223	10:54:26.021	821.53
1224	10:54:26.171	836.26
1225	10:54:26.320	822.02
1226	10:54:26.470	829.95
1227	10:54:26.620	839.01
1228	10:54:26.770	830.60
1229	10:54:26.920	833.51
1230	10:54:27.070	836.42
1231	10:54:27.220	823.96
1232	10:54:27.370	820.24
1233	10:54:27.520	837.88
1234	10:54:27.670	828.82
1235	10:54:27.820	849.37
1236	10:54:27.970	837.23
1237	10:54:28.120	837.39
1238	10:54:28.270	827.04
1239	10:54:28.419	830.60
1240	10:54:28.569	830.76
1241	10:54:28.719	820.72
1242	10:54:28.869	827.52
1243	10:54:29.019	831.73
1244	10:54:29.169	842.25
1245	10:54:29.319	840.47
1246	10:54:29.469	834.64
1247	10:54:29.619	842.08
1248	10:54:29.769	833.83
1249	10:54:29.919	825.58
1250	10:54:30.069	840.79

1251	10:54:30.218	843.06
1252	10:54:30.369	837.88
1253	10:54:30.519	847.10
1254	10:54:30.669	825.90
1255	10:54:30.819	833.02
1256	10:54:30.968	833.02
1257	10:54:31.118	843.70
1258	10:54:31.268	856.65
1259	10:54:31.418	840.79
1260	10:54:31.568	835.77
1261	10:54:31.718	837.88
1262	10:54:31.868	837.55
1263	10:54:32.018	826.55
1264	10:54:32.168	837.72
1265	10:54:32.318	833.67
1266	10:54:32.468	839.01
1267	10:54:32.618	838.20
1268	10:54:32.768	835.94
1269	10:54:32.917	810.53
1270	10:54:33.067	839.17
1271	10:54:33.217	827.36
1272	10:54:33.367	832.05
1273	10:54:33.517	845.16
1274	10:54:33.667	833.51
1275	10:54:33.817	837.55
1276	10:54:33.967	826.87
1277	10:54:34.117	843.86
1278	10:54:34.267	842.89
1279	10:54:34.417	850.34
1280	10:54:34.567	842.41
1281	10:54:34.717	831.89
1282	10:54:34.866	838.85
1283	10:54:35.016	824.93
1284	10:54:35.166	847.26
1285	10:54:35.322	833.99
1286	10:54:35.472	836.74
1287	10:54:35.622	845.97
1288	10:54:35.772	837.55
1289	10:54:35.922	844.19
1290	10:54:36.071	838.69
1291	10:54:36.221	820.89
1292	10:54:36.371	828.01
1293	10:54:36.521	838.69
1294	10:54:36.671	851.96
1295	10:54:36.821	830.11
1296	10:54:36.971	824.45
1297	10:54:37.121	828.65

1298	10:54:37.271	841.11
1299	10:54:37.421	833.02
1300	10:54:37.571	852.60
1301	10:54:37.721	824.12
1302	10:54:37.871	834.96
1303	10:54:38.020	852.60
1304	10:54:38.170	844.67
1305	10:54:38.320	846.62
1306	10:54:38.470	846.13
1307	10:54:38.620	839.33
1308	10:54:38.770	839.33
1309	10:54:38.920	845.64
1310	10:54:39.070	831.89
1311	10:54:39.220	840.30
1312	10:54:39.370	832.05
1313	10:54:39.520	827.84
1314	10:54:39.670	829.14
1315	10:54:39.820	836.42
1316	10:54:39.970	843.70
1317	10:54:40.120	829.62
1318	10:54:40.270	838.85
1319	10:54:40.420	841.92
1320	10:54:40.570	829.62
1321	10:54:40.719	854.71
1322	10:54:40.870	841.76
1323	10:54:41.020	830.43
1324	10:54:41.170	823.80
1325	10:54:41.320	825.09
1326	10:54:41.470	848.88
1327	10:54:41.620	840.79
1328	10:54:41.770	828.17
1329	10:54:41.920	844.67
1330	10:54:42.070	858.91
1331	10:54:42.220	835.45
1332	10:54:42.370	839.17
1333	10:54:42.520	826.39
1334	10:54:42.670	832.38
1335	10:54:42.819	834.48
1336	10:54:42.969	833.67
1337	10:54:43.119	836.74
1338	10:54:43.269	850.50
1339	10:54:43.419	832.38
1340	10:54:43.569	825.09
1341	10:54:43.719	833.02
1342	10:54:43.869	841.28
1343	10:54:44.019	829.30
1344	10:54:44.169	841.44

1345	10:54:44.319	839.82
1346	10:54:44.469	832.38
1347	10:54:44.618	845.00
1348	10:54:44.768	846.78
1349	10:54:44.919	845.16
1350	10:54:45.069	838.20
1351	10:54:45.219	844.03
1352	10:54:45.363	835.29
1353	10:54:45.513	843.06
1354	10:54:45.662	838.52
1355	10:54:45.812	843.54
1356	10:54:45.963	849.04
1357	10:54:46.113	844.35
1358	10:54:46.263	840.63
1359	10:54:46.412	833.67
1360	10:54:46.562	834.96
1361	10:54:46.715	841.44
1362	10:54:46.865	837.72
1363	10:54:47.015	837.39
1364	10:54:47.164	831.24
1365	10:54:47.314	820.72
1366	10:54:47.464	836.10
1367	10:54:47.614	828.49
1368	10:54:47.764	828.65
1369	10:54:47.914	840.79
1370	10:54:48.064	833.35
1371	10:54:48.214	829.62
1372	10:54:48.364	830.92
1373	10:54:48.514	827.52
1374	10:54:48.664	841.76
1375	10:54:48.814	832.54
1376	10:54:48.964	843.86
1377	10:54:49.113	836.42
1378	10:54:49.263	807.94
1379	10:54:49.413	839.82
1380	10:54:49.563	829.14
1381	10:54:49.713	838.52
1382	10:54:49.863	849.85
1383	10:54:50.013	833.67
1384	10:54:50.163	837.23
1385	10:54:50.313	822.67
1386	10:54:50.463	849.20
1387	10:54:50.613	844.67
1388	10:54:50.763	836.58
1389	10:54:50.913	832.86
1390	10:54:51.063	821.21
1391	10:54:51.213	819.27

1392	10:54:51.363	829.46
1393	10:54:51.513	845.81
1394	10:54:51.663	831.73
1395	10:54:51.813	831.89
1396	10:54:51.963	830.76
1397	10:54:52.113	841.60
1398	10:54:52.263	856.00
1399	10:54:52.413	836.26
1400	10:54:52.562	838.69
1401	10:54:52.712	830.76
1402	10:54:52.862	845.81
1403	10:54:53.012	841.28
1404	10:54:53.162	848.88
1405	10:54:53.312	857.78
1406	10:54:53.462	829.62
1407	10:54:53.612	835.13
1408	10:54:53.762	844.03
1409	10:54:53.912	847.91
1410	10:54:54.062	836.10
1411	10:54:54.212	848.23
1412	10:54:54.361	831.57
1413	10:54:54.511	832.86
1414	10:54:54.661	850.18
1415	10:54:54.811	858.91
1416	10:54:54.961	847.75
1417	10:54:55.111	852.76
1418	10:54:55.266	830.76
1419	10:54:55.416	845.00
1420	10:54:55.565	849.37
1421	10:54:55.715	840.79
1422	10:54:55.865	844.84
1423	10:54:56.016	841.60
1424	10:54:56.166	852.60
1425	10:54:56.316	840.63
1426	10:54:56.466	823.48
1427	10:54:56.615	833.02
1428	10:54:56.765	834.32
1429	10:54:56.920	839.50
1430	10:54:57.065	847.42
1431	10:54:57.215	852.60
1432	10:54:57.365	849.20
1433	10:54:57.515	834.16
1434	10:54:57.665	828.65
1435	10:54:57.815	825.26
1436	10:54:57.965	836.74
1437	10:54:58.115	847.59
1438	10:54:58.265	851.15

1439	10:54:58.414	839.17
1440	10:54:58.564	830.11
1441	10:54:58.714	843.86
1442	10:54:58.864	848.23
1443	10:54:59.014	833.67
1444	10:54:59.164	842.73
1445	10:54:59.314	829.30
1446	10:54:59.464	841.28
1447	10:54:59.614	845.00
1448	10:54:59.764	831.73
1449	10:54:59.914	837.07
1450	10:55:00.064	842.08
1451	10:55:00.214	844.03
1452	10:55:00.363	832.05
1453	10:55:00.513	835.13
1454	10:55:00.663	859.40
1455	10:55:00.813	836.26
1456	10:55:00.964	832.86
1457	10:55:01.114	839.33
1458	10:55:01.264	844.84
1459	10:55:01.414	844.84
1460	10:55:01.564	850.01
1461	10:55:01.714	846.94
1462	10:55:01.863	844.03
1463	10:55:02.013	843.70
1464	10:55:02.163	849.20
1465	10:55:02.313	845.97
1466	10:55:02.463	870.56
1467	10:55:02.613	856.16
1468	10:55:02.763	836.10
1469	10:55:02.913	847.91
1470	10:55:03.063	840.47
1471	10:55:03.213	842.25
1472	10:55:03.363	839.82
1473	10:55:03.513	835.61
1474	10:55:03.662	845.16
1475	10:55:03.812	828.33
1476	10:55:03.962	846.62
1477	10:55:04.112	858.43
1478	10:55:04.262	832.38
1479	10:55:04.412	849.04
1480	10:55:04.562	853.73
1481	10:55:04.712	855.68
1482	10:55:04.862	841.28
1483	10:55:05.012	851.31
1484	10:55:05.162	826.55
1485	10:55:05.314	843.86

1486	10:55:05.464	847.91
1487	10:55:05.614	834.16
1488	10:55:05.764	852.76
1489	10:55:05.914	825.90
1490	10:55:06.065	834.80
1491	10:55:06.215	842.73
1492	10:55:06.364	839.66
1493	10:55:06.514	831.08
1494	10:55:06.664	836.58
1495	10:55:06.814	854.71
1496	10:55:06.964	860.69
1497	10:55:07.114	837.07
1498	10:55:07.264	854.71
1499	10:55:07.414	853.73
1500	10:55:07.564	844.67
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2008	10:56:23.783	950.66
2009	10:56:23.932	955.68
2010	10:56:24.082	964.42
2011	10:56:24.232	955.84
2012	10:56:24.382	948.07
2013	10:56:24.532	952.12
2014	10:56:24.682	935.78
2015	10:56:24.832	966.84
2016	10:56:24.982	959.89
2017	10:56:25.132	981.41
2018	10:56:25.282	976.71
2019	10:56:25.435	960.05
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2022	10:56:25.885	958.27
2023	10:56:26.035	989.82
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2025	10:56:26.335	962.47
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2028	10:56:26.785	981.73
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2489	10:57:35.924	1125.75
2490	10:57:36.074	1115.55
2491	10:57:36.224	1117.01
2492	10:57:36.374	1130.60
2493	10:57:36.524	1095.97
2494	10:57:36.674	1108.11
2495	10:57:36.824	1139.18
2496	10:57:36.974	1125.10
2497	10:57:37.124	1110.86
2498	10:57:37.274	1108.59
2499	10:57:37.424	1113.45
2500	10:57:37.574	1147.75
2501	10:57:37.724	1112.48
2502	10:57:37.874	1112.80
2503	10:57:38.024	1105.68
2504	10:57:38.174	1124.45
2505	10:57:38.323	1104.06
2506	10:57:38.473	1127.69
2507	10:57:38.623	1117.82
2508	10:57:38.773	1110.86
2509	10:57:38.923	1121.54
2510	10:57:39.073	1114.90
2511	10:57:39.223	1120.24
2512	10:57:39.373	1122.51
2513	10:57:39.523	1099.05
2514	10:57:39.673	1134.32
2515	10:57:39.823	1122.51
2516	10:57:39.973	1114.90
2517	10:57:40.123	1107.14
2518	10:57:40.273	1126.72
2519	10:57:40.422	1126.07

2520	10:57:40.573	1113.45
2521	10:57:40.723	1122.99
2522	10:57:40.873	1129.14
2523	10:57:41.023	1131.57
2524	10:57:41.173	1137.72
2525	10:57:41.322	1133.35
2526	10:57:41.472	1140.31
2527	10:57:41.622	1135.45
2528	10:57:41.772	1121.38
2529	10:57:41.922	1118.95
2530	10:57:42.073	1122.99
2531	10:57:42.223	1148.40
2532	10:57:42.373	1144.19
2533	10:57:42.523	1136.26
2534	10:57:42.673	1126.07
2535	10:57:42.822	1146.30
2536	10:57:42.972	1125.75
2537	10:57:43.122	1111.67
2538	10:57:43.272	1117.82
2539	10:57:43.422	1143.06
2540	10:57:43.572	1137.56
2541	10:57:43.722	1136.75
2542	10:57:43.872	1125.10
2543	10:57:44.022	1137.23
2544	10:57:44.172	1122.02
2545	10:57:44.332	1138.04
2546	10:57:44.482	1147.11
2547	10:57:44.632	1133.51
2548	10:57:44.781	1141.60
2549	10:57:44.931	1121.86
2550	10:57:45.081	1130.92
2551	10:57:45.231	1139.50
2552	10:57:45.381	1127.53
2553	10:57:45.535	1126.39
2554	10:57:45.684	1142.25
2555	10:57:45.834	1141.44
2556	10:57:45.984	1147.59
2557	10:57:46.134	1155.36
2558	10:57:46.284	1137.56
2559	10:57:46.434	1144.52
2560	10:57:46.584	1149.05
2561	10:57:46.734	1144.03
2562	10:57:46.884	1155.52
2563	10:57:47.034	1130.92
2564	10:57:47.184	1131.73
2565	10:57:47.334	1132.70
2566	10:57:47.484	1140.63

2567	10:57:47.634	1139.66
2568	10:57:47.784	1157.46
2569	10:57:47.934	1138.04
2570	10:57:48.084	1163.29
2571	10:57:48.234	1145.33
2572	10:57:48.383	1150.34
2573	10:57:48.533	1148.40
2574	10:57:48.683	1150.34
2575	10:57:48.833	1156.65
2576	10:57:48.983	1150.83
2577	10:57:49.133	1153.42
2578	10:57:49.283	1159.73
2579	10:57:49.433	1151.64
2580	10:57:49.583	1148.08
2581	10:57:49.733	1156.81
2582	10:57:49.883	1138.69
2583	10:57:50.033	1170.41
2584	10:57:50.183	1167.49
2585	10:57:50.332	1138.04
2586	10:57:50.482	1159.40
2587	10:57:50.632	1160.37
2588	10:57:50.782	1154.55
2589	10:57:50.932	1151.80
2590	10:57:51.082	1150.34
2591	10:57:51.232	1165.07
2592	10:57:51.382	1145.65
2593	10:57:51.532	1155.36
2594	10:57:51.682	1161.02
2595	10:57:51.832	1157.95
2596	10:57:51.982	1154.39
2597	10:57:52.132	1155.84
2598	10:57:52.282	1149.69
2599	10:57:52.432	1173.00
2600	10:57:52.582	1154.55
2601	10:57:52.732	1158.76
2602	10:57:52.882	1159.56
2603	10:57:53.032	1152.93
2604	10:57:53.182	1140.15
2605	10:57:53.332	1165.23
2606	10:57:53.482	1152.93
2607	10:57:53.632	1160.05
2608	10:57:53.782	1142.74
2609	10:57:53.931	1154.87
2610	10:57:54.081	1143.87
2611	10:57:54.231	1159.24
2612	10:57:54.381	1159.89
2613	10:57:54.531	1165.07

2614	10:57:54.681	1169.92
2615	10:57:54.831	1155.20
2616	10:57:54.981	1147.59
2617	10:57:55.131	1162.15
2618	10:57:55.281	1166.52
2619	10:57:55.431	1164.74
2620	10:57:55.585	1176.39
2621	10:57:55.735	1163.29
2622	10:57:55.885	1178.66
2623	10:57:56.035	1168.95
2624	10:57:56.185	1160.54
2625	10:57:56.335	1183.19
2626	10:57:56.485	1189.82
2627	10:57:56.635	1151.47
2628	10:57:56.785	1143.38
2629	10:57:56.935	1164.10
2630	10:57:57.085	1172.02
2631	10:57:57.235	1166.68
2632	10:57:57.385	1164.58
2633	10:57:57.535	1153.25
2634	10:57:57.685	1160.54
2635	10:57:57.835	1156.17
2636	10:57:57.984	1167.98
2637	10:57:58.134	1168.46
2638	10:57:58.284	1153.74
2639	10:57:58.434	1172.02
2640	10:57:58.584	1161.34
2641	10:57:58.734	1155.20
2642	10:57:58.884	1172.67
2643	10:57:59.034	1169.76
2644	10:57:59.184	1162.80
2645	10:57:59.334	1173.97
2646	10:57:59.484	1149.53
2647	10:57:59.634	1156.65
2648	10:57:59.784	1185.13
2649	10:57:59.933	1172.83
2650	10:58:00.083	1165.07
2651	10:58:00.233	1150.34
2652	10:58:00.383	1163.77
2653	10:58:00.533	1154.71
2654	10:58:00.683	1166.68
2655	10:58:00.833	1166.85
2656	10:58:00.983	1175.58
2657	10:58:01.133	1174.61
2658	10:58:01.283	1167.01
2659	10:58:01.433	1166.20
2660	10:58:01.583	1153.90

10:58:01.733 1166.20
10:58:01.882 1165.39
10:58:02.038 1169.44
10:58:02.183 1170.89
10:58:02.333 1171.70
10:58:02.483 1189.99
10:58:02.633 1165.71
10:58:02.783 1172.35
10:58:02.933 1179.95
10:58:03.083 1179.14
10:58:03.232 1167.17
10:58:03.382 1169.76
10:58:03.532 1171.86
10:58:03.682 1182.38
10:58:03.832 1176.07
10:58:03.982 1184.16
10:58:04.132 1190.80
10:58:04.282 1180.28
10:58:04.433 1177.53
10:58:04.583 1194.84
10:58:04.732 1185.62
10:58:04.883 1177.04
10:58:05.033 1171.86
10:58:05.183 1172.67
10:58:05.333 1188.37
10:58:05.479 1206.33
10:58:05.629 1164.42
10:58:05.779 1174.29
10:58:05.929 1185.94
Trend type: Single 波长 at 339.35
积分时间（毫秒）: 10 (USB2+H14105)
平均次数: 15 (USB2+H14105)
平滑度: 0 (USB2+H14105)
去除暗噪声: 否 (USB2+H14105)
10:58:06.079 1164.58
10:58:06.229 1200.18
10:58:06.379 1197.75
10:58:06.529 1200.02
10:58:06.679 1176.72
10:58:06.829 1192.90
10:58:06.989 1180.28
10:58:07.129 1192.58
10:58:07.279 1206.17
10:58:07.429 1177.04
10:58:07.579 1202.61
10:58:07.729 1180.28
10:58:07.879 1185.78

2708	10:58:08.029	1175.58
2709	10:58:08.179	1190.63
2710	10:58:08.329	1194.36
2711	10:58:08.478	1198.56
2712	10:58:08.628	1186.10
2713	10:58:08.778	1200.50
2714	10:58:08.928	1188.04
2715	10:58:09.078	1178.50
2716	10:58:09.238	1195.00
2717	10:58:09.378	1215.39
2718	10:58:09.538	1202.12
2719	10:58:09.688	1179.31
2720	10:58:09.828	1185.78
2721	10:58:09.978	1208.76
2722	10:58:10.138	1188.69